What Does Reinforcement Learning Add to a Small Translator for African Languages?

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Abstract

Small language models translate many African languages poorly, and one common failure is that they do not finish their sentences: they stop midway or repeat themselves. We train a small English-to-African translation model with supervised fine-tuning, then with reinforcement learning that rewards outputs rated well by automatic quality scorers, and we ask what the second step adds. It improves every language we test, by about nine points of the standard chrF++ score on one benchmark. About half of that gain is finishing sentences and not repeating; a single decoding setting gives the fine-tuned model the same half without any extra training, while the rest of the gain remains. Spending the same computing time on more fine-tuning helps only the languages for which we supplied extra data, and by the rule we fixed beforehand the comparison is inconclusive. For scientific text, the extra fine-tuning step, not the reinforcement learning, carries the gain. On identical test sentences, our model stays about one point below NLLB-600M and about three below NLLB-3.3B (about 92% of its score), which is about 8.6 times larger, and a model distilled from NLLB-3.3B is ahead of ours on its own languages. We trained one model per setting, checked seed noise on a subset only, and used automatic scores only; no person judged the outputs.

1 Introduction

Machine translation into African languages is hard for small models. Training text is scarce, and small models have little capacity to spare. Large systems such as NLLB-200 (NLLB Team et al., 2022) help, but they are big, and the NLLB models and their outputs carry non-commercial licence terms.

Small models also fail in a characteristic way. After supervised fine-tuning (SFT), our model often does not finish its sentences: it stops midway, repeats a phrase until it runs out of space, writes in the wrong language, or copies the English input. Reinforcement learning (RL) with automatic quality scores as rewards is a natural remedy, and recent work reports gains for translation (Shao et al., 2024; Feng et al., 2025; Finkelstein et al., 2026). Those gains, however, may come from something simpler. Repeating-and-not-stopping is partly a property of greedy decoding, which a decoding setting can change; and RL costs computing time that more SFT could use instead.

We therefore ask four questions about a 385-million-parameter English-to-African translator, trained first with SFT and then with RL:

- **RQ1.** How much does RL improve translation, and for which languages?
- **RQ2.** How much of the gain is finishing and not repeating, rather than better translation, and can a decoding change provide it?
- **RQ3.** Does the same computing time spent on more SFT do as well?
- **RQ4.** How reliable is the gain, and how does the model compare with a much larger system?

Findings at a glance.

- RL improves every language we test on three standard benchmarks (RQ1).
- About half of the gain is finishing sentences and not repeating. A decoding setting recovers that half for the SFT model without training, but RL still adds clear gains beyond it (RQ2).
- Decoding tricks such as beam search and minimum Bayes risk (MBR) decoding help the SFT model a lot and the RL model little (RQ2).
- Matching the compute with more SFT is inconclusive by our pre-registered rule: it matches RL on the 19 languages where it had extra data, and is far behind on the others (RQ3).
- Almost all of the gain arrives in the first RL stage (RQ4). On identical sentences the model is still about one chrF++ point below NLLB-600M and about three below NLLB-3.3B, about 92% of its

score at about one ninth of its size; a model distilled from NLLB-3.3B is ahead of ours on its languages (Section 5.1).

Contributions. (1) A controlled decomposition of an RL gain in small-model translation: a decoding control, a matched-compute control, seed and ablation runs, and an independent audit of the reward's own repetition check. (2) A decontaminated SFT and RL pipeline for 125 languages, with an audit of the training text. (3) Negative results: RL aimed at weak languages, science fine-tuning on a quality-estimation metric, and whole-document translation. Model artefacts: model weights at `https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl` (see Ethical considerations). We do *not* claim that RL beats more fine-tuning in general, that our model beats NLLB, or any human-rated quality: all scoring is automatic.

2 Related Work

Translation for African languages. Community benchmarks and corpora made progress measurable: Masakhane (Nekoto et al., 2020), MAFAND-MT (Adelani et al., 2022), FLORES-200 and its community successor FLORES+ (Goyal et al., 2022; Open Language Data Initiative, 2025), NTREX-128 (Federmann et al., 2022), SALT (Akera et al., 2022) and SMOL (Caswell et al., 2025). NLLB-200 (NLLB Team et al., 2022) is the reference system. Recent language models for African languages are much larger than ours (Yu et al., 2026; Akera et al., 2025); TranslatePsy-AfriSLM (Gritta et al., 2026) is the nearest small competitor. For science text, AfriScience-MT (Abdulmumin et al., 2026) uses SSA-COMET (Li et al., 2025) as its main metric.

Reinforcement learning for translation. GRPO (Shao et al., 2024; Yu et al., 2025) powers recent translation systems with reasoning-style training (Feng et al., 2025; Wang et al., 2025; Cheng et al., 2025; Zheng et al., 2025), and TranslateGemma rewards with an ensemble of quality models (Finkelstein et al., 2026). Closest to us, other work trains NLLB models with round-trip or reference-free rewards (Attia and Aji, 2026; Garcia-Estrada et al., 2026) and studies reward hacks in quality-estimation rewards (Liu et al., 2026; Han et al., 2026). Whether RL generalises beyond what supervised training memorises has been asked for reasoning (Chu et al., 2025; Yue et al., 2025). For translation, we searched for a comparison at matched computing time and found none; the nearest compares reference-free RL with three epochs of supervised training (Garcia-Estrada et al., 2026).

Decoding and evaluation. Minimum Bayes risk and quality-aware decoding (Freitag et al., 2022; Fernandes et al., 2022) aim at similar gains without training, so we measure them as controls. AfriCOMET (Wang et al., 2024), chrF++ and sacreBLEU (Popović, 2017; Post, 2018) and the paired bootstrap (Koehn, 2004) are our metrics and tests. Learned rewards are easy to over-optimise (Gao et al., 2023; Coste et al., 2024), which is why we keep the metrics we report separate from the metrics we reward. We claim no priority for RL in translation, for small models, or for language coverage.

3 Approach

Figure 1 shows the pipeline. We describe the model and the two training steps, then the variants used as controls and extensions.

Model. We start from Gemma-3-270m-it (Gemma Team et al., 2025) and add layers to reach 385M parameters (39 layers) (cf. Wu et al., 2024; Kim et al., 2023). The task is English to one of 125 languages. A single extra token in the vocabulary names the target language, and the model answers with that token followed by the translation.

Step 1: supervised fine-tuning (the SFT baseline). We train on English-to-African sentence pairs and instruction data. Before training, we remove every pair that matches the text of any benchmark we later use, checked on both sides of the pair. Part of the data comes from NLLB-mined sentences and a collection of translated instructions; we report their share and licences in the Ethical considerations. Training uses the Muon optimiser (Liu et al., 2025) and takes about 45 hours on one A100 GPU (Appendix I).

Step 2: reinforcement learning (the RL model). We use GRPO (Shao et al., 2024). For each English text the model samples 16 translations. Each translation receives a reward, and the model is nudged towards translations that score above the average of their group and away from those below it; a penalty keeps the model close to its starting point. Three stages follow one another, and the first does most of the work (Section 5.5). The reward has three parts. (i) A trans-

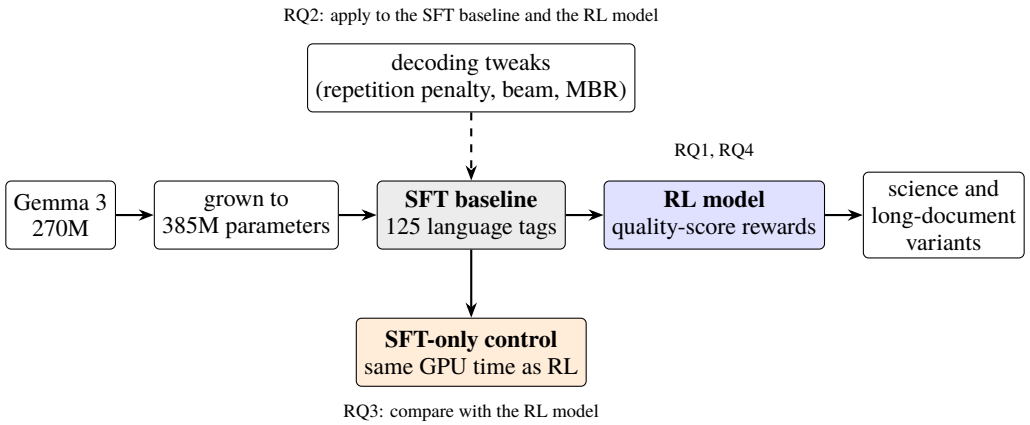


Figure 1: Overview. A 270M model is grown to 385M parameters and fine-tuned (the SFT baseline), then trained with reinforcement learning (the RL model). Two controls ask what else could give the same gain: decoding changes (RQ2) and more fine-tuning for the same GPU time (RQ3).

lation that fails a hard check scores the minimum: wrong or missing language tag, no end of text, a copy of the input, repetition, the wrong language, or an implausible length. (ii) Otherwise a quality score without a reference translation rates the output. We combine two quality models (Wang et al., 2024; Li et al., 2025), taking their mean unless they disagree strongly, and for languages without a validated quality model we reward agreement with the group's other samples. (iii) Small penalties discourage, among other things, unusual length, missing numbers or names, and missing diacritics. Exact settings are in Appendix I.

Variants. The *SFT-only control* continues SFT from the baseline for the same GPU time as the RL stages, using half replay data and half extra English-to-X pairs for 19 languages (Gritta et al., 2026). The *science variant* adds a short fine-tuning step on science text from AfriScience-MT (Abdulmumin et al., 2026) and then a second RL run; the *long-document variant* adds RL on long inputs. Internal names used in the appendix tables are listed in Appendix A.

4 Experimental Setup

Benchmarks. We translate English into African languages on the standard FLORES+ devtest set (Open Language Data Initiative, 2025) (1,012 sentences; 54 languages), NTREX-128 (Federmann et al., 2022) (24 languages) and MAFAND-MT (Adelani et al., 2022) (15 languages). Smaller sets, SmolSent (Caswell et al., 2025), SALT (Akera et al., 2022), GlotStoryBook (Kargaran et al., 2023), IbomMT (Kalejaiye et al., 2025) and FLORES-MOZ (MOZNLP, 2026), appear in the appendix, as does AfriScience-MT (Abdulmumin et al., 2026) for science text. Unless stated, we report the average over languages, with each language counting equally.

Metrics. *chrF++* compares character and word n-grams of the output with a human reference; higher is better. *Held-out AfriCOMET* is a learned score that compares source, output and reference; we never use it as a reward. The *finished rate* is the share of outputs that end properly; the *loop rate* is the share that repeat themselves. Both are the checks our reward also uses, so they are not independent evidence of better translation. We computed chrF++ with sacreBLEU 2.6.0 (word order 2, otherwise defaults) after Unicode normalisation; the full signature string was not saved.

Avoiding self-deception. First, *independence*: the metrics inside the reward (the two quality models, agreement between samples, and the finished, repetition and language checks) are not evidence that RL helped. Our evidence is chrF++ and held-out AfriCOMET against human references (Appendix N). Second, *paired tests*: every difference is a paired bootstrap over test sentences with a 95% interval, which covers sentence variance but not training randomness. Third, *contamination*: besides filtering the training pairs, we scanned 26 million training samples against all benchmark text and found no benchmark sentence of three or more words (Ethical considerations list the residuals). Fourth, *seeds*: every model is one training run; we measured seed noise only for the first RL stage on 20 languages.

Decoding. All main results use greedy decoding with up to 768 new tokens (vLLM, bf16). The *repetition penalty* used as a control is a decoding setting that discourages repeated tokens; we chose its strength on a development set by a rule fixed

in advance. Beam search and MBR decoding are described where they appear.

5 Results

5.1 How does the model compare with other systems?

Answer: the RL model beats the SFT baseline everywhere, but it stays below the NLLB systems. On identical FLORES+ sentences it scores about one chrF++ point below NLLB-600M and about three below NLLB-3.3B, which is about 92% of the larger system's score (Table 1, Figure 2). Held-out AfriCOMET gives the same order with a wider gap. The RL model is ahead of NLLB-600M in only some languages and of NLLB-3.3B in a few (last column). NTREX and MAFAND-MT show the same order against NLLB-3.3B; there, only language-level NLLB averages exist, so these rows have no sentence pairing and no AfriCOMET.

The 19 languages that TranslatePsy-0.8B covers show the limits of the comparison. TranslatePsy-0.8B is the strongest system there and clearly ahead of the RL model. It was distilled from NLLB-3.3B outputs, so its lead is partly a student–teacher effect and not independent evidence of a better design. The SFT-only control, which was trained on NLLB-3.3B translations for these languages, matches the RL model in chrF++ but not on the neural score. We decode greedily and NLLB with beam search (four beams); on a 20-language check, beam search or MBR moved the RL model by less than one chrF++ point either way (Section 5.3). SmolSent, SALT and three smaller benchmarks have no RL model; their per-language results are in Appendix B.

5.2 RQ1: How much does RL improve translation?

Answer: a lot, for every language we test. The RL model beats the SFT baseline on all three benchmarks (Table 2). Every language on FLORES+ and NTREX improves, and so does every MAFAND-MT language except Nigerian Pidgin, which does not change measurably. The gain is large where the baseline was weakest. On FLORES+ the baseline leaves about one output in six unfinished and repeats itself in about one in five; the RL model almost always finishes and rarely repeats (Table 2, lower rows).

Where RL does not help. For Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, quality stays very low (chrF++ below about 14) and RL makes one problem worse: the model increasingly answers in English. For Tunisian Arabic, English output rises from about one fifth of outputs to almost half (Appendix H). These languages were not among the RL prompts.

5.3 RQ2: Is the gain just finishing and not repeating?

Answer: about half of it is, and a decoding setting recovers that half for the SFT baseline; RL still adds clearly more. We tested this in three ways. First, we cut the baseline's output at the first repetition, which leaves the translation untouched; this recovers 49% of the chrF++ gain. Second, we gave the baseline a repetition penalty during decoding, with no further training. It recovers the same 49%. With the penalty on both models, RL is still about 4.4 chrF++ ahead (Table 3, Figure 3). This control is imperfect: the best penalty strength was at the edge of the range we searched, and stronger values looked better on a development set but were never evaluated fully, so the baseline may be undertuned.

Third, we tried stronger decoding. Beam search and MBR help the baseline much more than they help the RL model. MBR gives the baseline a large boost and the RL model a small one; for beam search, the RL result depends on how the search decides to stop, so we draw no conclusion from it. RL with plain greedy decoding stays about 1.6 points ahead of the best decoded baseline, which needs about 17 times the computation per sentence. The beam-search numbers of an earlier study that used 256 output tokens are not comparable, and we do not use them (Appendix D).

A caveat on our own checks. The repetition flag we report is also part of the reward, so we remeasured with a detector that shares none of its code. It finds about 1.8 times as many repetitive outputs as the reward's flag does. Rescoring the extra outputs at their language's average changes the RL model's chrF++ by at most 0.2, so the conclusion stands (Appendix H).

5.4 RQ3: Does the same compute on more SFT do as well?

Answer: inconclusive by the rule we wrote down beforehand. We continued SFT for the same GPU time as the three RL stages, on half replay data and half extra English-to-X pairs for 19 languages. The

system	chrF++	AfriCOMET	RL − system, chrF++	RL − system, AfriCOMET	RL ahead in
FLORES+, 44 languages, identical sentences (42,680)					
SFT baseline	25.6	0.550	+9.30 [+9.00, +9.55]	+0.073 [+0.071, +0.075]	44 of 44
SFT-only control[a]	28.6	0.586	+6.32 [+6.11, +6.52]	+0.038 [+0.036, +0.039]	34
RL model	**34.9**	**0.623**			
NLLB-600M	35.9	0.662	−1.05 [−1.24, −0.86]	−0.038 [−0.040, −0.036]	15
NLLB-3.3B	38.1	0.677	−3.21 [−3.38, −3.02]	−0.054 [−0.055, −0.052]	7
FLORES+, the 19 TranslatePsy languages (17,380 sentences)					
SFT baseline	34.0	0.589	+7.04 [+6.64, +7.41]	+0.054 [+0.051, +0.056]	
SFT-only control[a]	41.0	0.665	+0.08 [−0.13, +0.30]	−0.023 [−0.024, −0.021]	
RL model	**41.0**	**0.642**			
NLLB-600M	42.6	0.688	−1.54 [−1.78, −1.28]	−0.046 [−0.048, −0.043]	
NLLB-3.3B	44.7	0.701	−3.62 [−3.84, −3.39]	−0.059 [−0.061, −0.056]	
TranslatePsy-0.8B[b]	45.3	0.717	−4.23 [−4.39, −4.07]	−0.075 [−0.077, −0.073]	
NTREX (21 languages), MAFAND-MT (14); NLLB = language-level averages					
NTREX: SFT baseline	31.0	0.551	+7.97 [+7.73, +8.21]	+0.063 [+0.061, +0.065]	
NTREX: SFT-only control[a]	35.9	0.609	+3.14 [+2.97, +3.32]	+0.005 [+0.004, +0.006]	
NTREX: **RL model**	**39.0**	**0.614**			
NTREX: NLLB-3.3B	43.3	–	−4.26 [−5.22, −3.24][c]	–	1 of 21
MAFAND-MT: SFT baseline	34.5	0.614	+5.39 [+5.15, +5.63]	+0.040 [+0.038, +0.041]	
MAFAND-MT: SFT-only control[a]	39.5	0.670	+0.34 [+0.16, +0.52]	−0.016 [−0.017, −0.014]	
MAFAND-MT: **RL model**	**39.9**	**0.654**			
MAFAND-MT: NLLB-3.3B	44.6	–	−4.73 [−6.56, −3.12][c]	–	1 of 14

Table 1: Our models against other systems. chrF++ and held-out reference-based AfriCOMET: higher is better; averages over languages; 95% paired-bootstrap intervals for the RL model minus each system (positive = RL better), over test sentences. The 44 FLORES+ languages exclude the Arabic dialects, Tamazight, Dinka and Tamasheq, so averages are higher than elsewhere. [a]Trained on NLLB-3.3B translations for 19 languages (never released). [b]Distilled from NLLB-3.3B outputs. [c]Interval from a bootstrap over languages (no per-sentence NLLB output). Ours decode greedily; NLLB with 4 beams. Other benchmarks and per-language results: Appendix B.

Table 2: RL model against the SFT baseline ("baseline"). chrF++ (top rows, per benchmark with the number of languages) and AfriCOMET: higher is better. Averages over languages, greedy decoding, 95% paired-bootstrap intervals in brackets. One training run per model.

	baseline	RL model	gain
FLORES+ (54 lang.)	22.85	31.66	+8.81 [+8.59, +9.03]
NTREX (24)	29.25	37.17	+7.91 [+7.69, +8.12]
MAFAND-MT (15)	35.77	40.78	+5.01 [+4.76, +5.23]
all three, all lang.	26.59	34.55	+7.96 [+7.82, +8.12]
AfriCOMET, all three	0.565	0.628	+0.063 [+0.062, +0.064]
FLORES+: ends properly	83%	99.5%	
FLORES+: repeats	21%	5%	

extra pairs are NLLB-3.3B translations of web-text sentences (Gritta et al., 2026). We fixed the decision rule, the main comparison and the language groups before looking at any benchmark result for the control. The rule had four conditions; three were met. The fourth, that RL must also beat the control on the 19 languages that had extra data, was not met. So by our rule the result is inconclusive, and we did not change the rule afterwards.

In plain terms (Table 4, Figure 4): on the 19 languages with extra data, the control matches RL in chrF++ and finishes its sentences almost as reliably; it is even ahead on the learned metrics there, which were partly used to filter its data. On the other languages, the control barely improves on the baseline, while RL improves them all. Taking only the overall number, RL looks better; taking only the 19 languages, the two look equal. Neither alone is the registered verdict.

Caveats. The control is one run with one learning rate and one stopping point; a different mix or a longer run could change the picture. It is also not data-matched: its extra pairs come from a larger system (NLLB-3.3B). We therefore claim neither that RL is a better use of compute than SFT nor the opposite.

5.5 RQ4: How reliable is the gain, and how does the model compare with a larger system?

Answer: the gain is robust to random seeds, nearly all of it comes from the first RL stage, and the model reaches about 92% of NLLB-3.3B's chrF++ on shared languages (Section 5.1). The first RL stage produces 94% of the gain reached after the second stage, and the two later stages add little. Two extra random seeds of the first stage, run on 20 languages, differ by only 0.06 chrF++, far less than the gain itself. The last RL step is about the size of that seed noise, so we do not call it a gain (Figure 5).

We also tested four features that we added during the first stage. Together they are worth about

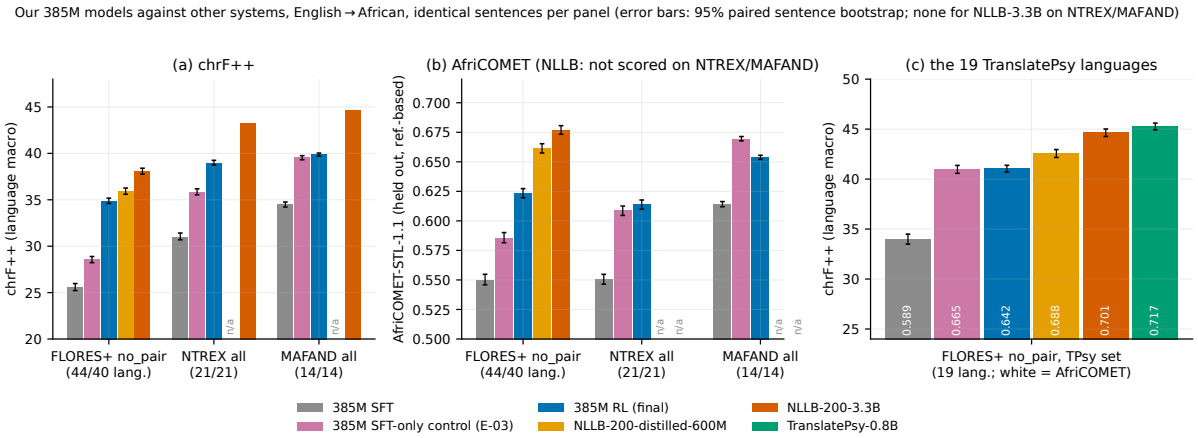


Figure 2: Our models against other systems (language averages on identical sentences; higher is better). (a) chrF++ and (b) held-out AfriCOMET on FLORES+, NTREX and MAFAND-MT; NLLB has no AfriCOMET score and no per-sentence output on NTREX and MAFAND-MT. (c) The 19 TranslatePsy languages, with AfriCOMET printed in the bars. **The vertical axes do not start at zero.** Error bars: 95% paired bootstrap over sentences; none for NLLB-3.3B on NTREX and MAFAND-MT. "SFT-only control" was trained on NLLB-3.3B translations for 19 languages.

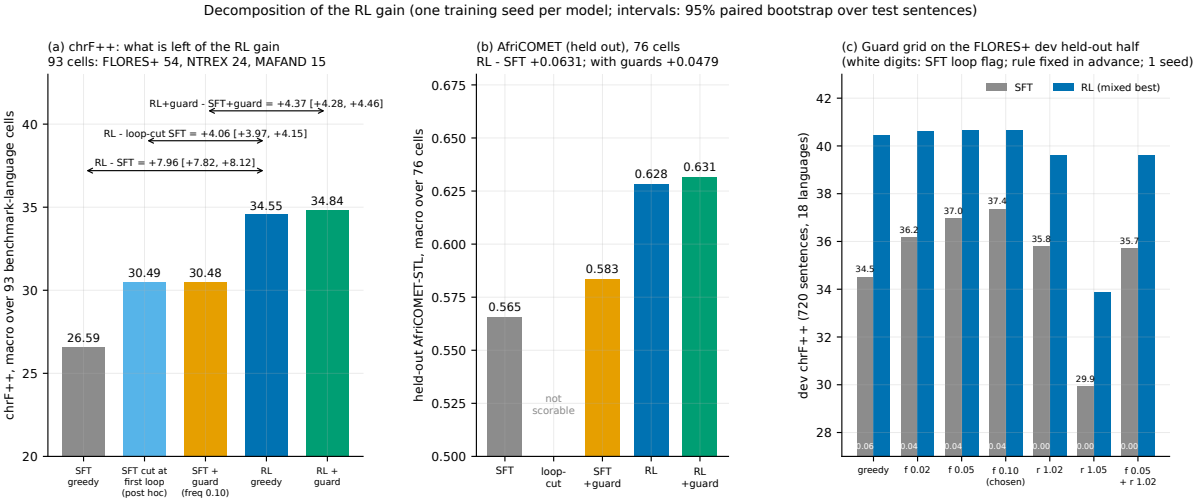


Figure 3: What is left of the RL gain once finishing is fixed. (a) chrF++, higher is better, averaged over all three benchmarks: the SFT baseline, the baseline with its repeated tails cut off, the baseline with a repetition penalty, the RL model, and the RL model with the same penalty. Arrows show paired differences with 95% intervals. (b) The same for held-out AfriCOMET. (c) How the penalty strength was chosen on a development set; the chosen value is at the edge of the grid.

0.3 chrF++ at 100 steps, mostly because of one of them, a term that rewards agreement with human reference translations for seven languages. A beam-search guide, which costs about a quarter more time per step, is not shown to be worth it, and none of the features causes the drift from the starting model.

NLLB-3.3B is about 8.6 times larger than our model, and Section 5.1 gives the comparison. The RL model closes about three quarters of the gap that the baseline leaves to NLLB-3.3B, and the remaining gap is not a matter of unfinished sentences. On the 43 languages shared with the published NLLB table the RL model scores about 92% of NLLB-3.3B's chrF++ (Appendix B). It matches or beats NLLB-3.3B on seven languages, all among those where NLLB scores lowest. Caveats: NLLB's published numbers use a slightly different test set and beam search; the hardest languages have no published NLLB number; and for a few languages the reward itself used NLLB outputs as a target. We therefore do not claim to beat NLLB. Compared with our earlier 270M model run in the same harness, the 385M SFT baseline is behind, the RL model is level on two benchmarks and behind on one, and it is behind at document level (Appendix B).

5.6 Further results: science text and long documents

Answer: for science text, the extra fine-tuning step carries the gain, and whole-document

Table 3: chrF++ with and without decoding changes (higher is better). Top: all three benchmarks, repetition penalty = a decoding setting that discourages repeated tokens (change without interval; per-benchmark intervals in Appendix D). Bottom: a 20-language subset with 200 sentences each; beam search (4 beams) stops when 4 hypotheses have ended, "HF rule" is a second stopping rule; MBR = choose the sample most similar to the others (16 samples). "Change" = difference to greedy decoding of the same model with a 95% paired-bootstrap interval. Per-language results: Appendix D.

decoding	SFT baseline	change	RL model	change	RL − SFT
greedy (all three benchmarks)	26.6		34.6		+7.96 [+7.82, +8.12]
+ repetition penalty	30.5	+3.9	34.8	+0.3	+4.37 [+4.28, +4.46]
greedy (20 languages)	32.3		39.9		+7.56 [+6.89, +8.19]
beam search, 4 beams	36.0	+3.71 [+2.78, +4.47]	39.1	−0.77 [−1.45, −0.25]	+3.07 [+2.67, +3.46]
beam search, HF rule	33.3	+0.95 [+0.28, +1.66]	40.7	+0.83 [+0.67, +1.00]	–
MBR, 16 samples	38.3	+5.96 [+5.37, +6.49]	40.8	+0.92 [+0.81, +1.03]	+2.52 [+2.26, +2.80]

E-03 matched-compute control: continued SFT (LR 1e-5, 6,000 steps, 50% replay + 50% TranslatePsy en->xx for 19 languages, equal pairs per language), one seed, one budget point; clean rows, language macro; the 3,100- and 4,800-step points were not evaluated

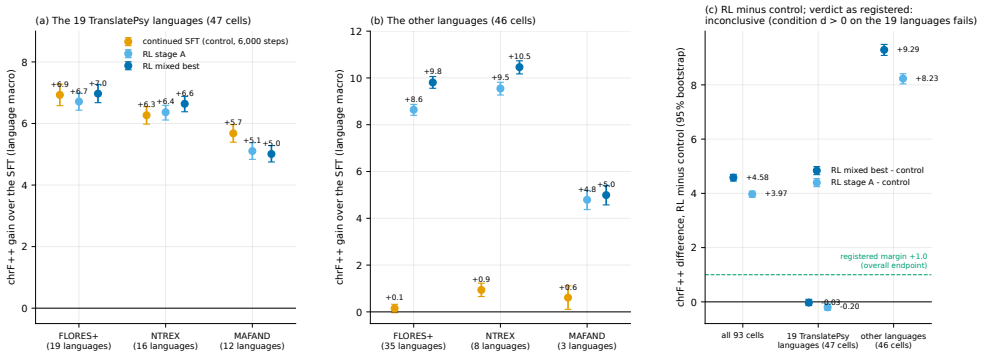


Figure 4: Gain over the SFT baseline in chrF++ (higher is better) for the SFT-only control, the first RL stage and the final RL model, on the 19 languages with extra control data (a) and on the others (b); (c) RL minus control, with the +1.0 margin of the registered rule. Error bars: 95% intervals. One run each; only the final control checkpoint was evaluated.

Table 4: SFT-only control against the RL model, chrF++ (higher is better), averaged over language–benchmark pairs, 95% paired-bootstrap intervals. The control received extra data for 19 languages.

languages	control	RL model	RL − control
all	29.97	34.55	+4.58 [+4.45, +4.70]
19 with extra data	41.37	41.34	−0.03 [−0.13, +0.09]
all others	18.32	27.61	+9.29 [+9.09, +9.49]

translation still does not work. A short fine-tuning step on science text raises chrF on AfriScience-MT but not the neural score the benchmark uses. An RL run started from that model adds a little more. To ask whether the science step is needed, we ran the same RL without it (Table 5): it ends about 1.4 chrF below the model that had the science step (one run each), and it gains little over its own start. The neural quality score rates the two models as equal, although chrF and held-out AfriCOMET rate the no-science-step model lower; because that score belongs to the family of scores the RL optimises, we do not treat it as evidence of quality. Translating whole documents in one pass fails for all our models. RL on long inputs raises the share of finished documents from 70% to 82%, but about three quarters of the documents still contain loops, so we translate sentence by sentence (Appendix G).

Table 5: Science text (AfriScience-MT test sentences, six languages, one run each). Higher is better. The last row is 1.44 chrF [1.25, 1.61] below the third row.

model	chrF	AfriCOMET
RL model	45.4	0.683
+ science fine-tuning	46.6	0.688
+ science fine-tuning, then RL	47.1	0.693
RL without science fine-tuning	45.7	0.685

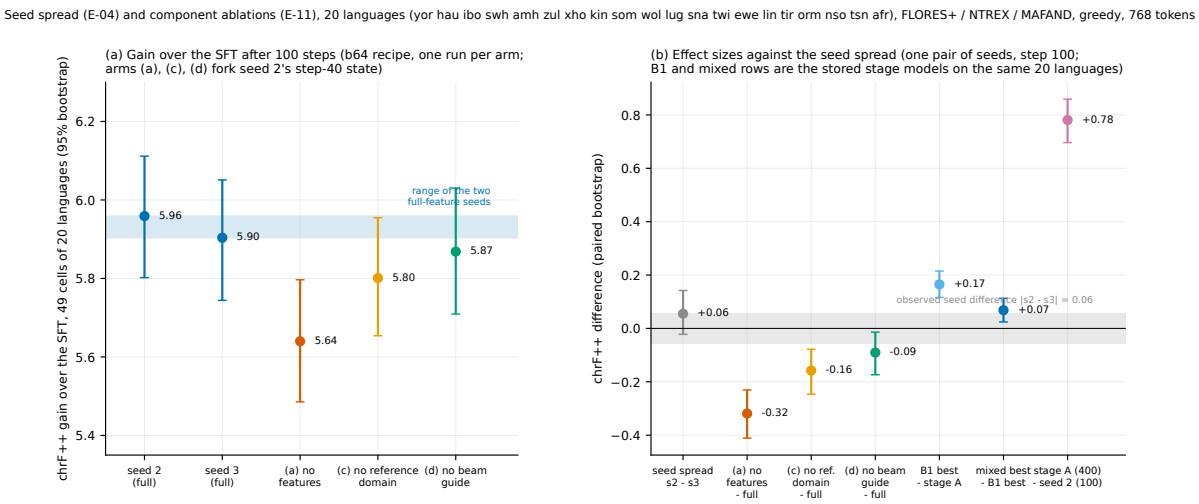


Figure 5: Seed noise and what four extra training features are worth (20 languages, chrF++ gain over the SFT baseline after 100 RL steps). (a) Two seeds of the full recipe (shaded band) and three runs that each remove a feature. (b) The same effects against the seed difference (shaded): removing all four features costs about 0.3 points, removing the beam-search guide is not separable from seed noise. Error bars: 95% intervals; one run per arm.

6 Discussion and Conclusion

What the evidence says. RL improves a small translator on every benchmark we use, but the gain has two parts that deserve separate names. About half is finishing sentences and not repeating, which a decoding setting also gives the fine-tuned model; the other half is better translation that the setting does not provide. Almost all of it arrives in the first RL stage. More fine-tuning for the same time is a serious alternative where it has extra in-language data, and a much weaker one where it does not; our registered comparison is therefore inconclusive, and what RL adds in our data is breadth, since it improved every language from the model's own samples and a quality score. For science text, the fine-tuning step, not RL, carried the gain.

What we do not claim. We do not claim that RL is a better use of compute than SFT in general, that the improvement beyond finishing is quality a human would confirm, that our model beats NLLB or our earlier 270M model, or that any reward ingredient other than the four tested features helped. The model is not usable for Arabic dialects, Tamazight, Dinka, Tamasheq or Kanuri.

What would settle the open points. Evaluating the stronger repetition penalties and the two unevaluated control checkpoints, more seeds for the later stages, and human judgement of outputs. Together with the licence work in the Ethical considerations, these are the next steps.

Limitations

Automatic scoring only. The references of FLORES+, NTREX-128 and MAFAND-MT are human translations, but all scoring is automatic: no human judgement of system outputs was collected. The neural scores belong to one family, and AfriCOMET has a length and entity bias. **One training run per model.** We measured seed noise only for the first RL stage, on 20 languages and 100 steps, with one pair of extra seeds. The last RL step and per-language differences below about half a point cannot be separated from seed noise. A second seed of the SFT baseline is impractical (about 45 GPU-hours). **The matched-compute control is one run:** one seed, one learning rate, one data mix with NLLB-3.3B targets and one evaluated stopping point. The repetition-penalty control chose its best value at the edge of the range searched. **Rewarded metrics are not evidence.** The finished, repetition and language checks, and the quality scores the RL optimises, are part of the reward. The independent repetition detector shares the idea of the reward's check and cannot see non-repetitive failures. **Ablations are limited:** three arms, 100 steps, one run each, 20 languages; later RL stages and most reward ingredients were not ablated, and mid-run changes (Ethical considerations) mean the stages are not one clean design. **Decoding and engine.** Headline numbers use one inference engine in bf16. The two engines differ in bf16 but agree in fp32; we did not run the RL model in fp32. Beam search and MBR cover 20 languages and one beam width. **Coverage and comparisons.** Fifty-one of the 125 languages have no reference test and are unscored. The model fails on Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri, where RL increases English output. Comparisons with the earlier 270M model and with NLLB mix differences in data, steps, test sets and decoding: our decoding is greedy, NLLB's is beam search, and on NTREX and MAFAND-MT only language-level NLLB averages exist (no sentence pairing, no AfriCOMET). Document results rest on 28 papers per language, and whole-document translation remains unusable. **Not run.** Human judgement of outputs; the NLLB-data ablation; scans of packed training shards for contamination; a 54-language decoding study; a science fine-tuning step at full strength followed by RL; a second seed of the science run; long-document RL from the no-science-step model.

Ethical considerations

Data and licences. Part of the fine-tuning data is derived from NLLB. Sentences mined with NLLB make up about 14% of the tokens of the main fine-tuning stage; adding NLLB-Seed and the Aya translations (NLLB-3.3B outputs) gives about 19%, and adding machine-translated Tumbuka instructions (system not stated, possibly NLLB) about 23%. NLLB models and outputs carry non-commercial (CC-BY-NC) terms, which limits commercial use of the NLLB-derived data and signals; a licence statement must accompany any release of the weights. Several sources have unspecified or unconfirmed licences (the `*_mt560` OPUS sets, which are mostly Bible and Jehovah's Witnesses text and 69% of the main pair data; `multi-open`; AfroLingu-MT; an Efik New Testament sheet) and need a legal decision. IbomMT is CC-BY-NC-4.0 and is a disclosed exception. Religious text dominates many low-resource languages. Reward-side data: NLLB-3.3B outputs (CC-BY-NC) served as a reward target for five languages only, and RL prompts (FineWeb-Edu, Wikipedia, news including BBC News, which declares no licence and carries BBC copyright) are input text only. The Gemma terms of use apply to the derived weights, and pre-training exposure cannot be excluded. Reward and monitor models, as declared on the Hub (read 2026-11-05): the two SSA-COMET and the two AfriCOMET checkpoints are apache-2.0; GlotLID declares `license: other` with `license_name: apache-2.0-plus-notices`.

Release and licence. The final RL model (branch `main`) is at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl, the SFT model at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-sft, and the seed, ablation and science-variant weights at https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl-experiments; the control checkpoint is not released. At the last check the Hub repositories declare no licence field and have no model card; the Gemma terms of use apply, and the NLLB-derived and non-commercial data above limit commercial use **[OWNER: confirm release licence]**. The data repositories named in our reports were private at the last check **[OWNER: which parts of the data are released]**.

The control's extra data. For the SFT-only control we used TranslatePsy English-to-X pairs (CC-BY-NC-4.0; NLLB-3.3B translations of web sentences, filtered by AfriCOMET, SSA-COMET and MetricX) for that experiment only. No TranslatePsy row is in the SFT baseline, any RL stage, the science or long-document variants, or any released weight; the control checkpoint is neither released nor used in production. A good control result partly means "distilling NLLB". TranslatePsy-AfriSLM-0.8B and -2B appear only as comparison systems; they were distilled from NLLB-3.3B outputs. The NLLB-218 language-identification model (CC-BY-NC) was used for measurement only.

Decontamination and its residuals (the clean 385M pipeline). The audit found no benchmark sentence of three or more words in the audited training text. The remaining benchmark matches are one- or two-word strings, numerals and names (for example 0.25% of NTREX rows and 0.47% of MAFAND-MT rows). The FLORES+ development half (440 sentence ids) is in training on purpose, and the held-out half is protected. A lexical fuzzy scan of the pair and instruction files found no benchmark string of eight or more tokens as a contiguous run, but strings of four to seven tokens occur inside longer training text, and 263 benchmark strings of eight or more tokens overlap training text by character 5-grams; these are candidates, and we did not measure their effect. The first RL stage drew 6 flagged prompts of 28,120 from seven unfiltered prompt files, and 0.9% of AfriScience-MT training rows have their African-language sentence in the earlier pair data (the test papers are untouched). We did not run paraphrase, machine-translation round-trip or packed-shard scans. These statements are about the clean 385M pipeline. Our earlier 270M pipeline trained on text that held the English side of 95% of FLORES+ devtest; its numbers appear only on rows clean for both pipelines.

Analysis decisions and process. We chose checkpoints on a 720-sentence FLORES+ development subset, chose the languages for the weak-language run from devtest-derived tables, ran the decomposition and family analyses after seeing results, and changed the science headline metric after writing the interpolation rule. The authors overrode two failed training gates. The first RL stage was stopped after 40 steps and resumed with four added features; later stages were restarted once or twice, and the hacking alarm was loosened once. The science RL step cap was extended by the trainer, one long-document run lost most of its time to a disk-full crash loop, and the no-science-step run had one out-of-memory restart (replayed steps use the later records). LibreTexts rows were removed from the long-document pool before it started.

Intended use, risks and dual use. Outputs are machine translations of unverified quality, with documented failure modes: Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri are not usable, RL increases English output there, and documents should be translated sentence by sentence. The model should not replace NLLB-class systems or be used where mistranslation is costly (health, legal, safety). As with any translation model, it can produce text at scale in under-served languages, including spam or misleading content, and we release no content filter.

Author contributions

[Placeholder, to be filled by the lead author: one sentence per person describing what that person contributed (for example design, data, training, evaluation, writing). Authorship is based on contribution.]

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A Glossary of internal names

The main text uses descriptive names. The appendix tables use the internal names of our experiment log; this table maps them.

internal name	name in the main text	meaning
A0	earlier 270M model (base of the grown model)	the 270M model with six inserted layers, trained before growth
SFT-1, SFT-2, stage B	SFT baseline (training steps)	SFT-1 trains only the new layers and language tags; SFT-2 (stage B) trains all weights
SFT, SFT final	SFT baseline	the 385M model after supervised fine-tuning
stage A	first RL stage	400 steps on short English windows
B1 best, B1 final	second RL stage (best checkpoint at step 50; final at step 135)	RL on longer windows; best and final differ by less than the noise (E-20)
mixed, mixed best	third RL stage; **RL model**	short and long windows; its best checkpoint is the final general RL model used throughout
weak-group RL	RL aimed at eight weak languages	85 extra steps; no measurable gain on its targets
science SFT, α	science fine-tuning step; + *science fine-tuning*	fine-tuning on science text, then weight interpolation between the starting model ($\alpha = 0$) and the fine-tuned model ($\alpha = 1$); $\alpha = 0.5$ is used
science RL	science RL	RL started from the science fine-tuned model
E-14	science RL without the science fine-tuning	the same RL started from the RL model
long-document RL	long-document variant	RL on inputs of up to 2,048 source tokens
clean rows	rows used for comparisons	benchmark rows minus 214 rows flagged by the contamination audit
cells	language–benchmark pairs	one language on one benchmark; 93 pairs on FLORES+, NTREX and MAFAND-MT; 76 with a held-out AfriCOMET score
TP languages	the 19 languages with extra control data	TranslatePsy languages of the SFT-only control
reward flag, independent flag	the reward's repetition check; the independent repetition detector	see Appendix H
E-01	repetition-penalty control	decoding setting that discourages repeated tokens
E-02	beam search and MBR study	decoding at 768 tokens on 20 languages
E-03	SFT-only control	matched-compute continued SFT
E-04, E-11	seed pair and ablations	extra seeds and feature-removal runs of the first RL stage
E-05	neural scores for all stages	SSA-COMET-QE, AfriCOMET and chrF on AfriScience-MT
E-09, E-09b	engine and precision checks	vLLM against Hugging Face, bf16 against fp32
E-13	earlier 270M models in the same harness	re-run on rows clean for both
E-15, E-16, E-20	long-document, science and B1-final evaluations	evaluation runs
E-10, E-17	human evaluation; NLLB-data ablation	not run
SFT-ctl, RL-A, RL-B1, RL	SFT-only control; first RL stage; second RL stage (best checkpoint); RL model	column labels of the full-results tables (Appendix B); [a] marks the control, which was trained on NLLB-3.3B translations for 19 languages
F+, NT, MF; no_pair; clean	FLORES+, NTREX-128, MAFAND-MT; row sets	*no_pair* = rows whose reference pair the earlier 270M SFT never saw; *clean* = rows whose English side is not in the 270M SFT text
TPsy-0.8B, TPsy-2B	TranslatePsy-AfriSLM models	small translation models for 19 (0.8B) and 9 (2B) of our languages, distilled from NLLB-3.3B outputs

Appendix map. **A** glossary of internal names (Appendix A); **B** full comparison with other systems and per-language results of all systems on FLORES+, NTREX, MAFAND-MT, SmolSent and SALT (Appendix B); **C** per-language RL gain over the SFT baseline with family and script labels (Appendix C); **D** decoding: loop guard (E-01), beam search and MBR per language (E-02), engine and precision checks, and the superseded 256-token study (Appendix D); **E** seeds and ablations (E-04, E-11) overall and per language (Appendix E); **F** the matched-compute control (E-03): decision rule, panels and per-language results (Appendix F); **G** science and long-document results, overall and per language, including the run

without the science step (E-14) (Appendix G); **H** the audit of the repetition check (Appendix H); **I** hyperparameters, the earlier 270M and A0 comparisons, and the stage chain (Appendix I, J, K); **J** family, script and resource analysis (Appendix L); then metric independence (Appendix N), disclosures, reproducibility, data statement and the Responsible NLP checklist (Appendix O, P, Q, R). Prose-only blocks (contamination follow-ups, NLLB-mined-data analysis, negative results, the full limitations list and parts of the narrative) are in the full version only; every result table is in both versions.

B Full results against other systems

Row sets and how each number was computed. chrF++ is sacreBLEU 2.6.0 `CHRF(word_order=2)` on NFC-normalised text, recomputed from the stored hypotheses for our systems and from stored per-sentence statistics for NLLB and TranslatePsy; corpus statistics are computed per language and then averaged over languages. AfriCOMET is the held-out, reference-based `africomet-stl-1.1` only; the two scoring runs (one for NLLB and TranslatePsy, one for our models) agree to a mean absolute difference of 2.3×10^{-4} on 15,400 identical triples. *no_pair* rows are FLORES+ rows whose reference pair the earlier 270M SFT never saw (42,680 of 44,528 rows); this is the set on which NLLB and TranslatePsy have AfriCOMET scores. *clean* rows are rows whose English side is not in the 270M SFT text (48 sentences per language on FLORES+; 25,221 NTREX and 14,276 MAFAND-MT sentences); only on them are the earlier 270M models not exposure-inflated. NTREX and MAFAND-MT use all rows (our flagged rows, 120 and 94, change the averages by at most 0.006 chrF++). The FLORES+ NLLB set has 44 languages (the published NLLB table has 43; the 44th is Kanuri), 40 of which have an AfriCOMET score in our harness; TranslatePsy-0.8B covers 19 languages and TranslatePsy-2B 9. Intervals are paired bootstrap over test sentences (1,000 resamples, Poisson weights per source sentence shared across systems and, for FLORES+ and NTREX, across languages; seed 20261104). Where NLLB exists only as a per-language corpus-level aggregate (NTREX, MAFAND-MT, SmolSent, SALT) the interval is a bootstrap over languages (10,000 resamples), there is no AfriCOMET, and no sentence pairing is possible. Decoding: ours greedy (768 tokens, vLLM, bf16); NLLB beam search with 4 beams; TranslatePsy greedy with its model-card prompt (256 tokens). The 385M beam-4 run of Section 5.3 covers 20 languages only, so no beam-matched comparison on the full set exists.

Independence and disclosures. chrF++, chrF and spBLEU against human references are independent of every reward. AfriCOMET-STL-1.1 is held out of every reward but belongs to the family of the quality model in the reward and was used to filter the TranslatePsy training rows, so it favours the SFT-only control and TranslatePsy. The SFT-only control was trained on NLLB-3.3B translations (TranslatePsy, CC-BY-NC-4.0) for 19 languages and its checkpoint is not released; TranslatePsy-AfriSLM-0.8B and -2B were distilled from NLLB-3.3B outputs, so their lead over NLLB-3.3B on their languages is a student–teacher effect and not independent evidence for a better model; NLLB models and outputs carry CC-BY-NC terms. The NLLB-derived share of our own SFT data and the NLLB-3.3B pseudo-reference reward term for five languages are listed in Appendix O and in the Ethical considerations. The five languages with that reward term (kea kbp nus bam kab) and the seven with a human-reference reward term are partly optimised towards chrF++. SmolSent, SALT, GlotStoryBook, IbomMT and FLORES-MOZ have no RL model evaluated anywhere. In the tables below SFT = SFT baseline, SFT-ctl = SFT-only control, RL = RL model, RL-A and RL-B1 = first and second RL stage, F+ = FLORES+, NT = NTREX-128, MF = MAFAND-MT, TPsy = TranslatePsy-AfriSLM (Appendix A).

Per-language results

Table 12: FLORES+ devtest, chrF++ per language on the no_pair rows (n sentences per language; identical for every system). "–": language not covered by the system. The last rows are macros over the languages with a value, so columns have different language sets (use the main tables for comparisons).

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	63.1	63.3	64.1	61.9	**64.3**	64.2
amh	Amharic	1012	16.6	27.0	26.6	37.3	**38.0**	37.6
bam	Bambara	1012	19.4	20.1	27.2	28.4	**31.0**	–

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
bem	Bemba	1012	19.3	20.0	33.9	34.7	**37.3**	–
dyu	Dyula	1012	9.9	9.8	**19.0**	16.6	15.4	–
ewe	Ewe	1012	26.0	27.4	31.2	35.2	**38.3**	–
fon	Fon	1012	20.2	19.8	**22.2**	17.9	20.3	–
hau	Hausa	170	35.9	44.5	44.1	45.4	47.3	**48.3**
ibo	Igbo	453	34.5	35.9	37.5	36.9	37.7	**38.8**
kab	Kabyle	1012	16.3	15.9	26.7	27.4	**31.8**	–
kam	Kamba	1012	17.2	16.7	25.9	**26.3**	22.9	–
kbp	Kabiye	1012	14.5	15.5	24.9	23.5	**28.8**	–
kea	Kabuverdianu	1012	18.9	19.0	39.2	44.1	**47.8**	–
kik	Kikuyu	1012	26.0	26.6	33.3	34.7	**37.0**	–
kin	Kinyarwanda	1012	25.8	39.7	38.7	43.5	**48.2**	48.0
kmb	Kimbundu	1012	16.0	16.3	**26.6**	24.8	23.7	–
knc	Kanuri	1012	3.8	3.0	14.2	14.7	**24.4**	–
lin	Lingala	1012	28.6	45.3	44.3	46.3	47.8	**48.8**
lua	Luba-Kasai	1012	17.2	16.0	33.4	33.4	**35.1**	–
lug	Luganda	1010	16.1	34.7	31.7	34.6	38.5	**39.0**
luo	Luo	1012	21.4	21.6	34.5	36.8	**38.4**	–
mlg	Malagasy	1012	39.0	44.2	44.7	47.8	**50.0**	49.6
mos	Mossi	1012	14.1	14.4	22.8	20.7	**23.7**	–
nso	Sepedi	1012	31.6	30.3	49.0	48.9	**50.4**	–
nus	Nuer	1012	13.5	11.4	24.8	25.3	**28.7**	–
nya	Chichewa	1012	36.9	40.2	42.2	44.0	44.8	**45.2**
orm	Oromo	988	12.1	31.3	30.2	31.1	35.6	**37.0**
run	Kirundi	1012	21.6	23.2	33.9	38.9	**41.6**	–
sag	Sango	1012	19.7	19.8	34.3	33.9	**36.4**	–
sna	Shona	1012	33.9	39.2	37.6	42.7	42.6	**43.2**
som	Somali	1012	35.5	38.5	39.7	41.4	42.6	**43.4**
sot	Sesotho	1012	37.9	43.3	42.7	45.4	46.2	**46.7**
ssw	Swati	1012	22.2	22.7	37.9	41.1	**43.1**	–
swh	Swahili	1012	56.2	56.3	58.4	58.0	59.8	**61.1**
tir	Tigrinya	1012	10.5	9.7	18.7	22.4	**24.7**	–
tsn	Setswana	982	37.8	45.6	45.0	46.3	**48.4**	48.3
tso	Tsonga	1012	32.9	29.8	46.9	48.9	**50.1**	–
tum	Tumbuka	1012	25.4	25.3	34.3	33.5	**35.8**	–
twi	Twi	1012	27.5	28.1	35.1	34.9	**38.1**	–
umb	Umbundu	1012	14.8	15.9	**25.5**	25.3	22.9	–
wol	Wolof	1012	26.9	29.1	**30.6**	23.3	27.6	29.8
xho	Xhosa	1012	43.6	44.4	45.4	46.4	48.5	**48.6**
yor	Yoruba	657	19.6	26.2	28.0	25.2	26.2	**28.4**
zul	Zulu	976	46.0	49.6	48.2	51.6	**54.4**	54.2
Macro (languages with a value)			25.6	28.6	34.9	35.9	38.1	45.3
Languages with a value			44	44	44	44	44	19

Table 13: FLORES+ devtest, held-out reference-based AfriCOMET-STL-1.1 per language, same sentences as the chrF++ table (languages outside the AfriCOMET encoder card are "–").

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	0.745	0.754	0.755	0.779	**0.797**	0.787
amh	Amharic	1012	0.504	0.629	0.592	0.712	0.719	**0.751**
bam	Bambara	1012	0.452	0.456	0.521	0.577	**0.596**	–
bem	Bemba	1012	0.450	0.455	0.561	0.620	**0.640**	–
dyu	Dyula	1012	–	–	–	–	–	–
ewe	Ewe	1012	0.562	0.571	0.608	0.668	**0.689**	–
fon	Fon	1012	0.573	0.569	**0.603**	0.549	0.573	–
hau	Hausa	170	0.529	0.592	0.568	0.621	0.618	**0.650**
ibo	Igbo	453	0.614	0.649	0.630	0.656	0.658	**0.678**
kab	Kabyle	1012	0.558	0.550	0.674	0.673	**0.708**	–
kam	Kamba	1012	0.440	0.437	0.503	**0.550**	0.528	–
kbp	Kabiye	1012	0.538	0.558	0.661	0.631	**0.679**	–
kea	Kabuverdianu	1012	–	–	–	–	–	–
kik	Kikuyu	1012	0.597	0.606	0.662	0.697	**0.718**	–

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
kin	Kinyarwanda	1012	0.564	0.687	0.656	0.724	0.746	**0.753**
kmb	Kimbundu	1012	0.419	0.421	0.500	0.506	**0.509**	–
knc	Kanuri	1012	–	–	–	–	–	–
lin	Lingala	1012	0.520	0.642	0.617	0.676	0.684	**0.701**
lua	Luba-Kasai	1012	0.430	0.420	0.556	0.598	**0.615**	–
lug	Luganda	1010	0.464	0.626	0.576	0.635	0.657	**0.671**
luo	Luo	1012	0.562	0.560	0.664	0.712	**0.722**	–
mlg	Malagasy	1012	0.593	0.658	0.634	0.708	**0.727**	0.725
mos	Mossi	1012	0.459	0.459	0.548	0.555	**0.595**	–
nso	Sepedi	1012	0.586	0.588	0.689	0.719	**0.727**	–
nus	Nuer	1012	0.474	0.427	0.624	0.639	**0.681**	–
nya	Chichewa	1012	0.625	0.674	0.651	0.701	0.702	**0.720**
orm	Oromo	988	0.420	0.652	0.610	0.651	0.696	**0.721**
run	Kirundi	1012	0.531	0.552	0.639	0.701	**0.716**	–
sag	Sango	1012	–	–	–	–	–	–
sna	Shona	1012	0.596	0.662	0.619	0.689	0.690	**0.707**
som	Somali	1012	0.590	0.643	0.620	0.669	0.684	**0.702**
sot	Sesotho	1012	0.615	0.670	0.642	0.691	0.695	**0.710**
ssw	Swati	1012	0.493	0.506	0.606	0.681	**0.697**	–
swh	Swahili	1012	0.693	0.718	0.709	0.745	0.753	**0.780**
tir	Tigrinya	1012	0.507	0.487	0.651	0.709	**0.727**	–
tsn	Setswana	982	0.633	0.696	0.674	0.710	0.723	**0.730**
tso	Tsonga	1012	0.597	0.570	0.667	0.715	**0.718**	–
tum	Tumbuka	1012	0.532	0.535	0.605	0.647	**0.673**	–
twi	Twi	1012	0.594	0.600	0.648	0.666	**0.686**	–
umb	Umbundu	1012	0.470	0.474	0.550	**0.580**	0.568	–
wol	Wolof	1012	0.628	0.654	0.653	0.606	0.650	**0.676**
xho	Xhosa	1012	0.627	0.653	0.638	0.686	0.699	**0.703**
yor	Yoruba	657	0.563	0.670	0.682	0.681	0.671	**0.699**
zul	Zulu	976	0.659	0.700	0.674	0.728	0.745	**0.752**
Macro (languages with a value)			0.550	0.586	0.623	0.662	0.677	0.717
Languages with a value			40	40	40	40	40	19

Table 14: FLORES+ devtest, plain chrF (word order 0) per language, same sentences as Table 12.

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	65.2	65.5	66.3	64.1	**66.6**	66.6
amh	Amharic	1012	18.2	29.4	29.0	40.1	**40.8**	40.4
bam	Bambara	1012	20.8	21.6	29.2	30.5	**33.3**	–
bem	Bemba	1012	22.2	23.0	37.9	38.9	**41.6**	–
dyu	Dyula	1012	10.7	10.7	**21.0**	18.5	17.3	–
ewe	Ewe	1012	27.8	29.4	33.4	37.6	**41.1**	–
fon	Fon	1012	21.6	21.2	**23.9**	19.1	21.7	–
hau	Hausa	170	37.7	47.3	46.9	48.1	50.0	**51.1**
ibo	Igbo	453	36.3	37.9	39.6	39.0	39.9	**40.9**
kab	Kabyle	1012	17.4	17.0	28.2	29.0	**33.8**	–
kam	Kamba	1012	19.4	18.8	28.6	**29.1**	25.8	–
kbp	Kabiye	1012	15.8	17.0	26.8	25.2	**30.7**	–
kea	Kabuverdianu	1012	20.2	20.2	40.8	46.3	**50.0**	–
kik	Kikuyu	1012	28.4	29.1	36.1	37.4	**39.9**	–
kin	Kinyarwanda	1012	28.6	43.5	42.6	47.5	**52.3**	52.0
kmb	Kimbundu	1012	18.5	18.9	**30.2**	28.1	26.9	–
knc	Kanuri	1012	4.7	3.8	16.6	16.9	**28.0**	–
lin	Lingala	1012	30.9	48.2	47.2	49.2	50.7	**51.8**
lua	Luba-Kasai	1012	19.7	18.5	37.3	37.4	**39.2**	–
lug	Luganda	1010	18.5	39.1	35.8	39.2	43.5	**43.9**
luo	Luo	1012	23.6	23.9	37.5	39.9	**41.7**	–
mlg	Malagasy	1012	42.4	48.1	48.7	51.8	**54.1**	53.7
mos	Mossi	1012	15.2	15.6	24.6	22.3	**25.6**	–
nso	Sepedi	1012	33.3	32.1	51.3	51.3	**52.8**	–
nus	Nuer	1012	15.0	12.8	27.1	27.5	**31.2**	–
nya	Chichewa	1012	40.7	44.6	46.6	48.6	49.5	**49.9**

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
orm	Oromo	988	14.4	36.3	35.2	36.4	41.2	**42.8**
run	Kirundi	1012	24.5	26.4	38.2	43.4	**46.2**	–
sag	Sango	1012	20.7	20.9	36.3	35.8	**38.5**	–
sna	Shona	1012	38.4	44.4	42.7	48.0	48.1	**48.7**
som	Somali	1012	39.4	42.6	44.0	45.7	47.1	**47.8**
sot	Sesotho	1012	40.4	46.2	45.5	48.3	49.3	**49.7**
ssw	Swati	1012	26.0	26.7	43.8	47.1	**49.2**	–
swh	Swahili	1012	58.6	58.8	60.8	60.5	62.4	**63.6**
tir	Tigrinya	1012	11.6	10.8	20.6	24.3	**26.6**	–
tsn	Setswana	982	40.0	48.1	47.5	48.9	**50.9**	50.8
tso	Tsonga	1012	35.5	32.4	50.1	52.1	**53.3**	–
tum	Tumbuka	1012	28.4	28.4	38.0	36.8	**39.2**	–
twi	Twi	1012	29.2	29.9	37.0	36.9	**40.1**	–
umb	Umbundu	1012	17.8	19.2	**30.1**	29.7	27.1	–
wol	Wolof	1012	28.5	30.9	**32.4**	24.7	29.4	31.6
xho	Xhosa	1012	49.2	50.3	51.4	52.5	54.6	**54.7**
yor	Yoruba	657	21.3	27.9	29.7	26.9	28.2	**30.3**
zul	Zulu	976	51.3	55.1	53.8	57.2	**60.0**	59.8
Macro (languages with a value)			27.9	31.2	37.9	39.0	41.4	48.9
Languages with a value			44	44	44	44	44	19

Table 15: FLORES+ devtest, spBLEU (flores200 SentencePiece tokenizer) per language, same sentences as Table 12.

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
afr	Afrikaans	1012	43.0	42.9	43.8	40.8	**44.2**	43.6
amh	Amharic	1012	4.7	14.5	15.4	28.4	**30.2**	28.6
bam	Bambara	1012	2.9	3.2	6.9	8.4	**10.0**	–
bem	Bemba	1012	2.2	2.4	9.6	10.6	**12.6**	–
dyu	Dyula	1012	0.7	0.6	**3.1**	2.5	2.1	–
ewe	Ewe	1012	5.8	7.1	9.7	13.3	**15.7**	–
fon	Fon	1012	4.1	3.8	**5.4**	3.4	5.0	–
hau	Hausa	170	11.5	20.5	19.6	21.9	**24.0**	23.9
ibo	Igbo	453	13.2	15.4	16.5	16.7	17.4	**18.1**
kab	Kabyle	1012	1.8	1.7	7.3	8.8	**12.7**	–
kam	Kamba	1012	2.1	1.9	6.6	**6.6**	4.8	–
kbp	Kabiye	1012	1.6	1.8	6.3	7.4	**12.2**	–
kea	Kabuverdianu	1012	2.9	3.1	17.4	22.0	**27.1**	–
kik	Kikuyu	1012	5.6	6.0	10.9	13.2	**15.3**	–
kin	Kinyarwanda	1012	4.3	14.4	14.7	19.9	**24.7**	24.0
kmb	Kimbundu	1012	1.2	1.3	4.6	**4.7**	4.2	–
knc	Kanuri	1012	0.2	0.1	1.6	2.0	**5.9**	–
lin	Lingala	1012	4.6	16.9	16.8	19.9	**20.4**	20.1
lua	Luba-Kasai	1012	1.2	1.0	8.3	8.7	**9.8**	–
lug	Luganda	1010	1.3	9.2	7.5	10.1	12.5	**12.8**
luo	Luo	1012	3.0	3.1	11.2	13.5	**14.7**	–
mlg	Malagasy	1012	12.1	17.3	17.7	22.3	**24.8**	23.4
mos	Mossi	1012	1.3	1.4	4.5	4.5	**5.5**	–
nso	Sepedi	1012	7.4	6.5	24.8	24.7	**26.2**	–
nus	Nuer	1012	2.1	1.5	9.3	10.1	**14.0**	–
nya	Chichewa	1012	10.6	12.2	15.2	17.4	**18.2**	17.6
orm	Oromo	988	0.6	6.6	5.7	6.9	10.8	**11.4**
run	Kirundi	1012	2.8	3.3	11.1	16.4	**18.3**	–
sag	Sango	1012	1.8	1.8	9.1	10.0	**11.1**	–
sna	Shona	1012	9.9	15.0	13.3	18.9	19.0	**19.1**
som	Somali	1012	10.1	13.3	14.8	16.6	18.3	**18.4**
sot	Sesotho	1012	12.6	18.1	17.4	20.0	20.9	**21.1**
ssw	Swati	1012	3.0	3.3	12.9	18.1	**20.2**	–
swh	Swahili	1012	33.1	32.2	35.7	35.5	38.1	**39.5**
tir	Tigrinya	1012	2.3	2.1	8.6	13.6	**16.5**	–
tsn	Setswana	982	11.9	21.3	21.2	22.8	**25.4**	24.5
tso	Tsonga	1012	7.8	5.7	22.4	25.4	**26.7**	–
tum	Tumbuka	1012	5.6	5.2	10.6	11.7	**13.7**	–

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-600M	NLLB-3.3B	TPsy-0.8B
twi	Twi	1012	6.2	6.8	12.4	12.9	**15.2**	–
umb	Umbundu	1012	0.6	0.8	3.2	**3.5**	2.6	–
wol	Wolof	1012	7.6	9.3	**11.0**	6.6	8.6	10.1
xho	Xhosa	1012	19.0	20.1	20.7	23.1	**25.0**	24.7
yor	Yoruba	657	5.2	10.2	12.2	9.8	11.2	**12.6**
zul	Zulu	976	22.6	26.0	23.8	29.2	**32.5**	32.1
Macro (languages with a value)			7.1	9.3	13.2	15.1	17.0	22.4
Languages with a value			44	44	44	44	44	19

Table 16: NTREX-128, chrF++ per language, all rows. NLLB-3.3B is a corpus-level aggregate on the same rows.

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-3.3B
afr	Afrikaans	1997	62.4	63.6	63.9	**68.5**
amh	Amharic	1997	10.6	18.1	18.0	**25.4**
bem	Bemba	1997	18.7	18.8	34.8	**38.6**
ewe	Ewe	1997	24.3	25.7	31.6	**36.9**
hau	Hausa	1997	42.1	49.5	48.6	**52.9**
ibo	Igbo	1997	39.9	43.3	44.9	**47.1**
kin	Kinyarwanda	1997	26.4	38.4	37.1	**45.8**
mlg	Malagasy	1997	36.9	41.5	40.8	**46.6**
nso	Sepedi	1997	20.1	19.7	32.7	**35.0**
nya	Chichewa	1997	37.9	43.2	44.2	**48.7**
orm	Oromo	1997	9.6	27.3	27.3	**31.2**
sna	Shona	1997	34.2	40.6	39.2	**45.1**
som	Somali	1997	35.8	40.6	44.2	**47.7**
ssw	Swati	1997	21.6	20.9	38.8	**45.3**
swh	Swahili	1997	55.2	57.6	58.7	**61.8**
tir	Tigrinya	1997	9.1	9.8	17.0	**23.8**
tsn	Setswana	1997	37.0	49.4	48.7	**54.8**
wol	Wolof	1997	25.3	29.5	**30.4**	29.4
xho	Xhosa	1997	39.8	40.7	42.4	**43.8**
yor	Yoruba	1997	23.7	30.9	32.1	**32.1**
zul	Zulu	1997	40.9	44.1	43.5	**47.9**
Macro (languages with a value)			31.0	35.9	39.0	43.3
Languages with a value			21	21	21	21

Table 17: NTREX-128, held-out AfriCOMET-STL-1.1 per language (no NLLB score exists on these rows).

ISO	Language	n	SFT	SFT-ctl[a]	RL
afr	Afrikaans	1997	0.682	**0.703**	0.698
amh	Amharic	1997	0.444	**0.558**	0.521
bem	Bemba	1997	0.433	0.436	**0.544**
ewe	Ewe	1997	0.519	0.535	**0.583**
hau	Hausa	1997	0.597	**0.662**	0.625
ibo	Igbo	1997	0.645	**0.689**	0.673
kin	Kinyarwanda	1997	0.551	**0.660**	0.623
mlg	Malagasy	1997	0.547	**0.613**	0.573
nso	Sepedi	1997	0.487	0.497	**0.588**
nya	Chichewa	1997	0.588	**0.655**	0.614
orm	Oromo	1997	0.395	**0.595**	0.562
sna	Shona	1997	0.573	**0.644**	0.597
som	Somali	1997	0.549	**0.623**	0.610
ssw	Swati	1997	0.471	0.474	**0.593**
swh	Swahili	1997	0.646	**0.679**	0.664
tir	Tigrinya	1997	0.488	0.493	**0.632**
tsn	Setswana	1997	0.577	**0.683**	0.650
wol	Wolof	1997	0.593	**0.635**	0.625
xho	Xhosa	1997	0.584	**0.618**	0.598
yor	Yoruba	1997	0.597	0.691	**0.701**
zul	Zulu	1997	0.600	**0.643**	0.616

ISO	Language	n	SFT	SFT-ctl[a]	RL
Macro (languages with a value)			0.551	0.609	0.614
Languages with a value			21	21	21

Table 18: MAFAND-MT test, chrF++ per language, all rows. NLLB-3.3B is a corpus-level aggregate on the same rows.

ISO	Language	n	SFT	SFT-ctl[a]	RL	NLLB-3.3B
amh	Amharic	1037	11.6	18.5	18.3	**23.7**
hau	Hausa	1500	32.0	33.7	32.8	**35.3**
ibo	Igbo	1500	43.6	44.7	44.9	**48.4**
kin	Kinyarwanda	1006	36.5	48.1	46.4	**56.3**
lug	Luganda	1500	27.0	45.4	39.6	**52.6**
luo	Luo	1500	23.0	25.0	32.6	**39.7**
nya	Chichewa	1004	38.1	42.1	41.0	**43.3**
sna	Shona	1005	30.4	34.3	32.9	**38.1**
swh	Swahili	1835	50.6	52.0	51.8	**55.3**
tsn	Setswana	1500	43.9	51.0	53.1	**56.3**
twi	Twi	1500	28.6	29.1	34.1	**36.5**
xho	Xhosa	1002	49.8	51.5	51.8	**56.7**
yor	Yoruba	1558	23.5	32.1	**33.4**	32.7
zul	Zulu	998	44.5	46.4	45.6	**49.5**
Macro (languages with a value)			34.5	39.5	39.9	44.6
Languages with a value			14	14	14	14

Table 19: MAFAND-MT test, held-out AfriCOMET-STL-1.1 per language (no NLLB score exists on these rows).

ISO	Language	n	SFT	SFT-ctl[a]	RL
amh	Amharic	1037	0.456	**0.556**	0.515
hau	Hausa	1500	0.630	**0.671**	0.644
ibo	Igbo	1500	0.700	**0.720**	0.706
kin	Kinyarwanda	1006	0.623	**0.716**	0.688
lug	Luganda	1500	0.558	**0.680**	0.623
luo	Luo	1500	0.579	0.587	**0.651**
nya	Chichewa	1004	0.627	**0.681**	0.636
sna	Shona	1005	0.583	**0.648**	0.601
swh	Swahili	1835	0.680	**0.711**	0.693
tsn	Setswana	1500	0.667	**0.716**	0.712
twi	Twi	1500	0.601	0.599	**0.643**
xho	Xhosa	1002	0.672	**0.706**	0.680
yor	Yoruba	1558	0.614	0.725	**0.740**
zul	Zulu	998	0.610	**0.657**	0.620
Macro (languages with a value)			0.614	0.670	0.654
Languages with a value			14	14	14

Table 20: SmolSent, chrF++ per language, all rows: 385M SFT and NLLB-3.3B (corpus-level aggregate). No RL model was evaluated on SmolSent.

ISO	Language	n	SFT	NLLB-3.3B
amh	Amharic	863	10.3	**30.6**
bam	Bambara	863	13.3	**26.5**
bem	Bemba	863	13.8	**32.3**
dyu	Dyula	863	10.5	**20.8**
ewe	Ewe	863	20.6	**36.0**
fon	Fon	863	15.5	**17.8**
hau	Hausa	863	28.2	**40.4**
ibo	Igbo	863	35.7	**48.4**
kab	Kabyle	862	11.6	**27.8**
kik	Kikuyu	863	21.2	**40.1**
kin	Kinyarwanda	863	18.8	**41.8**

ISO	Language	n	SFT	NLLB-3.3B
kon	Kongo	863	22.4	**47.8**
lin	Lingala	863	22.5	**45.1**
lug	Luganda	863	14.6	**39.4**
luo	Luo	863	16.0	**32.7**
mlg	Malagasy	863	31.6	**48.1**
mos	Mossi	863	11.5	**18.8**
nso	Sepedi	863	22.5	**45.0**
nya	Chichewa	863	25.6	**38.0**
orm	Oromo	863	8.5	**34.9**
run	Kirundi	863	15.1	**33.7**
sna	Shona	863	27.5	**40.2**
som	Somali	862	23.5	**35.4**
sot	Sesotho	863	25.9	**38.9**
ssw	Swati	863	18.2	**38.8**
swh	Swahili	863	48.6	**52.4**
tir	Tigrinya	863	6.2	**25.3**
tsn	Setswana	863	34.3	**50.4**
tso	Tsonga	863	29.9	**43.7**
tum	Tumbuka	863	19.5	**38.8**
twi	Twi	863	20.5	**31.3**
wol	Wolof	863	21.3	**23.3**
xho	Xhosa	863	36.6	**43.0**
yor	Yoruba	863	16.6	**28.6**
zul	Zulu	863	40.6	**51.3**
Macro (languages with a value)			21.7	36.8
Languages with a value			35	35

Table 21: SALT, chrF++ per language, all rows: 385M SFT and NLLB-3.3B (corpus-level aggregate). No RL model was evaluated on SALT.

ISO	Language	n	SFT	NLLB-3.3B
ibo	Igbo	500	32.4	**41.5**
lug	Luganda	500	29.3	**52.1**
swh	Swahili	500	56.8	**61.9**
Macro (languages with a value)			39.5	51.8
Languages with a value			3	3

System	F+ no_pair (44/40)		F+ clean (44/40)		NT all (21/21)		MF all (14/14)	
	chrF++	COMET	chrF++	COMET	chrF++	COMET	chrF++	COMET
SFT	25.6	0.550	22.5	0.468	31.0	0.551	34.5	0.614
SFT-ctl[a]	28.6	0.586	25.3	0.502	35.9	0.609	39.5	0.670
RL-A	34.2	0.620	31.2	0.538	38.6	0.613	39.9	0.656
RL-B1	34.5	0.621	31.8	0.541	38.8	0.613	39.9	0.655
RL	34.9	0.623	32.1	0.544	39.0	0.614	39.9	0.654
270M SFT	–	–	24.9	0.480	–	–	–	–
270M RL	–	–	31.8	0.549	–	–	–	–
NLLB-600M	35.9	0.662	32.5	0.579	–	–	–	–
NLLB-3.3B	**38.1**	**0.677**	**35.0**	**0.601**	**43.3**	–	**44.6**	–

Table 6: All systems, language macro, chrF++ and held-out AfriCOMET-STL-1.1 (per-sentence rows on identical sentences within a panel). RL-A = stage A final, RL-B1 = B1 best, RL = final (mixed best). Bold = best in column; "–" = not available.

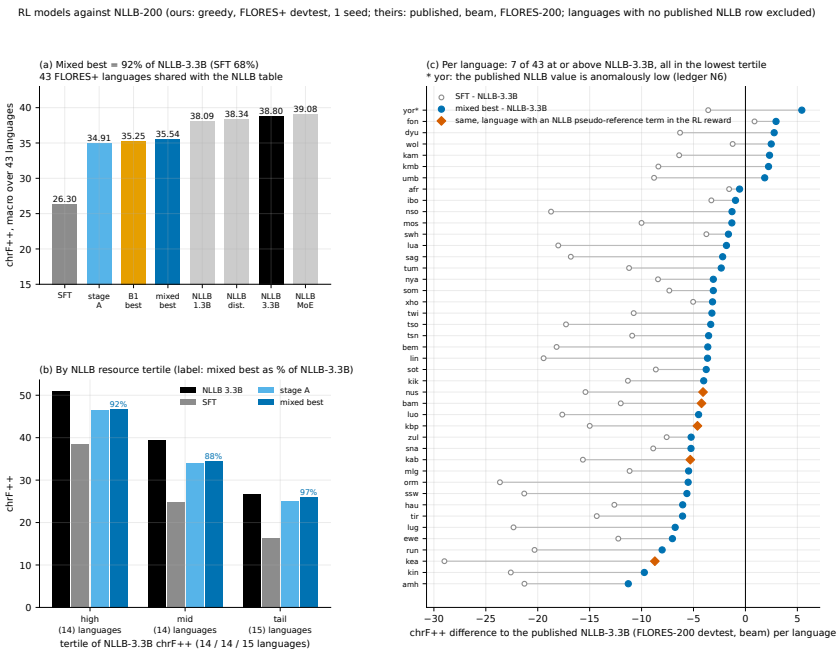


Figure 6: The SFT baseline and the RL model against NLLB-200 on the 43 languages shared with the published table (chrF++, higher is better). (a) Averages. (b) By how well NLLB-3.3B does on each language. (c) Difference per language; orange: NLLB outputs were part of the reward; yor*: anomalously low published value. Intervals over languages only.

Row set (langs)	RL − system	ΔchrF++	95% CI	ΔCOMET	95% CI
F+ no_pair (44/40)	SFT	9.30	[9.00, 9.55]	0.0732	[0.0711, 0.0752]
	SFT-ctl[a]	6.32	[6.11, 6.52]	0.0377	[0.0360, 0.0394]
	RL-A	0.72	[0.65, 0.81]	0.0039	[0.0031, 0.0046]
	RL-B1	0.34	[0.29, 0.41]	0.0024	[0.0017, 0.0030]
	NLLB-600M	-1.05	[-1.24, -0.86]	-0.0381	[-0.0399, -0.0362]
	NLLB-3.3B	-3.21	[-3.38, -3.02]	-0.0535	[-0.0553, -0.0518]
F+ TPsy-19 (19/19)	SFT	7.04	[6.64, 7.41]	0.0536	[0.0513, 0.0559]
	SFT-ctl[a]	0.08	[-0.13, 0.30]	-0.0226	[-0.0242, -0.0208]
	RL-A	0.29	[0.14, 0.43]	0.0008	[-0.0004, 0.0018]
	RL-B1	0.14	[0.04, 0.26]	0.0010	[-0.0000, 0.0020]
	NLLB-600M	-1.54	[-1.78, -1.28]	-0.0456	[-0.0480, -0.0434]
	NLLB-3.3B	-3.62	[-3.84, -3.39]	-0.0586	[-0.0609, -0.0563]
	TPsy-0.8B	-4.23	[-4.39, -4.07]	-0.0745	[-0.0765, -0.0727]
NT all (21/21)	SFT	7.97	[7.73, 8.21]	0.0630	[0.0614, 0.0646]
	SFT-ctl[a]	3.14	[2.97, 3.32]	0.0050	[0.0037, 0.0063]
	RL-A	0.41	[0.34, 0.48]	0.0013	[0.0006, 0.0020]
	RL-B1	0.16	[0.10, 0.23]	0.0004	[-0.0002, 0.0010]
	NLLB-3.3B	-4.26	[-5.22, -3.24][ℓ]	–	–
MF all (14/14)	SFT	5.39	[5.15, 5.63]	0.0395	[0.0377, 0.0413]
	SFT-ctl[a]	0.34	[0.16, 0.52]	-0.0158	[-0.0174, -0.0143]
	RL-A	-0.05	[-0.13, 0.05]	-0.0025	[-0.0035, -0.0015]
	RL-B1	-0.04	[-0.11, 0.03]	-0.0012	[-0.0021, -0.0003]
	NLLB-3.3B	-4.73	[-6.56, -3.12][ℓ]	–	–
F+ clean (44/40)	SFT	9.61	[8.34, 10.62]	0.0752	[0.0670, 0.0836]
	SFT-ctl[a]	6.87	[5.98, 7.61]	0.0416	[0.0335, 0.0495]
	RL-A	0.90	[0.58, 1.20]	0.0055	[0.0020, 0.0093]
	RL-B1	0.33	[0.10, 0.58]	0.0026	[-0.0006, 0.0057]
	270M SFT	7.20	[6.31, 7.94]	0.0639	[0.0574, 0.0709]
	270M RL	0.35	[0.08, 0.63]	-0.0052	[-0.0102, -0.0003]
	NLLB-600M	-0.39	[-1.13, 0.51]	-0.0353	[-0.0423, -0.0275]
	NLLB-3.3B	-2.85	[-3.57, -1.99]	-0.0576	[-0.0655, -0.0503]

Table 7: Final RL model minus each other system, language-macro difference with 95% paired bootstrap interval (1,000 resamples, Poisson weights per source sentence shared across systems and, for FLORES+ and NTREX, across languages; seed 20261104). Positive = RL better. Counts: languages for chrF++ / COMET. [ℓ]: no per-sentence output of NLLB-3.3B on NTREX/MAFAND is archived; the interval is a bootstrap over languages (language sampling only). SFT-ctl[a]: see Table 1.

	FLORES+ TPsy-19 set (19/19)			
System	chrF++	chrF	spBLEU	COMET
SFT	34.0	36.8	12.5	0.589
SFT-ctl[a]	41.0	44.4	17.7	0.665
RL	41.0	44.5	18.1	0.642
NLLB-600M	42.6	46.1	20.4	0.688
NLLB-3.3B	44.7	48.3	**22.4**	0.701
TPsy-0.8B	**45.3**	**48.9**	22.4	**0.717**

Table 8: FLORES+ devtest, the 19 languages of TranslatePsy-AfriSLM-0.8B (afr amh hau ibo kin lin lug mlg nya orm sna som sot swh tsn wol xho yor zul), same 17,380 sentences for every system (reference pair not seen by the earlier 270M SFT). Language macro; chrF++, chrF, spBLEU (flores200 tokenizer) and held-out reference-based AfriCOMET-STL-1.1. TPsy-0.8B was distilled from NLLB-3.3B outputs on these languages. SFT-ctl[a] as in Table 1. Bold = best in column.

	F+ (44/40)		NT (21/21)		MF (14/14)	
System	chrF++	COMET	chrF++	COMET	chrF++	COMET
270M SFT	24.9	0.480	33.2	0.546	36.6	0.618
270M RL	31.8	0.549	37.4	0.598	40.5	0.660
SFT	22.5	0.468	29.0	0.521	33.4	0.603
SFT-ctl[a]	25.3	0.502	33.8	0.579	39.1	0.659
RL	32.1	0.544	37.3	0.585	39.4	0.642
NLLB-600M	32.5	0.579	–	–	–	–
NLLB-3.3B	**35.0**	**0.601**	**41.5**	–	**44.0**	–

Table 9: The same comparison on the only rows that are clean for every system including the earlier 270M models (English side not in the 270M SFT text): FLORES+ 48 sentences per language (2,112 sentences), NTREX and MAFAND 25,221 and 14,276 sentences. 270M = the first research's SFT / RL-final models regenerated in the 385M harness (E-13). Language macro, counts in parentheses (chrF++ / COMET languages). Bold = best in column. NLLB-3.3B on NT/MF as in Table 1.

Table 10: SSA-COMET-MTL (reference-based, $\times 100$) on FLORES+ *no_pair* rows. **Reward family: not independent evidence.** No value exists for any 385M model; the 270M rows come from the earlier work's own run, not the re-run of Table 9.

system	languages	own languages	19 TPsy-0.8B languages	9 TPsy-2B languages
NLLB-600M	22	56.71	56.72	56.91
NLLB-3.3B	22	59.28	59.14	58.90
TranslatePsy-0.8B	19	60.75	60.75	60.31
TranslatePsy-2B	9	60.43	–	60.43
earlier 270M RL-final	22	51.92	52.81	50.29
earlier 270M SFT-final	22	47.54	48.78	46.19

Table 11: Against published whole-devtest NLLB numbers (FLORES-200 devtest chrF++; our systems rescored on all 1,012 FLORES+ rows; 43 shared languages; *not* sentence-paired; intervals over languages, 10,000 resamples).

system	chrF++	RL − system
RL model	35.54	
SFT baseline / SFT-only control / RL-A / RL-B1	26.30 / 29.35 / 34.91 / 35.25	$+9.24$ / $+6.19$ / $+0.63$ / $+0.29$
NLLB-600M, our rerun	36.62	$-1.08\ [-1.98,\ -0.20]$
NLLB-3.3B, our rerun	38.63	$-3.09\ [-4.13,\ -2.05]$
NLLB-3.3B, published	38.80	$-3.26\ [-4.30,\ -2.21]$
NLLB-1.3B, published	38.09	$-2.55\ [-3.46,\ -1.62]$
NLLB-distilled-1.3B, published	38.34	$-2.80\ [-3.72,\ -1.87]$
NLLB-MoE-54.5B, published	39.08	$-3.54\ [-4.59,\ -2.52]$

C RL gain over the SFT baseline per language, with family and script

Per language and benchmark: SFT baseline and RL model chrF++ and the RL gain in chrF++, plain chrF and held-out AfriCOMET with 95% paired-bootstrap intervals, with the Glottolog 5.3 top family and the script of each language from the family analysis (Appendix L). Sources: `reports/data/rl_vs_sft_evidence_2026-11-04/tables.md` and `reports/data/family_script_resource_analysis_2026-11-04/`.

Table 22: FLORES+ devtest, 54 languages: SFT baseline and RL model chrF++, and the RL gain in chrF++, plain chrF and held-out AfriCOMET with 95% paired-bootstrap intervals (all clean rows), with the Glottolog 5.3 top family and the script of each language from the family analysis.

ISO	Language	Glottolog family	Script	SFT	RL	RL − SFT, chrF++ [CI]	chrF [CI]	AfriCOMET [CI]
iso	iso	Atlantic-Congo	Latin	SFT	RL	Δ all rows [CI]	Δ chrF [CI]	Δ AfriCOMET [CI]
—	—			—:	—:	— — —	— — —	— — —
kea	Kabuverdianu	Indo-European	Latin	18.91	39.18	+20.27 [+18.91, +21.53]	+20.53 [+19.13, +21.81]	–
orm	Oromo	Afro-Asiatic	Latin	12.06	30.18	+18.12 [+17.49, +18.76]	+20.90 [+20.13, +21.64]	+0.191 [+0.181, +0.202]
nso	Sepedi	Atlantic-Congo	Latin	31.58	49.01	+17.43 [+15.75, +19.09]	+17.98 [+16.25, +19.66]	+0.103 [+0.092, +0.113]
lua	Luba-Kasai	Atlantic-Congo	Latin	17.18	33.37	+16.19 [+15.21, +17.13]	+17.57 [+16.50, +18.58]	+0.126 [+0.117, +0.134]
lin	Lingala	Atlantic-Congo	Latin	28.56	44.35	+15.78 [+14.19, +17.22]	+16.34 [+14.66, +17.84]	+0.097 [+0.088, +0.107]
ssw	Swati	Atlantic-Congo	Latin	22.20	37.86	+15.66 [+14.46, +16.85]	+17.72 [+16.34, +19.05]	+0.113 [+0.104, +0.123]
lug	Luganda	Atlantic-Congo	Latin	16.17	31.73	+15.57 [+14.66, +16.52]	+17.29 [+16.27, +18.36]	+0.112 [+0.104, +0.121]
sag	Sango	Atlantic-Congo	Latin	19.68	34.31	+14.62 [+13.64, +15.55]	+15.55 [+14.51, +16.54]	–
bem	Bemba	Atlantic-Congo	Latin	19.31	33.88	+14.56 [+13.63, +15.61]	+15.73 [+14.71, +16.86]	+0.111 [+0.103, +0.121]
tso	Tsonga	Atlantic-Congo	Latin	32.92	46.86	+13.93 [+12.43, +15.51]	+14.54 [+12.97, +16.19]	+0.070 [+0.062, +0.079]
luo	Luo	Nilotic	Latin	21.36	34.49	+13.13 [+11.91, +14.15]	+13.82 [+12.51, +14.92]	+0.101 [+0.092, +0.109]
kin	Kinyarwanda	Atlantic-Congo	Latin	25.81	38.66	+12.86 [+11.66, +14.03]	+13.94 [+12.61, +15.25]	+0.092 [+0.083, +0.101]
cjk	cjk	Atlantic-Congo	Latin	10.72	23.02	+12.31 [+11.70, +12.89]	+14.18 [+13.50, +14.85]	–
run	Kirundi	Atlantic-Congo	Latin	21.59	33.88	+12.29 [+11.21, +13.15]	+13.66 [+12.50, +14.65]	+0.107 [+0.098, +0.116]
nus	Nuer	Nilotic	Latin	13.49	24.82	+11.32 [+10.73, +11.91]	+12.16 [+11.51, +12.77]	+0.150 [+0.140, +0.159]
umb	Umbundu	Atlantic-Congo	Latin	14.80	25.46	+10.66 [+9.77, +11.44]	+12.21 [+11.17, +13.15]	+0.080 [+0.072, +0.087]
kmb	Kimbundu	Atlantic-Congo	Latin	16.02	26.62	+10.61 [+9.82, +11.37]	+11.70 [+10.84, +12.56]	+0.081 [+0.074, +0.087]
knc	Kanuri	Saharan	Latin	3.81	14.22	+10.41 [+9.95, +10.88]	+11.96 [+11.45, +12.48]	–
kbp	Kabiye	Atlantic-Congo	Latin	14.51	24.88	+10.37 [+9.70, +11.00]	+11.03 [+10.32, +11.69]	+0.123 [+0.115, +0.131]
kab	Kabyle	Afro-Asiatic	Latin	16.35	26.69	+10.34 [+9.50, +11.14]	+10.80 [+9.94, +11.62]	+0.116 [+0.107, +0.126]
dik	dik	Nilotic	Latin	3.76	14.06	+10.30 [+9.75, +10.81]	+11.37 [+10.80, +11.94]	–
amh	Amharic	Afro-Asiatic	Ethiopic	16.62	26.62	+10.00 [+9.04, +10.96]	+10.77 [+9.72, +11.82]	+0.088 [+0.078, +0.099]
vmw	vmw	Atlantic-Congo	Latin	17.45	27.14	+9.69 [+8.92, +10.48]	+10.87 [+10.02, +11.74]	–
mfe	mfe	Indo-European	Latin	28.51	37.93	+9.42 [+7.98, +10.94]	+9.57 [+8.10, +11.10]	–
dyu	Dyula	Mande	Latin	9.90	18.97	+9.06 [+8.47, +9.62]	+10.27 [+9.62, +10.88]	–
yor	Yoruba	Atlantic-Congo	Latin	21.32	30.33	+9.01 [+7.99, +10.00]	+9.02 [+7.99, +10.04]	+0.118 [+0.108, +0.128]
tum	Tumbuka	Atlantic-Congo	Latin	25.40	34.27	+8.87 [+7.89, +9.79]	+9.56 [+8.49, +10.58]	+0.073 [+0.064, +0.081]
kam	Kamba	Atlantic-Congo	Latin	17.22	25.92	+8.70 [+7.93, +9.50]	+9.19 [+8.35, +10.06]	+0.063 [+0.057, +0.069]
mos	Mossi	Atlantic-Congo	Latin	14.09	22.79	+8.70 [+8.13, +9.33]	+9.41 [+8.80, +10.05]	+0.088 [+0.081, +0.095]
tir	Tigrinya	Afro-Asiatic	Ethiopic	10.48	18.74	+8.26 [+7.69, +8.86]	+9.01 [+8.39, +9.66]	+0.144 [+0.134, +0.155]
taq	taq	Afro-Asiatic	Latin	2.81	11.06	+8.25 [+7.82, +8.67]	+9.14 [+8.68, +9.60]	–
seh	seh	Atlantic-Congo	Latin	21.67	29.65	+7.98 [+7.00, +8.86]	+9.04 [+7.93, +10.04]	–
bam	Bambara	Mande	Latin	19.39	27.17	+7.78 [+6.96, +8.59]	+8.34 [+7.50, +9.21]	+0.069 [+0.062, +0.076]
twi	Twi	Atlantic-Congo	Latin	27.54	35.07	+7.53 [+6.47, +8.61]	+7.81 [+6.72, +8.90]	+0.054 [+0.047, +0.061]
tsn	Setswana	Atlantic-Congo	Latin	37.69	45.05	+7.36 [+6.18, +8.65]	+7.59 [+6.35, +8.93]	+0.040 [+0.034, +0.048]
kik	Kikuyu	Atlantic-Congo	Latin	25.98	33.28	+7.30 [+6.39, +8.22]	+7.67 [+6.71, +8.66]	+0.065 [+0.058, +0.073]
hau	Hausa	Afro-Asiatic	Latin	39.88	46.46	+6.58 [+5.09, +8.34]	+7.24 [+5.62, +9.18]	+0.027 [+0.020, +0.035]
mlg	Malagasy	Austronesian	Latin	39.04	44.73	+5.68 [+4.54, +6.77]	+6.32 [+5.10, +7.50]	+0.041 [+0.034, +0.048]
nya	Chichewa	Atlantic-Congo	Latin	36.88	42.21	+5.33 [+4.23, +6.54]	+5.90 [+4.71, +7.21]	+0.026 [+0.019, +0.034]
ewe	Ewe	Atlantic-Congo	Latin	25.96	31.17	+5.21 [+4.46, +6.04]	+5.55 [+4.75, +6.43]	+0.046 [+0.039, +0.052]
sot	Sesotho	Atlantic-Congo	Latin	37.87	42.72	+4.85 [+3.70, +6.08]	+5.13 [+3.93, +6.45]	+0.027 [+0.021, +0.033]
arz	arz	Afro-Asiatic	Arabic	4.59	9.18	+4.59 [+4.01, +5.15]	+4.94 [+4.27, +5.58]	–
som	Somali	Afro-Asiatic	Latin	35.47	39.71	+4.24 [+3.14, +5.34]	+4.56 [+3.41, +5.71]	+0.030 [+0.023, +0.037]
wol	Wolof	Atlantic-Congo	Latin	26.87	30.58	+3.72 [+2.74, +4.63]	+3.99 [+2.91, +5.00]	+0.024 [+0.018, +0.030]
sna	Shona	Atlantic-Congo	Latin	33.92	37.56	+3.64 [+2.47, +4.85]	+4.29 [+3.00, +5.65]	+0.023 [+0.017, +0.030]
zul	Zulu	Atlantic-Congo	Latin	45.91	48.27	+2.36 [+1.56, +3.14]	+2.66 [+1.79, +3.52]	+0.014 [+0.009, +0.020]
ibo	Igbo	Atlantic-Congo	Latin	37.82	40.14	+2.33 [+1.57, +3.08]	+2.45 [+1.65, +3.23]	+0.014 [+0.009, +0.019]
ary	ary	Afro-Asiatic	Arabic	4.64	6.88	+2.24 [+1.76, +2.74]	+1.93 [+1.35, +2.51]	–
swh	Swahili	Atlantic-Congo	Latin	56.24	58.36	+2.12 [+1.10, +3.38]	+2.13 [+1.14, +3.33]	+0.016 [+0.010, +0.022]
fon	Fon	Atlantic-Congo	Latin	20.17	22.25	+2.07 [+1.60, +2.55]	+2.28 [+1.78, +2.77]	+0.029 [+0.024, +0.035]
xho	Xhosa	Atlantic-Congo	Latin	43.56	45.43	+1.87 [+1.05, +2.80]	+2.22 [+1.31, +3.24]	+0.011 [+0.006, +0.017]
zgh	zgh	Afro-Asiatic	Tifinagh	2.29	4.07	+1.78 [+1.30, +2.30]	+1.40 [+0.88, +1.96]	–
aeb	aeb	Afro-Asiatic	Arabic	2.75	4.12	+1.36 [+0.96, +1.78]	+1.08 [+0.62, +1.55]	–
afr	Afrikaans	Indo-European	Latin	63.13	64.13	+1.01 [+0.53, +1.61]	+1.05 [+0.56, +1.68]	+0.010 [+0.006, +0.015]

Table 23: NTREX, clean rows: SFT baseline and RL model chrF++ and the RL gain in chrF++, plain chrF and held-out AfriCOMET with 95% intervals, with family and script labels.

ISO	Language	Glottolog family	Script	n	SFT	RL	RL − SFT, chrF++ [CI]	chrF [CI]	AfriCOMET [CI]
iso	iso	Atlantic-Congo	Latin	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ chrF [CI]	Δ AfriCOMET [CI]

ISO	Language	Glottolog family	Script	n	SFT	RL	RL − SFT, chrF++ [CI]	chrF [CI]	AfriCOMET [CI]
—	—			—:	—:	—:	– – –	– – –	– – –
orm	Oromo	Afro-Asiatic	Latin	1992	9.63	27.33	+17.70 [+17.16, +18.25]	+20.84 [+20.20, +21.47]	+0.168 [+0.161, +0.175]
ven	ven	Atlantic-Congo	Latin	1992	21.30	38.81	+17.52 [+16.72, +18.31]	+18.28 [+17.47, +19.11]	–
ssw	Swati	Atlantic-Congo	Latin	1992	21.58	38.80	+17.22 [+16.40, +18.02]	+19.52 [+18.60, +20.40]	+0.122 [+0.115, +0.130]
bem	Bemba	Atlantic-Congo	Latin	1992	18.68	34.79	+16.11 [+15.43, +16.84]	+17.40 [+16.64, +18.21]	+0.111 [+0.105, +0.117]
nso	Sepedi	Atlantic-Congo	Latin	1992	20.14	32.73	+12.60 [+11.83, +13.35]	+13.33 [+12.50, +14.13]	+0.101 [+0.095, +0.108]
tsn	Setswana	Atlantic-Congo	Latin	1992	36.97	48.71	+11.74 [+10.68, +12.81]	+11.83 [+10.77, +12.95]	+0.073 [+0.067, +0.080]
kin	Kinyarwanda	Atlantic-Congo	Latin	1992	26.40	37.13	+10.73 [+9.95, +11.52]	+11.51 [+10.67, +12.35]	+0.072 [+0.066, +0.078]
som	Somali	Afro-Asiatic	Latin	1992	35.84	44.23	+8.39 [+7.44, +9.25]	+8.85 [+7.82, +9.79]	+0.061 [+0.055, +0.066]
yor	Yoruba	Atlantic-Congo	Latin	1992	23.77	32.10	+8.33 [+7.58, +9.10]	+8.53 [+7.79, +9.28]	+0.104 [+0.097, +0.110]
tir	Tigrinya	Afro-Asiatic	Ethiopic	1992	9.10	17.00	+7.90 [+7.52, +8.25]	+8.56 [+8.17, +8.94]	+0.144 [+0.136, +0.151]
amh	Amharic	Afro-Asiatic	Ethiopic	1992	10.58	18.05	+7.46 [+7.00, +7.93]	+8.16 [+7.66, +8.67]	+0.078 [+0.071, +0.084]
ewe	Ewe	Atlantic-Congo	Latin	1992	24.28	31.58	+7.30 [+6.64, +8.00]	+7.79 [+7.07, +8.55]	+0.064 [+0.059, +0.069]
hau	Hausa	Afro-Asiatic	Latin	1992	42.15	48.57	+6.42 [+5.24, +7.58]	+6.82 [+5.58, +8.06]	+0.028 [+0.023, +0.034]
nya	Chichewa	Atlantic-Congo	Latin	1992	37.87	44.23	+6.36 [+5.46, +7.34]	+6.84 [+5.86, +7.89]	+0.025 [+0.020, +0.030]
wol	Wolof	Atlantic-Congo	Latin	1992	25.29	30.37	+5.08 [+4.34, +5.86]	+5.46 [+4.68, +6.26]	+0.032 [+0.027, +0.037]
sna	Shona	Atlantic-Congo	Latin	1992	34.25	39.23	+4.99 [+4.17, +5.83]	+5.64 [+4.72, +6.61]	+0.024 [+0.019, +0.029]
ibo	Igbo	Atlantic-Congo	Latin	1992	39.90	44.87	+4.97 [+4.17, +5.82]	+5.14 [+4.29, +6.04]	+0.028 [+0.024, +0.032]
nde	nde	Atlantic-Congo	Latin	1992	28.83	32.95	+4.12 [+3.51, +4.74]	+4.67 [+3.95, +5.41]	–
mlg	Malagasy	Austronesian	Latin	1992	36.91	40.85	+3.94 [+3.19, +4.74]	+4.43 [+3.63, +5.33]	+0.026 [+0.021, +0.031]
swh	Swahili	Atlantic-Congo	Latin	1992	55.18	58.73	+3.55 [+2.55, +4.54]	+3.48 [+2.47, +4.48]	+0.017 [+0.012, +0.022]
zul	Zulu	Atlantic-Congo	Latin	1992	40.92	43.50	+2.58 [+1.89, +3.29]	+2.92 [+2.13, +3.73]	+0.015 [+0.010, +0.019]
xho	Xhosa	Atlantic-Congo	Latin	1992	39.85	42.36	+2.51 [+1.92, +3.16]	+2.95 [+2.25, +3.67]	+0.013 [+0.009, +0.017]
afr	Afrikaans	Indo-European	Latin	1992	62.42	63.92	+1.50 [+1.00, +2.15]	+1.50 [+0.97, +2.24]	+0.016 [+0.012, +0.020]
shi	shi	Afro-Asiatic	Arabic	1992	0.24	1.16	+0.93 [+0.85, +1.02]	+0.50 [+0.43, +0.60]	–

Table 24: MAFAND-MT, clean rows: SFT baseline and RL model chrF++ and the RL gain in chrF++, plain chrF and held-out AfriCOMET with 95% intervals, with family and script labels.

ISO	Language	Glottolog family	Script	n	SFT	RL	RL − SFT, chrF++ [CI]	chrF [CI]	AfriCOMET [CI]
iso	iso	Atlantic-Congo	Latin	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ chrF [CI]	Δ AfriCOMET [CI]
—	—			—:	—:	—:	– – –	– – –	– – –
lug	Luganda	Atlantic-Congo	Latin	1500	26.97	39.57	+12.60 [+11.60, +13.70]	+13.37 [+12.26, +14.53]	+0.065 [+0.058, +0.071]
yor	Yoruba	Atlantic-Congo	Latin	1550	23.47	33.45	+9.98 [+9.15, +10.70]	+10.12 [+9.28, +10.87]	+0.127 [+0.119, +0.133]
kin	Kinyarwanda	Atlantic-Congo	Latin	1005	36.51	46.41	+9.90 [+8.70, +11.20]	+10.47 [+9.18, +11.87]	+0.064 [+0.056, +0.073]
luo	Luo	Nilotic	Latin	1500	22.95	32.62	+9.67 [+8.76, +10.50]	+10.50 [+9.54, +11.38]	+0.072 [+0.065, +0.079]
tsn	Setswana	Atlantic-Congo	Latin	1500	43.85	53.10	+9.24 [+7.87, +10.70]	+9.32 [+7.93, +10.81]	+0.045 [+0.039, +0.052]
amh	Amharic	Afro-Asiatic	Ethiopic	1014	11.62	18.33	+6.71 [+5.91, +7.46]	+7.46 [+6.62, +8.31]	+0.061 [+0.052, +0.070]
twi	Twi	Atlantic-Congo	Latin	1500	28.58	34.14	+5.56 [+4.75, +6.29]	+5.99 [+5.14, +6.76]	+0.042 [+0.037, +0.047]
nya	Chichewa	Atlantic-Congo	Latin	1003	38.14	40.95	+2.81 [+1.96, +3.71]	+3.18 [+2.23, +4.12]	+0.009 [+0.002, +0.015]
sna	Shona	Atlantic-Congo	Latin	1004	30.35	32.91	+2.56 [+1.64, +3.39]	+3.11 [+2.05, +4.07]	+0.018 [+0.011, +0.025]
xho	Xhosa	Atlantic-Congo	Latin	1001	49.76	51.78	+2.02 [+1.06, +2.98]	+2.35 [+1.33, +3.40]	+0.008 [+0.002, +0.014]
ibo	Igbo	Atlantic-Congo	Latin	1458	43.55	44.89	+1.34 [+0.43, +2.32]	+1.46 [+0.50, +2.50]	+0.005 [+0.000, +0.010]
swh	Swahili	Atlantic-Congo	Latin	1822	50.68	51.79	+1.12 [+0.38, +1.89]	+1.15 [+0.41, +1.94]	+0.012 [+0.007, +0.017]
zul	Zulu	Atlantic-Congo	Latin	998	44.50	45.58	+1.08 [+0.49, +1.73]	+1.32 [+0.70, +2.04]	+0.011 [+0.005, +0.017]
hau	Hausa	Afro-Asiatic	Latin	1498	32.03	32.80	+0.77 [+0.21, +1.41]	+1.05 [+0.45, +1.75]	+0.014 [+0.009, +0.019]
pcm	pcm	Indo-European	Latin	1562	53.58	53.34	−0.25 [−0.69, +0.30]	−0.46 [−0.93, +0.14]	+0.007 [+0.004, +0.009]

Sources: `reports/data/rl_vs_sft_evidence_2026-11-04/tables.md` (Tables 6, 7a, 7b) and `reports/data/nllb_comparison_rl_models_2026-11-04/table2_per_language_43.csv`. Intervals are 95% paired bootstrap over sentences (Poisson weights, 1,000 resamples); NLLB columns are published values.

Table 25: RL minus SFT per language on FLORES+ devtest (54 languages, all rows; chrF++). Finished / loop are the reward's flags; "paired keeps rows where both models finished and neither loops.

iso	SFT	RL	Δ all rows [CI]	SFT finished / loop	RL finished / loop	Δ paired rows [CI]
kea	18.91	39.18	+20.27 [+18.91, +21.53]	0.83 / 0.20	1.00 / 0.00	+7.66 [+6.93, +8.42]
orm	12.06	30.18	+18.12 [+17.49, +18.76]	0.59 / 0.45	1.00 / 0.00	+2.99 [+2.51, +3.50]
nso	31.58	49.01	+17.43 [+15.75, +19.09]	0.86 / 0.15	1.00 / 0.00	+4.22 [+3.62, +4.91]
lua	17.18	33.37	+16.19 [+15.21, +17.13]	0.85 / 0.25	1.00 / 0.00	+5.45 [+4.94, +6.03]
lin	28.56	44.35	+15.78 [+14.19, +17.22]	0.90 / 0.13	1.00 / 0.00	+5.27 [+4.69, +5.84]
ssw	22.20	37.86	+15.66 [+14.46, +16.85]	0.87 / 0.17	1.00 / 0.00	+5.38 [+4.85, +5.93]
lug	16.17	31.73	+15.57 [+14.66, +16.52]	0.79 / 0.25	1.00 / 0.01	+5.32 [+4.78, +5.87]
sag	19.68	34.31	+14.62 [+13.64, +15.55]	0.87 / 0.20	1.00 / 0.00	+5.99 [+5.51, +6.47]
bem	19.31	33.88	+14.56 [+13.63, +15.61]	0.88 / 0.18	1.00 / 0.01	+7.16 [+6.60, +7.75]
tso	32.92	46.86	+13.93 [+12.43, +15.51]	0.91 / 0.11	1.00 / 0.00	+4.45 [+3.82, +5.07]
luo	21.36	34.49	+13.13 [+11.91, +14.15]	0.92 / 0.13	1.00 / 0.00	+6.44 [+5.88, +6.95]
kin	25.81	38.66	+12.86 [+11.66, +14.03]	0.89 / 0.12	1.00 / 0.00	+5.66 [+5.09, +6.21]
cjk	10.72	23.02	+12.31 [+11.70, +12.89]	0.71 / 0.31	1.00 / 0.01	+2.13 [+1.86, +2.42]
run	21.59	33.88	+12.29 [+11.21, +13.15]	0.88 / 0.15	1.00 / 0.01	+5.03 [+4.52, +5.57]
nus	13.49	24.82	+11.32 [+10.73, +11.91]	0.67 / 0.41	1.00 / 0.00	+1.97 [+1.56, +2.36]
umb	14.80	25.46	+10.66 [+9.77, +11.44]	0.89 / 0.15	1.00 / 0.01	+4.84 [+4.43, +5.26]
kmb	16.02	26.62	+10.61 [+9.82, +11.37]	0.88 / 0.23	1.00 / 0.00	+3.93 [+3.56, +4.29]

iso	SFT	RL	Δ all rows [CI]	SFT finished / loop	RL finished / loop	Δ paired rows [CI]
knc	3.81	14.22	+10.41 [+9.95, +10.88]	0.46 / 0.94	1.00 / 0.48	+2.49 [+0.82, +3.99]
kbp	14.51	24.88	+10.37 [+9.70, +11.00]	0.85 / 0.26	1.00 / 0.01	+5.48 [+4.98, +5.94]
kab	16.35	26.69	+10.34 [+9.50, +11.14]	0.84 / 0.20	1.00 / 0.00	+2.26 [+1.92, +2.61]
dik	3.76	14.06	+10.30 [+9.75, +10.81]	0.29 / 0.85	1.00 / 0.40	+1.80 [+1.10, +2.60]
amh	16.62	26.62	+10.00 [+9.04, +10.96]	0.87 / 0.13	1.00 / 0.00	+2.80 [+2.29, +3.29]
vmw	17.45	27.14	+9.69 [+8.92, +10.48]	0.84 / 0.18	1.00 / 0.00	+2.11 [+1.81, +2.44]
mfe	28.51	37.93	+9.42 [+7.98, +10.94]	0.94 / 0.07	1.00 / 0.01	+3.91 [+3.43, +4.37]
dyu	9.90	18.97	+9.06 [+8.47, +9.62]	0.83 / 0.25	1.00 / 0.00	+4.11 [+3.80, +4.44]
yor	21.32	30.33	+9.01 [+7.99, +10.00]	0.95 / 0.05	1.00 / 0.00	+5.83 [+5.09, +6.56]
tum	25.40	34.27	+8.87 [+7.89, +9.79]	0.94 / 0.08	1.00 / 0.00	+4.57 [+4.08, +5.06]
kam	17.22	25.92	+8.70 [+7.93, +9.50]	0.92 / 0.14	1.00 / 0.00	+4.42 [+4.01, +4.83]
mos	14.09	22.79	+8.70 [+8.13, +9.33]	0.90 / 0.19	1.00 / 0.01	+4.54 [+4.17, +4.89]
tir	10.48	18.74	+8.26 [+7.69, +8.86]	0.77 / 0.23	1.00 / 0.00	+2.23 [+1.83, +2.66]
taq	2.81	11.06	+8.25 [+7.82, +8.67]	0.14 / 0.91	0.99 / 0.56	+3.27 [+1.81, +4.58]
seh	21.67	29.65	+7.98 [+7.00, +8.86]	0.92 / 0.12	1.00 / 0.00	+2.01 [+1.70, +2.33]
bam	19.39	27.17	+7.78 [+6.96, +8.59]	0.89 / 0.13	1.00 / 0.00	+2.47 [+2.14, +2.80]
twi	27.54	35.07	+7.53 [+6.47, +8.61]	0.94 / 0.08	1.00 / 0.00	+2.47 [+2.02, +2.92]
tsn	37.69	45.05	+7.36 [+6.18, +8.65]	0.96 / 0.06	1.00 / 0.00	+2.09 [+1.67, +2.59]
kik	25.98	33.28	+7.30 [+6.39, +8.22]	0.93 / 0.08	1.00 / 0.00	+3.32 [+2.89, +3.81]
hau	39.88	46.46	+6.58 [+5.09, +8.34]	0.96 / 0.04	1.00 / 0.01	+0.90 [+0.50, +1.30]
mlg	39.04	44.73	+5.68 [+4.54, +6.77]	0.97 / 0.03	1.00 / 0.00	+2.03 [+1.59, +2.45]
nya	36.88	42.21	+5.33 [+4.23, +6.54]	0.97 / 0.04	1.00 / 0.00	+1.82 [+1.41, +2.24]
ewe	25.96	31.17	+5.21 [+4.46, +6.04]	0.95 / 0.06	1.00 / 0.00	+2.12 [+1.75, +2.51]
sot	37.87	42.72	+4.85 [+3.70, +6.08]	0.97 / 0.04	1.00 / 0.00	+1.15 [+0.82, +1.50]
arz	4.59	9.18	+4.59 [+4.01, +5.15]	0.49 / 0.52	0.87 / 0.19	+0.67 [+0.05, +1.25]
som	35.47	39.71	+4.24 [+3.14, +5.34]	0.97 / 0.03	1.00 / 0.00	+1.38 [+1.02, +1.75]
wol	26.87	30.58	+3.72 [+2.74, +4.63]	0.97 / 0.03	1.00 / 0.00	+1.14 [+0.82, +1.44]
sna	33.92	37.56	+3.64 [+2.47, +4.85]	0.97 / 0.04	1.00 / 0.00	+0.26 [−0.11, +0.66]
zul	45.91	48.27	+2.36 [+1.56, +3.14]	0.99 / 0.01	1.00 / 0.00	+1.26 [+0.84, +1.67]
ibo	37.82	40.14	+2.33 [+1.57, +3.08]	0.98 / 0.02	1.00 / 0.00	+0.88 [+0.53, +1.24]
ary	4.64	6.88	+2.24 [+1.76, +2.74]	0.54 / 0.48	0.97 / 0.11	−3.70 [−4.39, −3.03]
swh	56.24	58.36	+2.12 [+1.10, +3.38]	1.00 / 0.01	1.00 / 0.00	+0.88 [+0.45, +1.30]
fon	20.17	22.25	+2.07 [+1.60, +2.55]	0.98 / 0.04	1.00 / 0.00	+0.77 [+0.56, +0.97]
xho	43.56	45.43	+1.87 [+1.05, +2.80]	0.99 / 0.01	1.00 / 0.00	+0.67 [+0.28, +1.09]
zgh	2.29	4.07	+1.78 [+1.30, +2.30]	0.12 / 0.89	0.97 / 0.46	−6.69 [−9.11, −4.34]
aeb	2.75	4.12	+1.36 [+0.96, +1.78]	0.53 / 0.49	0.96 / 0.25	−2.46 [−3.19, −1.74]
afr	63.13	64.13	+1.01 [+0.53, +1.61]	1.00 / 0.00	1.00 / 0.00	+0.78 [+0.46, +1.11]

Table 26: RL minus SFT per language on NTREX (clean rows; chrF++ and held-out AfriCOMET).

iso	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ AfriCOMET [CI]
orm	1992	9.63	27.33	+17.70 [+17.16, +18.25]	+0.168 [+0.161, +0.175]
ven	1992	21.30	38.81	+17.52 [+16.72, +18.31]	-
ssw	1992	21.58	38.80	+17.22 [+16.40, +18.02]	+0.122 [+0.115, +0.130]
bem	1992	18.68	34.79	+16.11 [+15.43, +16.84]	+0.111 [+0.105, +0.117]
nso	1992	20.14	32.73	+12.60 [+11.83, +13.35]	+0.101 [+0.095, +0.108]
tsn	1992	36.97	48.71	+11.74 [+10.68, +12.81]	+0.073 [+0.067, +0.080]
kin	1992	26.40	37.13	+10.73 [+9.95, +11.52]	+0.072 [+0.066, +0.078]
som	1992	35.84	44.23	+8.39 [+7.44, +9.25]	+0.061 [+0.055, +0.066]
yor	1992	23.77	32.10	+8.33 [+7.58, +9.10]	+0.104 [+0.097, +0.110]
tir	1992	9.10	17.00	+7.90 [+7.52, +8.25]	+0.144 [+0.136, +0.151]
amh	1992	10.58	18.05	+7.46 [+7.00, +7.93]	+0.078 [+0.071, +0.084]
ewe	1992	24.28	31.58	+7.30 [+6.64, +8.00]	+0.064 [+0.059, +0.069]
hau	1992	42.15	48.57	+6.42 [+5.24, +7.58]	+0.028 [+0.023, +0.034]
nya	1992	37.87	44.23	+6.36 [+5.46, +7.34]	+0.025 [+0.020, +0.030]
wol	1992	25.29	30.37	+5.08 [+4.34, +5.86]	+0.032 [+0.027, +0.037]
sna	1992	34.25	39.23	+4.99 [+4.17, +5.83]	+0.024 [+0.019, +0.029]
ibo	1992	39.90	44.87	+4.97 [+4.17, +5.82]	+0.028 [+0.024, +0.032]
nde	1992	28.83	32.95	+4.12 [+3.51, +4.74]	-
mlg	1992	36.91	40.85	+3.94 [+3.19, +4.74]	+0.026 [+0.021, +0.031]
swh	1992	55.18	58.73	+3.55 [+2.55, +4.54]	+0.017 [+0.012, +0.022]
zul	1992	40.92	43.50	+2.58 [+1.89, +3.29]	+0.015 [+0.010, +0.019]
xho	1992	39.85	42.36	+2.51 [+1.92, +3.16]	+0.013 [+0.009, +0.017]
afr	1992	62.42	63.92	+1.50 [+1.00, +2.15]	+0.016 [+0.012, +0.020]
shi	1992	0.24	1.16	+0.93 [+0.85, +1.02]	-

Table 27: RL minus SFT per language on MAFAND-MT (clean rows; chrF++ and held-out AfriCOMET).

iso	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ AfriCOMET [CI]
lug	1500	26.97	39.57	+12.60 [+11.60, +13.70]	+0.065 [+0.058, +0.071]
yor	1550	23.47	33.45	+9.98 [+9.15, +10.70]	+0.127 [+0.119, +0.133]
kin	1005	36.51	46.41	+9.90 [+8.70, +11.20]	+0.064 [+0.056, +0.073]
luo	1500	22.95	32.62	+9.67 [+8.76, +10.50]	+0.072 [+0.065, +0.079]
tsn	1500	43.85	53.10	+9.24 [+7.87, +10.70]	+0.045 [+0.039, +0.052]
amh	1014	11.62	18.33	+6.71 [+5.91, +7.46]	+0.061 [+0.052, +0.070]
twi	1500	28.58	34.14	+5.56 [+4.75, +6.29]	+0.042 [+0.037, +0.047]
nya	1003	38.14	40.95	+2.81 [+1.96, +3.71]	+0.009 [+0.002, +0.015]

iso	n	SFT chrF++	RL chrF++	Δ all rows [CI]	Δ AfriCOMET [CI]
sna	1004	30.35	32.91	+2.56 [+1.64, +3.39]	+0.018 [+0.011, +0.025]
xho	1001	49.76	51.78	+2.02 [+1.06, +2.98]	+0.008 [+0.002, +0.014]
ibo	1458	43.55	44.89	+1.34 [+0.43, +2.32]	+0.005 [+0.000, +0.010]
swh	1822	50.68	51.79	+1.12 [+0.38, +1.89]	+0.012 [+0.007, +0.017]
zul	998	44.50	45.58	+1.08 [+0.49, +1.73]	+0.011 [+0.005, +0.017]
hau	1498	32.03	32.80	+0.77 [+0.21, +1.41]	+0.014 [+0.009, +0.019]
pcm	1562	53.58	53.34	−0.25 [−0.69, +0.30]	+0.007 [+0.004, +0.009]

Table 28: FLORES+ devtest chrF++ on the 43 languages shared with the published NLLB table (our values greedy; NLLB-3.3B published). Yoruba: the published value is anomalously low.

iso	language	NLLB-3.3B	SFT	RL mixed best	mixed minus NLLB-3.3B	NLLB tertile
yor	Yoruba	24.90	21.32	30.33	+5.43	tail (15)
fon	Fon	19.30	20.17	22.25	+2.95	tail (15)
dyu	Dyula	16.20	9.90	18.97	+2.77	tail (15)
wol	Wolof	28.10	26.87	30.58	+2.48	tail (15)
kam	Kamba	23.60	17.22	25.92	+2.32	tail (15)
kmb	Kimbundu	24.40	16.02	26.62	+2.22	tail (15)
umb	Umbundu	23.60	14.80	25.46	+1.86	tail (15)
afr	Afrikaans	64.70	63.13	64.13	−0.57	high (14)
ibo	Igbo	41.10	37.82	40.14	−0.96	mid (14)
nso	Sepedi	50.30	31.58	49.01	−1.29	high (14)
mos	Mossi	24.10	14.09	22.79	−1.31	tail (15)
swh	Swahili	60.00	56.24	58.36	−1.64	high (14)
lua	Luba-Kasai	35.20	17.18	33.37	−1.83	tail (15)
sag	Sango	36.50	19.68	34.31	−2.19	mid (14)
tum	Tumbuka	36.60	25.40	34.27	−2.33	mid (14)
nya	Chichewa	45.30	36.88	42.21	−3.09	high (14)
som	Somali	42.80	35.47	39.71	−3.09	mid (14)
xho	Xhosa	48.60	43.56	45.43	−3.17	high (14)
twi	Twi	38.30	27.54	35.07	−3.23	mid (14)
tso	Tsonga	50.20	32.92	46.86	−3.34	high (14)
tsn	Setswana	48.60	37.69	45.05	−3.55	high (14)
bem	Bemba	37.50	19.31	33.88	−3.62	mid (14)
lin	Lingala	48.00	28.56	44.35	−3.65	high (14)
sot	Sesotho	46.50	37.87	42.72	−3.78	high (14)
kik	Kikuyu	37.30	25.98	33.28	−4.02	mid (14)
nus	Nuer	28.90	13.49	24.82	−4.08	tail (15)
bam	Bambara	31.40	19.39	27.17	−4.23	tail (15)
luo	Luo	39.00	21.36	34.49	−4.51	mid (14)
kbp	Kabiye	29.50	14.51	24.88	−4.62	tail (15)
zul	Zulu	53.50	45.91	48.27	−5.23	high (14)
sna	Shona	42.80	33.92	37.56	−5.24	mid (14)
kab	Kabyle	32.00	16.35	26.69	−5.31	tail (15)
mlg	Malagasy	50.20	39.04	44.73	−5.47	high (14)
orm	Oromo	35.70	12.06	30.18	−5.52	tail (15)
ssw	Swati	43.50	22.20	37.86	−5.64	mid (14)
hau	Hausa	52.50	39.88	46.46	−6.04	high (14)
tir	Tigrinya	24.80	10.48	18.74	−6.06	tail (15)
lug	Luganda	38.50	16.17	31.73	−6.77	mid (14)
ewe	Ewe	38.20	25.96	31.17	−7.03	mid (14)
run	Kirundi	41.90	21.59	33.88	−8.02	mid (14)
kea	Kabuverdianu	47.90	18.91	39.18	−8.72	high (14)
kin	Kinyarwanda	48.40	25.81	38.66	−9.74	high (14)
amh	Amharic	37.90	16.62	26.62	−11.28	mid (14)

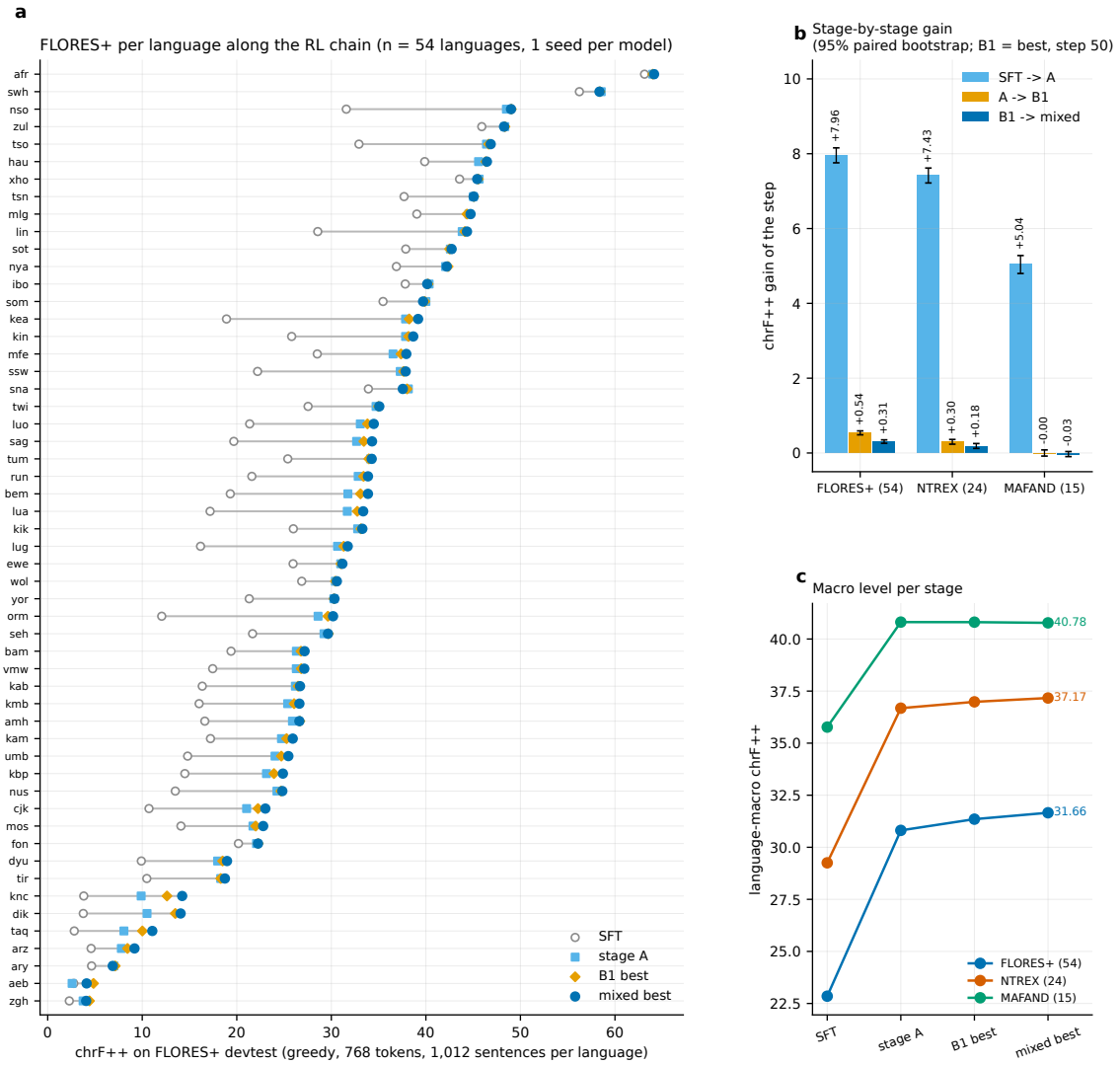


Figure 7: (a) FLORES+ devtest chrF++ per language (n = 54 languages, 1,012 sentences each; greedy, 768 tokens) for the SFT (open circle), RL stage A (square), B1 best (step 50, diamond) and the mixed best (filled circle), sorted by the mixed best; one training seed per model. (b) Gain of each RL step (language macro, 95% paired bootstrap over sentences, Poisson weights shared across parallel languages) on FLORES+ (54 languages), NTREX (24) and MAFAND (15). (c) Language-macro chrF++ after each stage. Sources: `reports/data/eval_weak_group_2026-11-03/results/results_{sft,stageA,b1best,mixedbest}.csv`, `reports/data/rl_vs_sft_evidence_2026-11-04/evidence.json`.

D Decoding: loop guard, beam search, MBR and engine checks

E-02: beam search and MBR against greedy decoding

Table 30: E-02 per language: chrF++ of the SFT baseline and the RL model under greedy decoding, beam search with 4 beams (primary: stops when 4 hypotheses have ended; "HF": Hugging Face rule) and MBR with 16 samples (utility chrF++). 20 FLORES+ languages, 200 sentences each, 768 new tokens, vLLM bf16. Higher is better.

ISO	Language	SFT greedy	SFT beam-4	SFT beam-4 (HF)	SFT MBR-16	RL greedy	RL beam-4	RL beam-4 (HF)	RL MBR-16
afr	Afrikaans	63.2	62.1	64.2	63.9	63.3	62.8	64.0	64.0
amh	Amharic	15.3	21.9	16.4	23.2	25.5	26.3	27.1	27.0
ewe	Ewe	27.6	27.1	22.2	31.4	32.4	31.6	32.7	33.2
hau	Hausa	43.4	45.7	42.0	47.3	46.7	45.8	47.6	47.7
ibo	Igbo	40.0	39.0	40.3	41.8	41.6	40.1	42.4	42.6
kin	Kinyarwanda	26.6	34.3	27.6	35.2	38.9	39.5	40.8	40.3
lin	Lingala	26.6	36.0	27.4	38.6	44.2	42.5	44.8	45.3
lug	Luganda	17.1	24.3	12.9	27.8	32.4	32.7	33.4	33.6
nso	Sepedi	31.3	43.5	37.2	45.6	49.5	47.8	49.6	50.1
orm	Oromo	12.8	22.2	11.4	27.2	29.9	29.0	29.7	31.3
sna	Shona	34.0	37.2	38.0	38.8	38.5	36.2	39.4	39.1
som	Somali	36.0	36.1	36.9	39.2	39.4	38.2	40.5	40.3
swh	Swahili	55.2	55.4	59.2	59.0	58.9	56.5	59.7	60.0
tir	Tigrinya	9.5	16.0	11.3	16.2	18.6	19.1	19.4	19.4

ISO	Language	SFT greedy	SFT beam-4	SFT beam-4 (HF)	SFT MBR-16	RL greedy	RL beam-4	RL beam-4 (HF)	RL MBR-16
tsn	Setswana	37.8	43.2	41.2	44.2	45.0	44.5	46.0	45.7
twi	Twi	29.1	32.4	28.3	34.3	35.9	36.0	36.5	36.6
wol	Wolof	27.2	28.8	28.0	30.8	30.8	31.0	31.4	31.5
xho	Xhosa	44.9	44.2	46.9	46.8	45.9	45.0	47.4	46.5
yor	Yoruba	21.0	24.4	24.2	24.8	30.7	30.0	31.3	31.4
zul	Zulu	47.7	46.9	49.8	49.1	49.1	47.3	50.3	50.3
Macro		32.3	36.0	33.3	38.3	39.9	39.1	40.7	40.8

Table 31: E-02 per language: held-out reference-based AfriCOMET-STL-1.1 (mean of per-sentence scores) for the same eight systems and sentences as Table 30. Higher is better.

ISO	Language	SFT greedy	SFT beam-4	SFT beam-4 (HF)	SFT MBR-16	RL greedy	RL beam-4	RL beam-4 (HF)	RL MBR-16
afr	Afrikaans	0.699	0.689	0.716	0.712	0.701	0.704	0.713	0.709
amh	Amharic	0.463	0.480	0.480	0.517	0.547	0.553	0.572	0.564
ewe	Ewe	0.545	0.520	0.514	0.554	0.580	0.577	0.594	0.588
hau	Hausa	0.559	0.572	0.581	0.575	0.576	0.572	0.592	0.583
ibo	Igbo	0.618	0.606	0.627	0.628	0.632	0.616	0.643	0.639
kin	Kinyarwanda	0.543	0.576	0.568	0.583	0.616	0.623	0.638	0.626
lin	Lingala	0.479	0.530	0.539	0.494	0.581	0.580	0.599	0.591
lug	Luganda	0.447	0.469	0.440	0.473	0.545	0.560	0.568	0.556
nso	Sepedi	0.562	0.593	0.611	0.603	0.646	0.638	0.658	0.652
orm	Oromo	0.421	0.436	0.416	0.495	0.575	0.567	0.577	0.580
sna	Shona	0.548	0.567	0.583	0.569	0.584	0.559	0.598	0.593
som	Somali	0.557	0.552	0.571	0.565	0.581	0.566	0.593	0.587
swh	Swahili	0.600	0.593	0.627	0.617	0.619	0.599	0.631	0.631
tir	Tigrinya	0.471	0.524	0.521	0.566	0.627	0.643	0.651	0.643
tsn	Setswana	0.606	0.616	0.633	0.612	0.634	0.628	0.646	0.639
twi	Twi	0.568	0.587	0.580	0.598	0.615	0.620	0.626	0.622
wol	Wolof	0.594	0.592	0.601	0.611	0.615	0.615	0.617	0.616
xho	Xhosa	0.574	0.569	0.595	0.588	0.584	0.578	0.603	0.592
yor	Yoruba	0.548	0.582	0.596	0.568	0.674	0.670	0.685	0.681
zul	Zulu	0.624	0.626	0.648	0.633	0.632	0.622	0.650	0.645
Macro		0.551	0.564	0.572	0.578	0.608	0.605	0.623	0.617

Table 32: E-02 per language, SFT baseline: chrF++ difference of each decoder to greedy decoding with 95% paired-bootstrap interval over sentences (200 sentences per language; positive = better than greedy).

ISO	Language	beam-4 − greedy	beam-4 (HF) − greedy	MBR-16 − greedy
afr	Afrikaans	−1.15 [−2.41, −0.01]	+0.92 [+0.31, +1.60]	+0.66 [+0.20, +1.18]
amh	Amharic	+6.59 [+4.40, +8.48]	+1.06 [−1.08, +3.28]	+7.90 [+6.14, +9.50]
ewe	Ewe	−0.47 [−2.29, +1.37]	−5.38 [−7.53, −3.01]	+3.87 [+2.23, +5.42]
hau	Hausa	+2.23 [−0.04, +4.57]	−1.40 [−4.51, +1.58]	+3.87 [+1.13, +6.79]
ibo	Igbo	−1.05 [−2.87, +0.90]	+0.27 [−2.18, +2.36]	+1.84 [+0.65, +3.23]
kin	Kinyarwanda	+7.62 [+4.80, +10.41]	+0.91 [−2.64, +4.44]	+8.56 [+6.29, +10.88]
lin	Lingala	+9.41 [+6.45, +12.17]	+0.79 [−2.95, +4.56]	+12.02 [+9.15, +14.58]
lug	Luganda	+7.12 [+4.90, +9.04]	−4.27 [−6.44, −2.41]	+10.69 [+8.76, +12.50]
nso	Sepedi	+12.18 [+8.43, +15.53]	+5.90 [+1.44, +10.93]	+14.34 [+10.97, +17.37]
orm	Oromo	+9.42 [+8.01, +10.70]	−1.37 [−2.84, −0.02]	+14.42 [+12.91, +15.70]
sna	Shona	+3.20 [+0.42, +5.93]	+4.08 [+1.69, +6.61]	+4.85 [+2.40, +7.42]
som	Somali	+0.13 [−2.41, +2.97]	+0.91 [−1.79, +3.87]	+3.26 [+1.19, +5.77]
swh	Swahili	+0.18 [−2.85, +3.86]	+3.98 [+1.44, +7.25]	+3.81 [+1.47, +7.19]
tir	Tigrinya	+6.49 [+5.13, +7.80]	+1.87 [+0.61, +3.30]	+6.72 [+5.71, +7.64]
tsn	Setswana	+5.40 [+2.21, +8.45]	+3.39 [−0.37, +6.96]	+6.43 [+3.39, +9.40]
twi	Twi	+3.35 [+1.07, +5.79]	−0.80 [−3.69, +2.20]	+5.19 [+3.35, +7.34]
wol	Wolof	+1.61 [−0.46, +3.53]	+0.80 [−1.28, +2.98]	+3.62 [+1.95, +5.41]
xho	Xhosa	−0.66 [−2.47, +1.16]	+2.02 [+0.56, +3.62]	+1.91 [+0.90, +3.20]
yor	Yoruba	+3.42 [+0.80, +5.95]	+3.24 [+0.53, +6.07]	+3.82 [+1.49, +5.92]
zul	Zulu	−0.82 [−2.81, +1.05]	+2.03 [+0.37, +3.79]	+1.42 [+0.46, +2.72]

Table 33: E-02 per language, SFT baseline: held-out AfriCOMET difference of each decoder to greedy decoding with 95% interval (positive = better than greedy).

ISO	Language	beam-4 − greedy	beam-4 (HF) − greedy	MBR-16 − greedy
afr	Afrikaans	−0.010 [−0.023, +0.004]	+0.017 [+0.008, +0.026]	+0.013 [+0.005, +0.021]
amh	Amharic	+0.017 [−0.009, +0.041]	+0.018 [−0.010, +0.043]	+0.054 [+0.036, +0.072]
ewe	Ewe	−0.025 [−0.041, −0.008]	−0.030 [−0.049, −0.012]	+0.009 [−0.001, +0.021]
hau	Hausa	+0.013 [−0.002, +0.026]	+0.022 [+0.007, +0.035]	+0.016 [+0.006, +0.027]
ibo	Igbo	−0.013 [−0.029, +0.003]	+0.009 [−0.004, +0.021]	+0.009 [+0.001, +0.019]
kin	Kinyarwanda	+0.034 [+0.009, +0.057]	+0.025 [+0.000, +0.050]	+0.041 [+0.025, +0.057]
lin	Lingala	+0.051 [+0.030, +0.071]	+0.059 [+0.039, +0.081]	+0.014 [−0.003, +0.031]
lug	Luganda	+0.022 [+0.002, +0.039]	−0.008 [−0.030, +0.014]	+0.025 [+0.008, +0.041]

ISO	Language	beam-4 − greedy	beam-4 (HF) − greedy	MBR-16 − greedy
nso	Sepedi	+0.031 [+0.006, +0.055]	+0.049 [+0.025, +0.072]	+0.041 [+0.022, +0.061]
orm	Oromo	+0.015 [−0.007, +0.036]	−0.005 [−0.028, +0.019]	+0.074 [+0.053, +0.096]
sna	Shona	+0.019 [−0.002, +0.037]	+0.034 [+0.016, +0.051]	+0.021 [+0.010, +0.032]
som	Somali	−0.006 [−0.027, +0.014]	+0.014 [−0.004, +0.031]	+0.008 [−0.002, +0.019]
swh	Swahili	−0.006 [−0.026, +0.012]	+0.027 [+0.011, +0.043]	+0.017 [+0.006, +0.032]
tir	Tigrinya	+0.053 [+0.024, +0.081]	+0.050 [+0.025, +0.077]	+0.095 [+0.072, +0.116]
tsn	Setswana	+0.011 [−0.006, +0.027]	+0.027 [+0.012, +0.044]	+0.006 [−0.004, +0.017]
twi	Twi	+0.019 [−0.000, +0.038]	+0.012 [−0.009, +0.032]	+0.031 [+0.018, +0.045]
wol	Wolof	−0.002 [−0.021, +0.016]	+0.007 [−0.010, +0.023]	+0.017 [+0.008, +0.027]
xho	Xhosa	−0.005 [−0.020, +0.010]	+0.021 [+0.008, +0.033]	+0.014 [+0.005, +0.023]
yor	Yoruba	+0.034 [+0.008, +0.061]	+0.048 [+0.023, +0.074]	+0.019 [+0.002, +0.037]
zul	Zulu	+0.002 [−0.015, +0.018]	+0.024 [+0.009, +0.038]	+0.009 [+0.001, +0.019]

Table 34: E-02 per language, RL model: chrF++ difference of each decoder to greedy decoding with 95% paired-bootstrap interval over sentences (200 sentences per language; positive = better than greedy).

ISO	Language	beam-4 − greedy	beam-4 (HF) − greedy	MBR-16 − greedy
afr	Afrikaans	−0.49 [−1.17, +0.18]	+0.64 [+0.17, +1.17]	+0.70 [+0.30, +1.16]
amh	Amharic	+0.77 [−0.72, +2.06]	+1.61 [+0.42, +2.74]	+1.46 [+0.78, +2.23]
ewe	Ewe	−0.80 [−1.83, +0.10]	+0.32 [−0.55, +1.07]	+0.79 [+0.40, +1.22]
hau	Hausa	−0.91 [−2.32, +0.27]	+0.90 [+0.16, +1.62]	+0.98 [+0.43, +1.55]
ibo	Igbo	−1.54 [−2.90, −0.37]	+0.82 [+0.29, +1.33]	+0.99 [+0.51, +1.52]
kin	Kinyarwanda	+0.58 [−0.63, +1.66]	+1.94 [+0.91, +2.89]	+1.34 [+0.77, +1.91]
lin	Lingala	−1.69 [−3.03, −0.59]	+0.53 [−0.12, +1.21]	+1.04 [+0.52, +1.66]
lug	Luganda	+0.29 [−0.67, +1.20]	+1.05 [+0.39, +1.80]	+1.23 [+0.70, +1.86]
nso	Sepedi	−1.67 [−3.00, −0.59]	+0.04 [−0.59, +0.67]	+0.60 [+0.19, +1.04]
orm	Oromo	−0.88 [−1.71, −0.14]	−0.22 [−1.00, +0.46]	+1.36 [+0.93, +1.83]
sna	Shona	−2.26 [−3.64, −1.08]	+0.95 [+0.36, +1.49]	+0.61 [+0.22, +1.01]
som	Somali	−1.23 [−2.30, −0.25]	+1.06 [+0.55, +1.58]	+0.81 [+0.44, +1.21]
swh	Swahili	−2.45 [−4.09, −1.01]	+0.79 [+0.28, +1.39]	+1.09 [+0.66, +1.63]
tir	Tigrinya	+0.55 [−0.15, +1.23]	+0.79 [+0.17, +1.43]	+0.84 [+0.41, +1.29]
tsn	Setswana	−0.58 [−1.51, +0.27]	+0.92 [+0.37, +1.48]	+0.63 [+0.25, +1.09]
twi	Twi	+0.09 [−0.75, +0.85]	+0.60 [−0.02, +1.19]	+0.71 [+0.24, +1.19]
wol	Wolof	+0.13 [−0.65, +0.78]	+0.56 [+0.09, +1.04]	+0.71 [+0.34, +1.10]
xho	Xhosa	−0.94 [−2.26, +0.25]	+1.51 [+0.89, +2.16]	+0.60 [+0.11, +1.14]
yor	Yoruba	−0.67 [−1.58, +0.14]	+0.66 [+0.19, +1.15]	+0.74 [+0.42, +1.09]
zul	Zulu	−1.76 [−3.17, −0.52]	+1.17 [+0.49, +1.76]	+1.19 [+0.65, +1.75]

Table 35: E-02 per language, RL model: held-out AfriCOMET difference of each decoder to greedy decoding with 95% interval (positive = better than greedy).

ISO	Language	beam-4 − greedy	beam-4 (HF) − greedy	MBR-16 − greedy
afr	Afrikaans	+0.003 [−0.006, +0.011]	+0.013 [+0.006, +0.019]	+0.009 [+0.003, +0.015]
amh	Amharic	+0.006 [−0.014, +0.024]	+0.025 [+0.012, +0.039]	+0.017 [+0.008, +0.027]
ewe	Ewe	−0.004 [−0.018, +0.010]	+0.014 [+0.003, +0.025]	+0.008 [+0.001, +0.017]
hau	Hausa	−0.004 [−0.019, +0.009]	+0.016 [+0.007, +0.025]	+0.007 [−0.002, +0.015]
ibo	Igbo	−0.016 [−0.028, −0.005]	+0.011 [+0.005, +0.018]	+0.007 [+0.002, +0.013]
kin	Kinyarwanda	+0.007 [−0.008, +0.021]	+0.022 [+0.009, +0.035]	+0.010 [+0.004, +0.018]
lin	Lingala	−0.001 [−0.014, +0.010]	+0.017 [+0.008, +0.025]	+0.010 [+0.003, +0.016]
lug	Luganda	+0.016 [+0.004, +0.028]	+0.023 [+0.013, +0.034]	+0.012 [+0.004, +0.020]
nso	Sepedi	−0.008 [−0.021, +0.002]	+0.012 [+0.004, +0.018]	+0.006 [−0.000, +0.011]
orm	Oromo	−0.008 [−0.021, +0.006]	+0.002 [−0.012, +0.014]	+0.005 [−0.003, +0.014]
sna	Shona	−0.025 [−0.041, −0.012]	+0.014 [+0.006, +0.021]	+0.008 [+0.003, +0.014]
som	Somali	−0.015 [−0.027, −0.004]	+0.012 [+0.005, +0.019]	+0.006 [+0.000, +0.012]
swh	Swahili	−0.020 [−0.034, −0.007]	+0.012 [+0.006, +0.019]	+0.011 [+0.006, +0.017]
tir	Tigrinya	+0.016 [+0.002, +0.030]	+0.024 [+0.012, +0.036]	+0.016 [+0.007, +0.025]
tsn	Setswana	−0.006 [−0.016, +0.004]	+0.012 [+0.006, +0.019]	+0.005 [−0.001, +0.011]
twi	Twi	+0.005 [−0.005, +0.015]	+0.012 [+0.004, +0.019]	+0.007 [+0.001, +0.014]
wol	Wolof	−0.000 [−0.008, +0.008]	+0.002 [−0.004, +0.010]	+0.002 [−0.005, +0.008]
xho	Xhosa	−0.007 [−0.020, +0.005]	+0.018 [+0.011, +0.026]	+0.007 [+0.001, +0.013]
yor	Yoruba	−0.004 [−0.014, +0.005]	+0.011 [+0.005, +0.018]	+0.007 [+0.002, +0.013]
zul	Zulu	−0.010 [−0.023, +0.002]	+0.018 [+0.010, +0.025]	+0.013 [+0.006, +0.019]

Table 36: E-02 per language: RL model minus SFT baseline in chrF++ under the same decoder, with 95% paired-bootstrap intervals (positive = RL better). The gain shrinks when the SFT baseline is decoded with beam search or MBR.

ISO	Language	RL − SFT, greedy	RL − SFT, beam-4	RL − SFT, MBR-16	RL − SFT, beam-4 (HF)
afr	Afrikaans	+0.1 [−0.6, +0.9]	+0.7 [−0.4, +2.0]	+0.1 [−0.6, +1.0]	
amh	Amharic	+10.2 [+8.1, +12.1]	+4.4 [+2.9, +5.8]	+3.7 [+2.6, +5.0]	
ewe	Ewe	+4.8 [+3.1, +6.5]	+4.5 [+3.2, +5.7]	+1.7 [+1.0, +2.5]	
hau	Hausa	+3.3 [+0.4, +6.4]	+0.1 [−1.2, +1.5]	+0.4 [−0.4, +1.2]	
ibo	Igbo	+1.6 [+0.4, +3.1]	+1.1 [−0.0, +2.4]	+0.8 [+0.1, +1.4]	
kin	Kinyarwanda	+12.3 [+9.7, +14.8]	+5.2 [+3.7, +6.7]	+5.0 [+3.9, +6.1]	

ISO	Language	RL − SFT, greedy	RL − SFT, beam-4	RL − SFT, MBR-16	RL − SFT, beam-4 (HF)
lin	Lingala	+17.6 [+14.5, +20.6]	+6.5 [+4.8, +8.1]	+6.7 [+5.6, +7.8]	
lug	Luganda	+15.2 [+13.2, +17.0]	+8.4 [+7.1, +9.7]	+5.8 [+4.9, +6.7]	
nso	Sepedi	+18.2 [+14.6, +21.9]	+4.4 [+2.7, +6.2]	+4.5 [+3.5, +5.5]	
orm	Oromo	+17.2 [+15.5, +18.6]	+6.9 [+5.6, +8.2]	+4.1 [+3.3, +4.9]	
sna	Shona	+4.5 [+2.1, +7.2]	−0.9 [−2.5, +0.6]	+0.3 [−0.4, +1.0]	
som	Somali	+3.5 [+1.3, +6.1]	+2.1 [+1.0, +3.3]	+1.0 [+0.3, +1.8]	
swh	Swahili	+3.7 [+1.2, +7.0]	+1.1 [−0.2, +2.4]	+1.0 [+0.1, +1.7]	
tir	Tigrinya	+9.1 [+7.9, +10.3]	+3.2 [+2.1, +4.3]	+3.2 [+2.4, +4.1]	
tsn	Setswana	+7.2 [+4.1, +10.3]	+1.3 [+0.3, +2.3]	+1.4 [+0.6, +2.3]	
twi	Twi	+6.8 [+4.9, +9.0]	+3.6 [+2.5, +4.6]	+2.3 [+1.4, +3.2]	
wol	Wolof	+3.7 [+1.8, +5.5]	+2.2 [+1.3, +3.2]	+0.7 [+0.1, +1.4]	
xho	Xhosa	+1.0 [−0.1, +2.4]	+0.7 [−0.4, +2.0]	−0.3 [−1.0, +0.5]	
yor	Yoruba	+9.7 [+7.2, +12.0]	+5.6 [+4.0, +7.3]	+6.6 [+5.1, +8.2]	
zul	Zulu	+1.4 [+0.1, +3.1]	+0.4 [−0.9, +1.8]	+1.1 [+0.2, +2.0]	

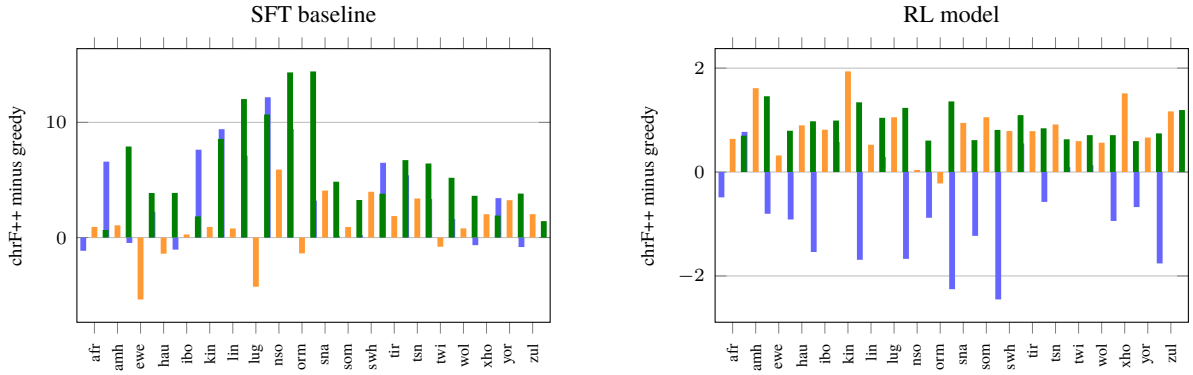


Figure 8: E-02 per language: chrF++ gain of beam search (4 beams, primary and HF rule) and MBR (16 samples) over greedy decoding, for the SFT baseline (left) and the RL model (right); 20 FLORES+ languages, 200 sentences each. Bars: blue = beam-4, orange = beam-4 (HF rule), green = MBR-16; bars above zero help. The baseline gains a lot in many languages and the RL model little; intervals per language are in Tables 32 and 34.

E-01: decode-time loop guard

Table 40: E-01 per language and benchmark (clean rows): SFT baseline and RL model both decoded with the repetition guard (vLLM frequency penalty 0.10); chrF++ and plain chrF difference with 95% paired-bootstrap intervals. The language averages of the difference are the primary E-01 result (+4.37 chrF++ over all 93 pairs).

benchmark	ISO	Language	SFT + guard	RL + guard	RL − SFT, chrF++	RL − SFT, chrF
FLORES+	aeb	aeb	4.9	4.8	−0.1 [−0.5, +0.5]	−0.6 [−1.2, −0.0]
FLORES+	afr	Afrikaans	63.3	64.1	+0.8 [+0.5, +1.1]	+0.8 [+0.5, +1.1]
FLORES+	amh	Amharic	21.2	26.9	+5.8 [+5.0, +6.5]	+6.1 [+5.3, +7.0]
FLORES+	ary	ary	7.2	7.8	+0.6 [+0.1, +1.1]	−0.1 [−0.7, +0.5]
FLORES+	arz	arz	7.2	12.6	+5.4 [+4.9, +5.9]	+5.7 [+5.2, +6.3]
FLORES+	bam	Bambara	24.0	27.4	+3.4 [+3.0, +3.9]	+3.8 [+3.3, +4.2]
FLORES+	bem	Bemba	25.2	34.2	+9.0 [+8.4, +9.7]	+9.5 [+8.9, +10.3]
FLORES+	cjk	cjk	18.9	23.4	+4.5 [+4.0, +5.0]	+5.1 [+4.6, +5.7]
FLORES+	dik	dik	9.9	17.2	+7.3 [+6.9, +7.8]	+7.9 [+7.4, +8.4]
FLORES+	dyu	Dyula	13.1	19.1	+6.0 [+5.6, +6.4]	+6.9 [+6.5, +7.4]
FLORES+	ewe	Ewe	28.6	31.4	+2.8 [+2.3, +3.2]	+2.9 [+2.4, +3.4]
FLORES+	fon	Fon	21.1	22.3	+1.2 [+0.9, +1.5]	+1.4 [+1.1, +1.8]
FLORES+	hau	Hausa	44.9	46.5	+1.6 [+1.1, +2.2]	+1.8 [+1.2, +2.4]
FLORES+	ibo	Igbo	38.9	40.3	+1.4 [+0.9, +1.9]	+1.5 [+1.0, +1.9]
FLORES+	kab	Kabyle	21.9	26.8	+4.9 [+4.4, +5.5]	+5.3 [+4.7, +5.8]
FLORES+	kam	Kamba	21.0	26.2	+5.2 [+4.7, +5.7]	+5.4 [+4.9, +6.0]
FLORES+	kbp	Kabiye	19.2	25.3	+6.1 [+5.7, +6.6]	+6.5 [+6.0, +7.0]
FLORES+	kea	Kabuverdianu	25.9	39.1	+13.2 [+12.1, +14.3]	+13.5 [+12.4, +14.6]
FLORES+	kik	Kikuyu	29.3	33.5	+4.1 [+3.6, +4.7]	+4.3 [+3.7, +4.9]
FLORES+	kin	Kinyarwanda	31.6	39.0	+7.4 [+6.7, +8.2]	+7.7 [+6.9, +8.5]
FLORES+	kmb	Kimbundu	20.7	26.8	+6.1 [+5.6, +6.6]	+6.7 [+6.1, +7.3]
FLORES+	knc	Kanuri	9.9	18.0	+8.1 [+7.7, +8.6]	+9.0 [+8.5, +9.5]
FLORES+	lin	Lingala	35.8	44.1	+8.3 [+7.5, +9.2]	+8.6 [+7.6, +9.5]
FLORES+	lua	Luba-Kasai	25.3	33.4	+8.2 [+7.5, +8.8]	+8.6 [+7.9, +9.2]
FLORES+	lug	Luganda	23.6	32.1	+8.4 [+7.7, +9.1]	+9.2 [+8.4, +9.9]
FLORES+	luo	Luo	26.8	34.6	+7.8 [+7.2, +8.4]	+8.2 [+7.6, +8.9]
FLORES+	mfe	mfe	32.7	38.5	+5.8 [+5.1, +6.7]	+5.9 [+5.1, +6.8]
FLORES+	mlg	Malagasy	42.3	44.8	+2.5 [+2.0, +3.0]	+2.8 [+2.3, +3.3]
FLORES+	mos	Mossi	18.1	23.2	+5.2 [+4.8, +5.6]	+5.8 [+5.4, +6.2]
FLORES+	nso	Sepedi	41.4	49.2	+7.9 [+6.9, +8.8]	+8.0 [+7.0, +9.0]

benchmark	ISO	Language	SFT + guard	RL + guard	RL − SFT, chrF++	RL − SFT, chrF
FLORES+	nus	Nuer	19.5	24.9	+5.4 [+4.8, +5.9]	+5.7 [+5.2, +6.3]
FLORES+	nya	Chichewa	39.9	42.2	+2.3 [+1.7, +2.8]	+2.5 [+2.0, +3.0]
FLORES+	orm	Oromo	20.0	30.3	+10.4 [+9.6, +11.1]	+11.8 [+11.0, +12.6]
FLORES+	run	Kirundi	27.3	34.4	+7.0 [+6.4, +7.7]	+7.5 [+6.8, +8.2]
FLORES+	sag	Sango	26.1	34.4	+8.3 [+7.6, +9.0]	+9.0 [+8.3, +9.7]
FLORES+	seh	seh	26.4	29.8	+3.5 [+3.0, +3.9]	+4.0 [+3.4, +4.5]
FLORES+	sna	Shona	37.0	37.9	+0.9 [+0.4, +1.4]	+1.1 [+0.6, +1.6]
FLORES+	som	Somali	37.3	39.8	+2.5 [+1.9, +3.2]	+2.8 [+2.1, +3.5]
FLORES+	sot	Sesotho	41.0	42.6	+1.5 [+1.1, +2.0]	+1.6 [+1.2, +2.2]
FLORES+	ssw	Swati	29.7	38.0	+8.3 [+7.6, +9.1]	+9.2 [+8.4, +10.1]
FLORES+	swh	Swahili	57.5	58.4	+0.9 [+0.4, +1.4]	+0.9 [+0.5, +1.4]
FLORES+	taq	taq	6.6	13.0	+6.4 [+6.0, +6.9]	+6.8 [+6.3, +7.3]
FLORES+	tir	Tigrinya	13.4	18.8	+5.4 [+4.9, +6.0]	+6.0 [+5.5, +6.6]
FLORES+	tsn	Setswana	42.2	45.2	+3.1 [+2.4, +3.8]	+3.2 [+2.5, +3.9]
FLORES+	tso	Tsonga	39.9	47.0	+7.1 [+6.2, +8.1]	+7.3 [+6.4, +8.4]
FLORES+	tum	Tumbuka	28.6	34.3	+5.7 [+5.1, +6.3]	+6.0 [+5.3, +6.7]
FLORES+	twi	Twi	31.9	35.3	+3.4 [+2.9, +3.9]	+3.5 [+3.0, +4.1]
FLORES+	umb	Umbundu	19.4	25.6	+6.1 [+5.7, +6.6]	+6.9 [+6.4, +7.5]
FLORES+	vmw	vmw	22.7	27.3	+4.7 [+4.1, +5.2]	+5.2 [+4.7, +5.8]
FLORES+	wol	Wolof	28.7	30.6	+1.9 [+1.5, +2.4]	+2.1 [+1.6, +2.5]
FLORES+	xho	Xhosa	44.7	45.4	+0.7 [+0.3, +1.2]	+1.0 [+0.5, +1.4]
FLORES+	yor	Yoruba	23.6	30.4	+6.7 [+6.0, +7.5]	+6.6 [+5.8, +7.4]
FLORES+	zgh	zgh	5.7	4.5	−1.1 [−1.7, −0.4]	−1.8 [−2.5, −1.1]
FLORES+	zul	Zulu	46.8	48.4	+1.6 [+1.1, +2.0]	+1.7 [+1.2, +2.2]
NTREX	afr	Afrikaans	62.8	63.9	+1.1 [+0.8, +1.4]	+1.1 [+0.8, +1.4]
NTREX	amh	Amharic	13.5	18.2	+4.7 [+4.3, +5.1]	+5.1 [+4.6, +5.5]
NTREX	bem	Bemba	24.7	35.0	+10.3 [+9.8, +10.9]	+10.9 [+10.3, +11.5]
NTREX	ewe	Ewe	27.1	31.7	+4.6 [+4.2, +5.0]	+5.0 [+4.5, +5.4]
NTREX	hau	Hausa	46.7	48.6	+1.9 [+1.5, +2.3]	+2.0 [+1.6, +2.4]
NTREX	ibo	Igbo	42.8	45.1	+2.3 [+2.0, +2.7]	+2.3 [+1.9, +2.7]
NTREX	kin	Kinyarwanda	30.9	37.2	+6.3 [+5.8, +6.8]	+6.5 [+6.0, +7.1]
NTREX	mlg	Malagasy	39.1	40.9	+1.8 [+1.5, +2.2]	+2.0 [+1.7, +2.4]
NTREX	nde	nde	30.9	33.0	+2.1 [+1.8, +2.4]	+2.3 [+1.9, +2.7]
NTREX	nso	Sepedi	27.0	32.8	+5.8 [+5.2, +6.3]	+6.0 [+5.4, +6.5]
NTREX	nya	Chichewa	41.2	44.5	+3.3 [+2.9, +3.7]	+3.5 [+3.1, +4.0]
NTREX	orm	Oromo	16.4	27.8	+11.4 [+10.9, +11.9]	+13.3 [+12.7, +13.9]
NTREX	shi	shi	0.7	1.3	+0.6 [+0.6, +0.7]	+0.3 [+0.3, +0.4]
NTREX	sna	Shona	37.6	39.3	+1.8 [+1.4, +2.2]	+2.0 [+1.6, +2.4]
NTREX	som	Somali	39.3	44.4	+5.2 [+4.5, +5.9]	+5.4 [+4.8, +6.2]
NTREX	ssw	Swati	28.9	39.0	+10.1 [+9.5, +10.7]	+11.2 [+10.6, +11.9]
NTREX	swh	Swahili	57.2	59.0	+1.7 [+1.3, +2.2]	+1.7 [+1.3, +2.1]
NTREX	tir	Tigrinya	12.4	17.0	+4.6 [+4.3, +5.0]	+5.1 [+4.7, +5.5]
NTREX	tsn	Setswana	42.8	48.7	+5.9 [+5.3, +6.5]	+5.9 [+5.2, +6.5]
NTREX	ven	ven	32.2	39.1	+6.9 [+6.3, +7.5]	+7.1 [+6.6, +7.7]
NTREX	wol	Wolof	27.7	30.5	+2.9 [+2.5, +3.2]	+3.0 [+2.7, +3.4]
NTREX	xho	Xhosa	41.2	42.5	+1.2 [+0.9, +1.6]	+1.5 [+1.1, +1.8]
NTREX	yor	Yoruba	25.9	32.1	+6.2 [+5.6, +6.9]	+6.4 [+5.8, +7.0]
NTREX	zul	Zulu	42.3	43.7	+1.5 [+1.1, +1.8]	+1.6 [+1.3, +2.0]
MAFAND-MT	amh	Amharic	14.4	18.4	+4.0 [+3.3, +4.6]	+4.4 [+3.6, +5.1]
MAFAND-MT	hau	Hausa	33.0	32.8	−0.2 [−0.4, +0.0]	−0.0 [−0.3, +0.2]
MAFAND-MT	ibo	Igbo	45.0	44.9	−0.1 [−0.5, +0.3]	−0.1 [−0.5, +0.3]
MAFAND-MT	kin	Kinyarwanda	40.9	46.9	+6.0 [+5.2, +6.7]	+6.1 [+5.3, +6.8]
MAFAND-MT	lug	Luganda	31.4	39.9	+8.5 [+7.9, +9.1]	+8.7 [+8.1, +9.4]
MAFAND-MT	luo	Luo	27.8	32.8	+5.0 [+4.5, +5.5]	+5.5 [+4.9, +6.0]
MAFAND-MT	nya	Chichewa	40.3	41.1	+0.8 [+0.5, +1.2]	+1.0 [+0.7, +1.4]
MAFAND-MT	pcm	pcm	53.9	53.4	−0.5 [−0.8, −0.2]	−0.7 [−1.0, −0.4]
MAFAND-MT	sna	Shona	32.5	33.0	+0.5 [+0.2, +0.9]	+0.7 [+0.3, +1.1]
MAFAND-MT	swh	Swahili	51.6	51.9	+0.3 [−0.1, +0.6]	+0.3 [−0.1, +0.7]
MAFAND-MT	tsn	Setswana	49.1	53.3	+4.2 [+3.4, +4.9]	+4.2 [+3.4, +4.9]
MAFAND-MT	twi	Twi	31.9	34.1	+2.2 [+1.8, +2.7]	+2.5 [+2.0, +3.0]
MAFAND-MT	xho	Xhosa	50.7	51.9	+1.2 [+0.6, +1.8]	+1.4 [+0.8, +2.0]
MAFAND-MT	yor	Yoruba	26.2	33.4	+7.2 [+6.6, +7.8]	+7.3 [+6.7, +7.9]
MAFAND-MT	zul	Zulu	44.8	45.7	+0.9 [+0.5, +1.3]	+1.1 [+0.7, +1.5]

Engine and precision checks (E-09, E-09b)

Earlier 256-token study (superseded)

Table 29: E-02: greedy, beam-4 and MBR-16 at 768 new tokens, full table (20 languages, 4,000 sentences per system, 95% paired bootstrap; plain chrF, spBLEU, finished rate, loop rate and length ratio included).

system	chrF++	spBLEU	AfriCOMET-STL	finished	loop (independent)	output / reference length
SFT greedy	32.31 [31.43, 33.38]	11.47	0.5513 [0.5403, 0.5621]	0.915	0.147	2.05
SFT beam-4 (stop at 4 ended)	36.02 [35.13, 36.95]	16.49	0.5639 [0.5521, 0.5750]	1.000	0.108	0.96
SFT beam-4 (HF early_stopping=False)	33.26 [32.22, 34.46]	13.07	0.5722 [0.5603, 0.5845]	0.897	0.139	2.15
SFT MBR-16 (metric-matched)	38.27 [37.60, 39.01]	15.25	0.5780 [0.5687, 0.5870]	0.9998	0.016	1.16
RL greedy	39.87 [39.17, 40.62]	17.39	0.6082 [0.5991, 0.6172]	1.000	0.007	1.05
RL beam-4 (stop at 4 ended)	39.10 [38.20, 39.96]	17.83	0.6045 [0.5944, 0.6148]	1.000	0.007	0.98
RL beam-4 (HF early_stopping=False)	40.70 [40.00, 41.42]	18.47	0.6227 [0.6139, 0.6317]	0.9998	0.007	1.03
RL MBR-16 (metric-matched)	40.79 [40.09, 41.54]	17.74	0.6168 [0.6079, 0.6258]	1.000	0.003	1.08

Table 37: E-01 guard grid on the FLORES+ dev held-out half (720 sentences, 18 languages): dev chrF++, finished rate and loop rate of the SFT baseline and the RL model for seven decoding settings (freq = vLLM frequency penalty, rep = repetition penalty; settings fixed in advance; freq 0.10 was chosen).

config	SFT dev chrF++	SFT finished	SFT loop	RL dev chrF++	RL finished	RL loop
greedy	34.50	0.946	0.058	40.43	0.999	0.003
freq002	36.17	0.974	0.040	40.62	1.000	0.000
freq005	36.95	0.992	0.039	40.64	1.000	0.000
freq010	37.36	0.996	0.040	40.65	1.000	0.000
rep102	35.79	0.999	0.001	39.60	1.000	0.000
rep105	29.92	0.997	0.003	33.85	1.000	0.001
freq005_rep102	35.71	1.000	0.000	39.59	1.000	0.000

Table 38: E-01 full evaluation: the SFT baseline and the RL model with and without the chosen decoding guard (frequency penalty 0.10), language averages per benchmark, held-out AfriCOMET over 76 pairs and FLORES+ finished / loop / length ratio.

system	FLORES+ chrF++ (54)	NTREX (24)	MAFAND (15)	all 93 cells chrF++	all cells chrF	AfriCOMET-STL (76 cells)	FLORES+ finished / loop / length ratio
SFT, greedy	22.85	29.25	35.77	26.59	29.06	0.565	0.829 / 0.209 / 3.03
SFT + guard (freq 0.10)	27.22	32.96	38.22	**30.48**	33.30	0.583	0.9945 / 0.136 / 1.41
RL (mixed best), greedy	31.66	37.17	40.78	34.55	37.65	0.628	0.9954 / 0.048 / 1.14

Table 39: E-01 paired differences (language averages, 95% bootstrap intervals).

comparison	FLORES+	NTREX	MAFAND	all 93 cells chrF++	all cells AfriCOMET
guard effect on SFT (SFT + guard minus SFT)	+4.37 [+4.23, +4.50]	+3.71 [+3.54, +3.86]	+2.45 [+2.24, +2.66]	+3.89	+0.0224 / +0.0157 / +0.0106 by benchmark (+0.018 on all cells)
guard effect on RL	+0.41 [+0.38, +0.44]	+0.14 [+0.10, +0.18]	+0.11 [+0.07, +0.16]	+0.29	+0.0038 / +0.0026 / +0.0019
RL + guard minus SFT + guard	see primary line			**+4.37 [+4.28, +4.46]**	**+0.0479 [+0.0469, +0.0490]**
RL minus SFT, both greedy (EVID)	+8.81 [+8.59, +9.03]	+7.91 [+7.69, +8.12]	+5.01 [+4.76, +5.23]	+7.96 [+7.82, +8.12]	+0.063

Table 41: E-09 engine and precision check on 240 FLORES+ sentences (10 languages $\times$ 24), greedy, 256 new tokens: chrF++ of vLLM and Hugging Face (HF) in bf16 and fp32, and the share of identical outputs between the two engines. The engines differ in bf16 and agree in fp32.

	vLLM bf16	HF bf16	vLLM fp32	HF fp32	identical bf16	identical fp32
SFT baseline	36.21	36.80	38.19	38.18	9.6%	98.8%
RL model	40.47	40.28	40.59	40.61	16.7%	97.1%

Table 42: E-09 paired differences in chrF++ on the same 240 sentences, with 95% paired-bootstrap intervals over sentences. The $+1.81$ for the SFT baseline is a small-sample result; the full-set check (Table 43) does not reproduce it.

model	comparison	chrF++ difference
SFT baseline	vLLM fp32 − bf16	+1.81 [+0.59, +3.12]
SFT baseline	HF fp32 − bf16	+1.25 [+0.32, +2.19]
SFT baseline	HF bf16 − vLLM bf16	+0.54 [−0.67, +1.79]
SFT baseline	HF fp32 − vLLM fp32	−0.01 [−0.08, +0.06]
RL model	vLLM fp32 − bf16	+0.09 [−0.41, +0.59]
RL model	HF fp32 − bf16	+0.34 [−0.27, +0.96]
RL model	HF bf16 − vLLM bf16	−0.23 [−0.73, +0.23]
RL model	HF fp32 − vLLM fp32	+0.02 [−0.02, +0.08]

Table 43: E-09b full-set precision check: the SFT baseline generated in fp32 against bf16 on all rows (214 flagged rows removed; greedy, 768 tokens). Columns: language averages in bf16 and fp32, then fp32 minus bf16 for chrF++, plain chrF and held-out AfriCOMET with 95% intervals (AfriCOMET only where the run was scored). Right: RL gain in chrF++ over the bf16 and over the fp32 SFT baseline. The RL model was not re-run in fp32.

benchmark	bf16	fp32	chrF++ diff	chrF diff	AfriCOMET diff	RL gain over bf16 SFT	RL gain over fp32 SFT
FLORES+	22.85	22.69	−0.16 [−0.29, −0.02]	−0.21 [−0.35, −0.05]	+0.0029 [+0.0018, +0.0042]	+8.81 [+8.59, +9.03]	+8.97 [+8.71, +9.20]
NTREX	29.25	29.33	+0.08 [−0.08, +0.25]	+0.03 [−0.13, +0.22]	+0.0129 [+0.0108, +0.0152]	+7.91 [+7.69, +8.12]	+7.84 [+7.59, +8.05]
MAFAND-MT	35.77	35.99	+0.22 [−0.03, +0.50]	+0.17 [−0.09, +0.48]	–	+5.01 [+4.76, +5.23]	+4.78 [+4.53, +5.01]

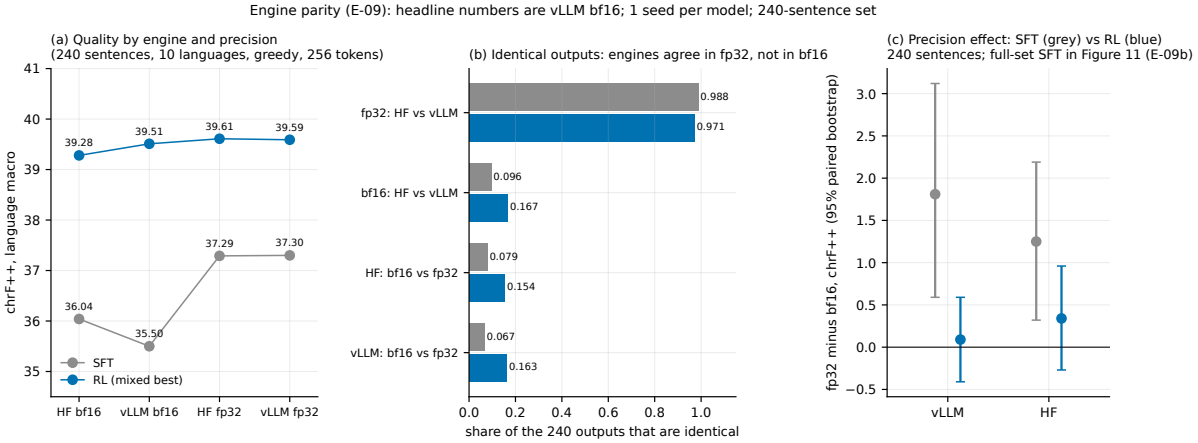


Figure 9: Engine parity (E-09), 240 sentences (10 languages × 24 FLORES+ devtest sentences), greedy, 256 new tokens, SFT and RL (mixed best), one seed. (a) Language-macro chrF++ by engine and precision. (b) Share of identical outputs: vLLM and HF agree in fp32 (0.988, 0.971) and not in bf16 (0.096, 0.167). (c) fp32 minus bf16 chrF++ with 95% paired bootstrap over sentences: negligible for the RL model, +1.81 [+0.59, +3.12] (vLLM) for the SFT on this small set, which the full-set run of Figure 11 does not reproduce. Headline numbers of the paper are vLLM bf16. Sources: `reports/data/e09_engine_parity_2026-11-04/{parity_bootstrap,parity_summary}.json`.

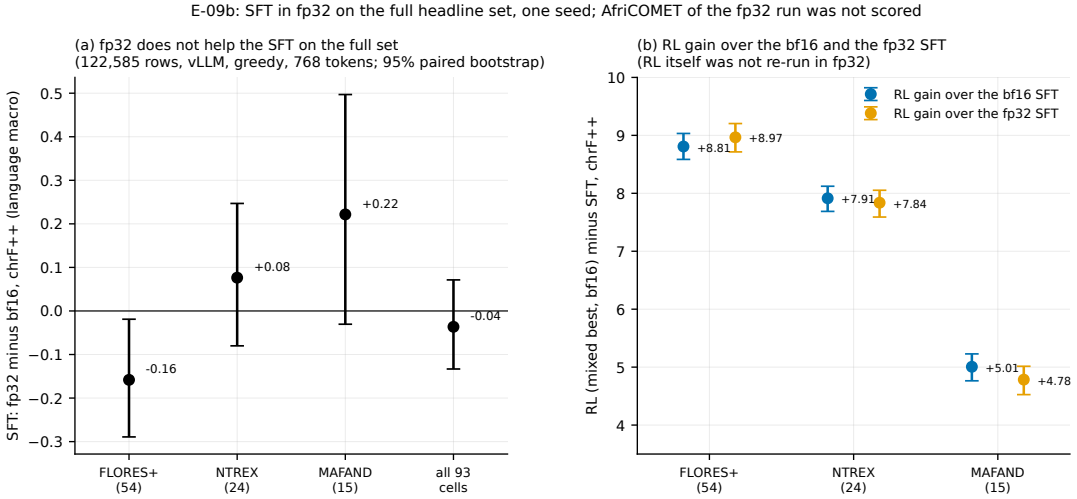


Figure 10: Full-set precision check (E-09b): the SFT generated in fp32 instead of bf16 (vLLM, greedy, 768 new tokens, 122,585 rows, 214 flagged rows removed; one seed). (a) fp32 minus bf16 chrF++ (language macro, 95% paired bootstrap over sentences) on FLORES+ (54 languages), NTREX (24), MAFAND (15) and over all 93 cells (−0.04 [−0.13, +0.07]). (b) RL gain (mixed best, bf16) over the bf16 SFT and over the fp32 SFT. The RL model was not re-run in fp32 and the AfriCOMET of the fp32 run was not scored. Source: `reports/data/e09b_fp32_2026-11-04/evidence_e09b.json`.

Table 44: **Superseded.** Earlier 256-token study (SFT baseline only; 20 languages × 200 FLORES+ sentences). Its beam rows came from a beam loop that ran past the end token and are not comparable with E-02; they are kept only as the record of what that study measured.

decoding	chrF++	delta vs greedy [95% CI]	spBLEU	delta spBLEU
greedy	35.00	-	13.64	-
beam 4	37.53	+2.53 [+2.18, +2.87]	16.38	+2.73 [+2.39, +3.08]
MBR-16 (+ greedy)	38.29	+3.30 [+3.00, +3.58]	15.30	+1.65 [+1.38, +1.90]

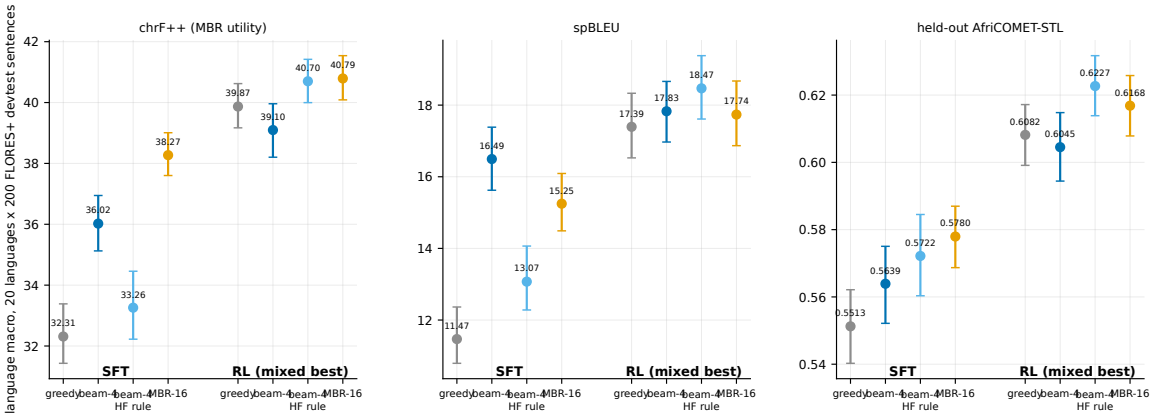


Figure 11: Greedy, beam-4 and MBR-16 at 768 new tokens (E-02) for the SFT and the RL model (mixed best): chrF++, spBLEU and held-out AfriCOMET-STL, language macro over 20 languages × 200 FLORES+ devtest sentences (4,000 per system), 95% paired bootstrap over source sentences; beam-4 stops when 4 hypotheses have ended, the HF rule is early_stopping=False, MBR is metric-matched (utility chrF++). One SFT and one RL checkpoint, vLLM bf16. Not comparable with the earlier 256-token study. Source: `reports/data/e02_beam_mbr_768_2026-11-05/cmp_all.json`.

E Seeds and ablations (E-04, E-11)

Table 46: E-04 per language and benchmark (20 languages, 100 RL steps of the first-stage recipe, two extra seeds): chrF++ gain over the SFT baseline for each seed and the difference between the seeds, with 95% paired-bootstrap intervals; last column: chrF++ of seed 2.

benchmark	ISO	seed 2 − SFT	seed 3 − SFT	seed 2 − seed 3	seed 2
FLORES+	afr	+0.91 [+0.46, +1.46]	+0.96 [+0.51, +1.53]	−0.05 [−0.23, +0.14]	64.0
FLORES+	amh	+7.40 [+6.37, +8.49]	+7.35 [+6.27, +8.34]	+0.05 [−0.81, +0.92]	24.0
FLORES+	ewe	+4.62 [+3.76, +5.54]	+4.50 [+3.66, +5.46]	+0.13 [−0.30, +0.55]	30.6
FLORES+	hau	+5.27 [+3.91, +6.80]	+5.26 [+3.81, +6.89]	+0.01 [−0.79, +0.94]	45.1
FLORES+	ibo	+2.24 [+1.52, +2.96]	+2.25 [+1.54, +2.99]	−0.01 [−0.29, +0.29]	40.1
FLORES+	kin	+10.82 [+9.57, +12.12]	+11.18 [+9.94, +12.37]	−0.36 [−1.11, +0.37]	36.6
FLORES+	lin	+13.64 [+11.94, +15.23]	+13.75 [+12.14, +15.24]	−0.11 [−0.98, +0.80]	42.2
FLORES+	lug	+12.18 [+11.11, +13.33]	+12.60 [+11.52, +13.68]	−0.41 [−1.41, +0.62]	28.3
FLORES+	nso	+15.91 [+14.24, +17.68]	+15.42 [+13.69, +17.16]	+0.49 [−0.57, +1.53]	47.5
FLORES+	orm	+13.15 [+12.17, +14.15]	+12.86 [+11.85, +13.84]	+0.29 [−0.66, +1.34]	25.2
FLORES+	sna	+3.65 [+2.50, +4.84]	+3.59 [+2.39, +4.88]	+0.06 [−0.63, +0.74]	37.6
FLORES+	som	+4.19 [+3.17, +5.21]	+3.87 [+2.89, +4.86]	+0.33 [+0.06, +0.67]	39.7
FLORES+	swh	+2.54 [+1.58, +3.65]	+2.46 [+1.45, +3.60]	+0.08 [−0.16, +0.32]	58.8
FLORES+	tir	+6.88 [+6.20, +7.56]	+6.94 [+6.23, +7.61]	−0.06 [−0.55, +0.37]	17.4
FLORES+	tsn	+7.08 [+5.87, +8.32]	+6.94 [+5.72, +8.30]	+0.14 [−0.45, +0.72]	44.8
FLORES+	twi	+6.17 [+4.97, +7.23]	+6.27 [+5.10, +7.44]	−0.10 [−0.79, +0.55]	33.7
FLORES+	wol	+3.37 [+2.43, +4.25]	+3.10 [+2.13, +4.08]	+0.26 [−0.11, +0.68]	30.2
FLORES+	xho	+2.29 [+1.43, +3.27]	+2.41 [+1.50, +3.37]	−0.11 [−0.62, +0.44]	45.9
FLORES+	yor	+7.91 [+6.95, +8.87]	+7.83 [+6.88, +8.77]	+0.08 [−0.28, +0.42]	29.2
FLORES+	zul	+2.48 [+1.72, +3.29]	+1.95 [+1.10, +2.79]	+0.54 [+0.03, +1.14]	48.4
NTREX	afr	+1.49 [+0.99, +2.09]	+1.38 [+0.91, +2.01]	+0.11 [−0.06, +0.26]	63.9
NTREX	amh	+5.64 [+5.17, +6.10]	+5.53 [+5.03, +6.03]	+0.11 [−0.27, +0.49]	16.2
NTREX	ewe	+6.22 [+5.46, +6.91]	+6.04 [+5.33, +6.68]	+0.18 [−0.23, +0.56]	30.5
NTREX	hau	+4.99 [+3.80, +6.18]	+4.54 [+3.38, +5.62]	+0.45 [−0.25, +1.23]	47.1
NTREX	ibo	+4.25 [+3.44, +5.12]	+4.37 [+3.56, +5.22]	−0.12 [−0.58, +0.29]	44.1
NTREX	kin	+8.62 [+7.67, +9.53]	+8.47 [+7.61, +9.25]	+0.14 [−0.50, +0.77]	35.0
NTREX	nso	+10.69 [+9.85, +11.44]	+10.31 [+9.49, +11.09]	+0.38 [−0.10, +0.82]	30.8
NTREX	orm	+11.84 [+11.11, +12.56]	+11.30 [+10.62, +12.03]	+0.53 [−0.27, +1.28]	21.5
NTREX	sna	+4.69 [+3.86, +5.63]	+4.68 [+3.80, +5.65]	+0.01 [−0.38, +0.39]	38.9
NTREX	som	+7.09 [+6.14, +8.03]	+6.88 [+5.91, +7.88]	+0.21 [−0.23, +0.68]	42.9
NTREX	swh	+3.89 [+3.00, +4.83]	+3.68 [+2.76, +4.69]	+0.21 [−0.21, +0.69]	59.1
NTREX	tir	+6.38 [+5.95, +6.79]	+6.19 [+5.76, +6.60]	+0.19 [−0.16, +0.56]	15.5
NTREX	tsn	+10.25 [+9.12, +11.35]	+10.45 [+9.26, +11.57]	−0.21 [−0.82, +0.40]	47.2
NTREX	wol	+4.47 [+3.77, +5.23]	+4.88 [+4.16, +5.63]	−0.41 [−0.77, −0.06]	29.8
NTREX	xho	+2.57 [+1.97, +3.18]	+2.48 [+1.93, +3.12]	+0.08 [−0.19, +0.36]	42.4
NTREX	yor	+7.30 [+6.55, +8.02]	+7.28 [+6.52, +7.99]	+0.02 [−0.30, +0.33]	31.0
NTREX	zul	+2.16 [+1.56, +2.80]	+2.09 [+1.43, +2.72]	+0.07 [−0.31, +0.47]	43.1
MAFAND-MT	amh	+5.21 [+4.39, +6.02]	+5.27 [+4.49, +6.01]	−0.06 [−0.59, +0.51]	16.9
MAFAND-MT	hau	+0.98 [+0.46, +1.54]	+1.03 [+0.46, +1.64]	−0.04 [−0.32, +0.24]	33.0
MAFAND-MT	ibo	+2.43 [+1.53, +3.35]	+2.17 [+1.28, +3.09]	+0.25 [−0.07, +0.67]	46.0
MAFAND-MT	kin	+7.86 [+6.61, +9.21]	+8.31 [+7.07, +9.56]	−0.45 [−1.18, +0.30]	44.4
MAFAND-MT	lug	+10.40 [+9.25, +11.51]	+10.63 [+9.43, +11.81]	−0.23 [−0.93, +0.51]	37.4
MAFAND-MT	sna	+2.79 [+1.90, +3.70]	+2.67 [+1.81, +3.56]	+0.12 [−0.21, +0.49]	33.1
MAFAND-MT	swh	+1.50 [+0.80, +2.26]	+1.66 [+1.04, +2.42]	−0.16 [−0.51, +0.16]	52.1
MAFAND-MT	tsn	+8.71 [+7.21, +10.22]	+8.71 [+7.18, +10.18]	−0.00 [−0.57, +0.56]	52.6
MAFAND-MT	twi	+4.50 [+3.71, +5.26]	+4.85 [+3.98, +5.63]	−0.36 [−0.81, +0.08]	33.1
MAFAND-MT	xho	+3.03 [+2.10, +3.97]	+3.19 [+2.34, +4.10]	−0.16 [−0.57, +0.24]	52.8
MAFAND-MT	yor	+8.03 [+7.26, +8.84]	+7.90 [+7.03, +8.74]	+0.12 [−0.35, +0.64]	31.5
MAFAND-MT	zul	+1.34 [+0.74, +1.99]	+0.86 [+0.22, +1.54]	+0.47 [−0.10, +1.04]	45.8

Table 47: E-11 per language and benchmark: chrF++ difference of each ablation arm to the full-feature run (seed 2; paired, same prefix), with 95% paired-bootstrap intervals (negative = the removed feature helped); each column is the arm minus seed 2.

benchmark	ISO	no features	no reference domain	no beam guide
FLORES+	afr	−0.05 [−0.26, +0.15]	+0.19 [+0.00, +0.38]	+0.04 [−0.16, +0.24]
FLORES+	amh	−0.38 [−1.23, +0.41]	−0.19 [−0.89, +0.47]	−0.09 [−0.84, +0.71]
FLORES+	ewe	+0.11 [−0.41, +0.61]	+0.48 [+0.04, +0.98]	−0.05 [−0.54, +0.47]
FLORES+	hau	−0.39 [−1.24, +0.49]	+0.10 [−0.85, +0.95]	−0.11 [−0.99, +0.80]
FLORES+	ibo	−0.32 [−0.78, +0.13]	−0.10 [−0.54, +0.30]	−0.08 [−0.39, +0.28]
FLORES+	kin	−0.90 [−1.85, +0.02]	−0.96 [−1.92, −0.03]	−0.61 [−1.36, +0.16]
FLORES+	lin	−0.37 [−1.15, +0.37]	+0.09 [−0.92, +1.02]	−0.03 [−0.93, +0.84]
FLORES+	lug	−0.40 [−1.37, +0.55]	−0.68 [−1.80, +0.40]	−0.35 [−1.39, +0.64]
FLORES+	nso	−0.46 [−1.47, +0.53]	−0.40 [−1.49, +0.59]	−0.41 [−1.25, +0.41]
FLORES+	orm	−1.17 [−2.07, −0.25]	−0.69 [−1.73, +0.27]	+0.08 [−0.85, +1.01]
FLORES+	sna	+0.09 [−0.43, +0.66]	−0.10 [−0.73, +0.49]	−0.01 [−0.61, +0.56]
FLORES+	som	−0.58 [−1.12, −0.12]	−0.42 [−0.85, −0.07]	−0.34 [−0.72, −0.01]
FLORES+	swh	−0.20 [−0.45, +0.05]	−0.38 [−0.92, +0.05]	−0.13 [−0.36, +0.12]
FLORES+	tir	−0.08 [−0.62, +0.46]	+0.22 [−0.22, +0.70]	−0.18 [−0.66, +0.34]
FLORES+	tsn	+0.07 [−0.45, +0.60]	+0.07 [−0.47, +0.66]	−0.07 [−0.66, +0.53]
FLORES+	twi	−0.47 [−1.13, +0.20]	+0.51 [−0.08, +1.16]	−0.11 [−0.65, +0.46]

benchmark	ISO	no features	no reference domain	no beam guide
FLORES+	wol	−0.27 [−0.67, +0.12]	+0.05 [−0.29, +0.42]	−0.21 [−0.53, +0.04]
FLORES+	xho	−0.10 [−0.60, +0.39]	+0.25 [−0.13, +0.70]	−0.32 [−0.98, +0.23]
FLORES+	yor	−1.09 [−1.56, −0.64]	−0.77 [−1.31, −0.29]	−0.21 [−0.59, +0.12]
FLORES+	zul	−0.60 [−0.96, −0.24]	−0.33 [−0.75, +0.04]	−0.51 [−1.07, −0.03]
NTREX	afr	−0.04 [−0.20, +0.13]	−0.22 [−0.42, −0.02]	−0.14 [−0.35, +0.05]
NTREX	amh	−0.37 [−0.75, −0.04]	−0.10 [−0.49, +0.27]	+0.02 [−0.30, +0.38]
NTREX	ewe	−0.15 [−0.49, +0.21]	+0.04 [−0.32, +0.42]	−0.06 [−0.46, +0.35]
NTREX	hau	−0.36 [−1.27, +0.41]	−0.49 [−1.22, +0.19]	+0.07 [−0.63, +0.82]
NTREX	ibo	−0.27 [−0.73, +0.15]	+0.14 [−0.23, +0.49]	−0.23 [−0.63, +0.10]
NTREX	kin	−0.61 [−1.26, +0.07]	−0.31 [−0.89, +0.26]	−0.29 [−0.85, +0.24]
NTREX	nso	−0.21 [−0.67, +0.22]	−0.43 [−0.96, +0.10]	−0.29 [−0.77, +0.17]
NTREX	orm	−0.97 [−1.74, −0.20]	−0.15 [−0.80, +0.59]	−0.30 [−1.01, +0.43]
NTREX	sna	+0.20 [−0.19, +0.58]	+0.16 [−0.14, +0.45]	−0.07 [−0.46, +0.35]
NTREX	som	−0.21 [−0.66, +0.19]	−0.20 [−0.69, +0.27]	−0.30 [−0.77, +0.13]
NTREX	swh	−0.25 [−0.74, +0.17]	−0.17 [−0.65, +0.25]	+0.09 [−0.29, +0.49]
NTREX	tir	−0.25 [−0.62, +0.10]	+0.16 [−0.11, +0.44]	+0.19 [−0.14, +0.51]
NTREX	tsn	+0.11 [−0.44, +0.65]	+0.03 [−0.55, +0.58]	−0.11 [−0.60, +0.36]
NTREX	wol	−0.11 [−0.53, +0.32]	+0.03 [−0.32, +0.41]	+0.12 [−0.30, +0.56]
NTREX	xho	−0.12 [−0.32, +0.15]	−0.18 [−0.53, +0.14]	−0.06 [−0.39, +0.27]
NTREX	yor	−0.81 [−1.21, −0.41]	−0.48 [−0.82, −0.13]	+0.05 [−0.27, +0.38]
NTREX	zul	+0.05 [−0.35, +0.44]	+0.02 [−0.35, +0.37]	+0.18 [−0.14, +0.51]
MAFAND-MT	amh	−0.33 [−0.96, +0.29]	−0.02 [−0.56, +0.53]	−0.01 [−0.60, +0.54]
MAFAND-MT	hau	+0.06 [−0.17, +0.30]	+0.02 [−0.17, +0.22]	+0.06 [−0.19, +0.25]
MAFAND-MT	ibo	−0.13 [−0.55, +0.17]	−0.27 [−0.66, +0.07]	+0.10 [−0.12, +0.34]
MAFAND-MT	kin	−0.21 [−1.07, +0.66]	+0.61 [−0.10, +1.33]	+0.75 [−0.05, +1.64]
MAFAND-MT	lug	−0.82 [−1.67, −0.01]	−0.86 [−1.67, −0.02]	+0.00 [−0.86, +0.78]
MAFAND-MT	sna	−0.18 [−0.52, +0.14]	−0.08 [−0.46, +0.32]	−0.28 [−0.66, +0.07]
MAFAND-MT	swh	−0.11 [−0.60, +0.32]	−0.29 [−0.75, +0.17]	+0.26 [+0.01, +0.59]
MAFAND-MT	tsn	−0.65 [−1.35, +0.05]	−0.78 [−1.50, −0.06]	+0.15 [−0.55, +0.84]
MAFAND-MT	twi	+0.09 [−0.42, +0.62]	+0.35 [−0.09, +0.81]	+0.13 [−0.29, +0.57]
MAFAND-MT	xho	−0.67 [−1.28, −0.04]	−0.27 [−0.79, +0.23]	−0.55 [−1.09, +0.06]
MAFAND-MT	yor	−0.53 [−1.02, −0.03]	−0.79 [−1.24, −0.37]	−0.13 [−0.68, +0.38]
MAFAND-MT	zul	−0.24 [−0.81, +0.31]	−0.18 [−0.66, +0.29]	−0.08 [−0.51, +0.36]

Table 45: Ablation arms at 100 steps, chrF++ over 49 cells (20 languages); arm minus the full-feature seed 2 (paired) and minus seed 3 (E-11).

arm	chrF++ (49 cells)	arm minus seed 2	arm minus seed 3	AfriCOMET vs seed 2
(b) full, seed 2	38.51 [38.30, 38.69]	-	-	-
(b) full, seed 3	38.46 [38.26, 38.64]	-	-	-
(a) no features	38.19 [37.99, 38.38]	−0.32 [−0.41, −0.23]	−0.26 [−0.35, −0.18]	−0.0028 [−0.0033, −0.0022]
(c) no reference domain	38.35 [38.16, 38.54]	−0.16 [−0.25, −0.08]	−0.10 [−0.19, −0.02]	−0.0017 [−0.0023, −0.0011]
(d) no beam guide	38.42 [38.23, 38.61]	−0.09 [−0.17, −0.01]	−0.04 [−0.11, +0.04]	−0.0005 [−0.0010, +0.0001]

F Matched-compute control (E-03): tables and per-language results

Table 56: E-03 per language and benchmark (clean rows): SFT-only control (6,000 steps) against the RL model (mixed best), chrF++ and plain chrF difference with 95% paired-bootstrap intervals; TP = one of the 19 TranslatePsy languages that the control received extra data for. Positive = RL better.

benchmark	ISO	Language	TP	control	RL model	RL − control, chrF++	RL − control, chrF
FLORES+	aeb	aeb		2.0	4.1	+2.1 [+1.7, +2.5]	+2.0 [+1.6, +2.5]
FLORES+	afr	Afrikaans	yes	63.3	64.1	+0.8 [+0.4, +1.2]	+0.7 [+0.4, +1.1]
FLORES+	amh	Amharic	yes	27.0	26.6	−0.4 [−1.2, +0.5]	−0.4 [−1.3, +0.5]
FLORES+	ary	ary		5.3	6.9	+1.6 [+1.1, +2.1]	+1.2 [+0.6, +1.8]
FLORES+	arz	arz		5.3	9.2	+3.9 [+3.3, +4.5]	+4.2 [+3.5, +4.8]
FLORES+	bam	Bambara		20.1	27.2	+7.1 [+6.2, +7.9]	+7.6 [+6.8, +8.5]
FLORES+	bem	Bemba		20.0	33.9	+13.9 [+12.9, +14.9]	+15.0 [+13.9, +16.0]
FLORES+	cjk	cjk		11.2	23.0	+11.8 [+11.2, +12.4]	+13.6 [+12.9, +14.3]
FLORES+	dik	dik		3.8	14.1	+10.3 [+9.8, +10.8]	+11.3 [+10.8, +11.9]
FLORES+	dyu	Dyula		9.8	19.0	+9.1 [+8.5, +9.8]	+10.2 [+9.6, +10.9]
FLORES+	ewe	Ewe		27.4	31.2	+3.7 [+3.0, +4.5]	+4.0 [+3.2, +4.8]
FLORES+	fon	Fon		19.8	22.2	+2.5 [+1.9, +3.0]	+2.7 [+2.1, +3.3]
FLORES+	hau	Hausa	yes	47.2	46.5	−0.8 [−1.5, +0.1]	−0.6 [−1.3, +0.4]
FLORES+	ibo	Igbo	yes	39.2	40.1	+0.9 [+0.3, +1.6]	+0.9 [+0.2, +1.6]
FLORES+	kab	Kabyle		15.9	26.7	+10.8 [+9.9, +11.6]	+11.2 [+10.3, +12.0]
FLORES+	kam	Kamba		16.7	25.9	+9.3 [+8.5, +9.9]	+9.7 [+8.9, +10.5]
FLORES+	kbp	Kabiye		15.5	24.9	+9.3 [+8.6, +10.0]	+9.9 [+9.1, +10.6]
FLORES+	kea	Kabuverdianu		19.0	39.2	+20.2 [+18.9, +21.5]	+20.5 [+19.2, +21.8]
FLORES+	kik	Kikuyu		26.6	33.3	+6.7 [+5.8, +7.5]	+7.0 [+6.0, +7.9]
FLORES+	kin	Kinyarwanda	yes	39.7	38.7	−1.1 [−2.0, −0.2]	−0.9 [−1.9, +0.1]
FLORES+	kmb	Kimbundu		16.3	26.6	+10.3 [+9.5, +11.0]	+11.2 [+10.3, +12.0]
FLORES+	knc	Kanuri		3.0	14.2	+11.2 [+10.7, +11.7]	+12.9 [+12.3, +13.4]
FLORES+	lin	Lingala	yes	45.3	44.3	−1.0 [−1.5, −0.4]	−1.0 [−1.6, −0.4]
FLORES+	lua	Luba-Kasai		16.0	33.4	+17.3 [+16.4, +18.2]	+18.7 [+17.6, +19.8]
FLORES+	lug	Luganda	yes	34.7	31.7	−3.0 [−3.6, −2.2]	−3.3 [−4.0, −2.4]
FLORES+	luo	Luo		21.6	34.5	+12.9 [+11.8, +14.0]	+13.5 [+12.3, +14.7]
FLORES+	mfe	mfe		29.9	37.9	+8.1 [+6.7, +9.6]	+8.1 [+6.7, +9.6]
FLORES+	mlg	Malagasy	yes	44.2	44.7	+0.5 [−0.1, +1.3]	+0.6 [−0.1, +1.4]
FLORES+	mos	Mossi		14.4	22.8	+8.4 [+7.8, +9.0]	+9.0 [+8.4, +9.6]
FLORES+	nso	Sepedi		30.3	49.0	+18.7 [+17.0, +20.2]	+19.2 [+17.4, +20.8]
FLORES+	nus	Nuer		11.4	24.8	+13.4 [+12.9, +13.9]	+14.3 [+13.8, +14.9]
FLORES+	nya	Chichewa	yes	40.2	42.2	+2.0 [+1.1, +2.9]	+2.0 [+1.1, +2.9]
FLORES+	orm	Oromo	yes	31.2	30.2	−1.0 [−1.8, −0.1]	−1.1 [−2.0, −0.1]
FLORES+	run	Kirundi		23.2	33.9	+10.7 [+9.7, +11.7]	+11.8 [+10.7, +12.9]
FLORES+	sag	Sango		19.8	34.3	+14.5 [+13.6, +15.5]	+15.4 [+14.4, +16.4]
FLORES+	seh	seh		20.5	29.6	+9.2 [+8.1, +10.1]	+10.4 [+9.2, +11.4]
FLORES+	sna	Shona	yes	39.2	37.6	−1.7 [−2.4, −0.9]	−1.7 [−2.5, −0.9]
FLORES+	som	Somali	yes	38.5	39.7	+1.3 [+0.5, +2.2]	+1.3 [+0.5, +2.3]
FLORES+	sot	Sesotho	yes	43.3	42.7	−0.6 [−1.1, −0.1]	−0.7 [−1.2, −0.2]
FLORES+	ssw	Swati		22.7	37.9	+15.1 [+13.9, +16.3]	+17.1 [+15.7, +18.4]
FLORES+	swh	Swahili	yes	56.3	58.4	+2.1 [+1.2, +3.1]	+2.0 [+1.1, +3.1]
FLORES+	taq	taq		3.1	11.1	+8.0 [+7.5, +8.4]	+8.9 [+8.4, +9.3]
FLORES+	tir	Tigrinya		9.7	18.7	+9.0 [+8.3, +9.6]	+9.8 [+9.1, +10.5]
FLORES+	tsn	Setswana	yes	45.6	45.1	−0.6 [−1.1, −0.1]	−0.6 [−1.1, +0.0]
FLORES+	tso	Tsonga		29.8	46.9	+17.1 [+15.6, +18.6]	+17.7 [+16.2, +19.3]
FLORES+	tum	Tumbuka		25.3	34.3	+9.0 [+8.0, +10.0]	+9.6 [+8.4, +10.6]
FLORES+	twi	Twi		28.1	35.1	+6.9 [+5.8, +8.0]	+7.2 [+6.0, +8.2]
FLORES+	umb	Umbundu		15.9	25.5	+9.5 [+8.7, +10.3]	+10.8 [+10.0, +11.7]
FLORES+	vmw	vmw		18.7	27.1	+8.5 [+7.6, +9.3]	+9.5 [+8.6, +10.4]
FLORES+	wol	Wolof	yes	29.1	30.6	+1.5 [+1.0, +2.1]	+1.5 [+1.0, +2.1]
FLORES+	xho	Xhosa	yes	44.4	45.4	+1.1 [+0.5, +1.6]	+1.1 [+0.5, +1.7]
FLORES+	yor	Yoruba	yes	28.4	30.3	+1.9 [+1.3, +2.6]	+1.9 [+1.3, +2.6]
FLORES+	zgh	zgh		6.1	4.1	−2.0 [−2.5, −1.4]	−3.0 [−3.6, −2.4]
FLORES+	zul	Zulu	yes	49.6	48.3	−1.3 [−1.8, −0.8]	−1.3 [−1.8, −0.7]
NTREX	afr	Afrikaans	yes	63.6	63.9	+0.3 [−0.1, +0.9]	+0.4 [−0.1, +1.0]
NTREX	amh	Amharic	yes	18.1	18.0	−0.1 [−0.5, +0.4]	−0.1 [−0.5, +0.4]
NTREX	bem	Bemba		18.8	34.8	+16.0 [+15.3, +16.7]	+17.2 [+16.5, +18.0]

benchmark	ISO	Language	TP	control	RL model	RL − control, chrF++	RL − control, chrF
NTREX	ewe	Ewe		25.7	31.6	+5.9 [+5.2, +6.5]	+6.2 [+5.5, +6.9]
NTREX	hau	Hausa	yes	49.5	48.6	−0.9 [−1.5, −0.3]	−0.9 [−1.5, −0.2]
NTREX	ibo	Igbo	yes	43.3	44.9	+1.6 [+0.9, +2.5]	+1.6 [+0.8, +2.8]
NTREX	kin	Kinyarwanda	yes	38.4	37.1	−1.3 [−1.9, −0.7]	−1.3 [−2.0, −0.7]
NTREX	mlg	Malagasy	yes	41.5	40.9	−0.6 [−1.1, −0.1]	−0.6 [−1.1, −0.0]
NTREX	nde	nde		30.7	33.0	+2.2 [+1.7, +2.7]	+2.7 [+2.1, +3.3]
NTREX	nso	Sepedi		19.7	32.7	+13.0 [+12.2, +13.8]	+13.8 [+12.9, +14.6]
NTREX	nya	Chichewa	yes	43.2	44.2	+1.0 [+0.4, +1.7]	+0.9 [+0.3, +1.7]
NTREX	orm	Oromo	yes	27.3	27.3	+0.1 [−0.6, +0.8]	+0.1 [−0.8, +0.9]
NTREX	shi	shi		0.3	1.2	+0.9 [+0.8, +1.0]	+0.5 [+0.4, +0.6]
NTREX	sna	Shona	yes	40.6	39.2	−1.3 [−1.9, −0.8]	−1.4 [−1.9, −0.8]
NTREX	som	Somali	yes	40.6	44.2	+3.6 [+2.8, +4.5]	+3.8 [+2.9, +4.7]
NTREX	ssw	Swati		20.9	38.8	+18.0 [+17.0, +18.8]	+20.4 [+19.3, +21.3]
NTREX	swh	Swahili	yes	57.6	58.7	+1.1 [+0.5, +1.9]	+0.9 [+0.2, +1.7]
NTREX	tir	Tigrinya		9.8	17.0	+7.2 [+6.8, +7.6]	+7.8 [+7.4, +8.2]
NTREX	tsn	Setswana	yes	49.4	48.7	−0.7 [−1.3, −0.1]	−0.6 [−1.3, +0.0]
NTREX	ven	ven		25.7	38.8	+13.1 [+12.3, +13.9]	+13.6 [+12.8, +14.4]
NTREX	wol	Wolof	yes	29.5	30.4	+0.9 [+0.4, +1.4]	+0.8 [+0.3, +1.3]
NTREX	xho	Xhosa	yes	40.7	42.4	+1.7 [+1.2, +2.2]	+1.9 [+1.3, +2.4]
NTREX	yor	Yoruba	yes	30.9	32.1	+1.2 [+0.8, +1.8]	+1.2 [+0.7, +1.7]
NTREX	zul	Zulu	yes	44.1	43.5	−0.6 [−1.0, −0.1]	−0.6 [−1.1, −0.1]
MAFAND-MT	amh	Amharic	yes	18.4	18.3	−0.1 [−0.7, +0.5]	−0.1 [−0.7, +0.6]
MAFAND-MT	hau	Hausa	yes	33.7	32.8	−0.9 [−1.2, −0.6]	−0.7 [−1.0, −0.4]
MAFAND-MT	ibo	Igbo	yes	44.7	44.9	+0.2 [−0.5, +1.2]	+0.3 [−0.5, +1.3]
MAFAND-MT	kin	Kinyarwanda	yes	48.1	46.4	−1.7 [−2.6, −0.9]	−1.6 [−2.5, −0.7]
MAFAND-MT	lug	Luganda	yes	45.4	39.6	−5.8 [−6.3, −5.3]	−5.9 [−6.4, −5.4]
MAFAND-MT	luo	Luo		25.0	32.6	+7.6 [+6.8, +8.5]	+8.3 [+7.3, +9.2]
MAFAND-MT	nya	Chichewa	yes	42.1	41.0	−1.1 [−1.6, −0.6]	−1.0 [−1.5, −0.4]
MAFAND-MT	pcm	pcm		52.9	53.3	+0.4 [+0.1, +0.8]	+0.3 [−0.1, +0.6]
MAFAND-MT	sna	Shona	yes	34.3	32.9	−1.3 [−1.8, −0.9]	−1.4 [−1.9, −0.9]
MAFAND-MT	swh	Swahili	yes	52.0	51.8	−0.2 [−0.8, +0.6]	−0.2 [−0.8, +0.6]
MAFAND-MT	tsn	Setswana	yes	51.0	53.1	+2.1 [+1.4, +3.0]	+2.1 [+1.4, +3.0]
MAFAND-MT	twi	Twi		29.1	34.1	+5.1 [+4.3, +5.8]	+5.4 [+4.6, +6.2]
MAFAND-MT	xho	Xhosa	yes	51.5	51.8	+0.3 [−0.4, +1.0]	+0.4 [−0.3, +1.0]
MAFAND-MT	yor	Yoruba	yes	32.1	33.4	+1.4 [+0.8, +1.9]	+1.3 [+0.8, +1.9]
MAFAND-MT	zul	Zulu	yes	46.4	45.6	−0.8 [−1.3, −0.4]	−0.9 [−1.4, −0.4]

Table 48: E-03 headline: RL model (three checkpoints) minus the SFT-only control, clean rows; chrF++ and AfriCOMET over language–benchmark pairs; 95% paired bootstrap.

RL model	cells	control chrF++	RL chrF++	RL - control chrF++	RL - control chrF (plain)	AfriCOMET cells	control AfriCOMET	RL AfriCOMET	RL - control AfriCOMET
mixed best	93	29.97	34.55	+4.58 [+4.45, +4.70]	+4.91 [+4.78, +5.04]	76	0.6101	0.6284	+0.0183 [+0.0172, +0.0192]
stage A	93	29.97	33.94	+3.97 [+3.86, +4.08]	+4.22 [+4.10, +4.34]	76	0.6101	0.6265	+0.0164 [+0.0154, +0.0173]
B1 best	93	29.97	34.33	+4.36 [+4.24, +4.47]	+4.66 [+4.54, +4.78]	76	0.6101	0.6274	+0.0173 [+0.0163, +0.0182]

Table 49: E-03 by benchmark (RL model minus control).

benchmark	cells	control	RL	RL - control chrF++	AfriCOMET control	RL	diff
flores_plus	54	25.38	31.66	+6.27 [+6.07, +6.45]	0.5894	0.6267	+0.0373 [+0.0355, +0.0389]
ntrex	24	33.74	37.17	+3.42 [+3.26, +3.59]	0.6088	0.6139	+0.0051 [+0.0037, +0.0064]
mafand	15	40.44	40.78	+0.34 [+0.17, +0.50]	0.6672	0.6531	−0.0141 [−0.0154, −0.0128]

Table 50: E-03 by language panel: the 19 TranslatePsy languages (TP19) against the other languages.

RL model	panel	cells	control chrF++	RL chrF++	RL - control	AfriCOMET diff
mixed best	TP19	47	41.37	41.34	−0.03 [−0.13, +0.09]	−0.0249 [−0.0257, −0.0241] (n=47)
mixed best	OTHERS	46	18.32	27.61	+9.29 [+9.09, +9.49]	+0.0883 [+0.0861, +0.0903] (n=29)
stage A	TP19	47	41.37	41.17	−0.20 [−0.31, −0.08]	−0.0244 [−0.0252, −0.0235] (n=47)
stage A	OTHERS	46	18.32	26.55	+8.23 [+8.04, +8.41]	+0.0824 [+0.0803, +0.0843] (n=29)

Table 51: E-03: SFT baseline to control and to RL model, language-average chrF++ per benchmark.

benchmark	SFT	control	control - SFT	mixed	mixed - SFT
flores_plus	22.85	25.38	+2.53 [+2.35, +2.71]	31.66	+8.81 [+8.59, +9.03]
ntrex	29.25	33.74	+4.49 [+4.29, +4.70]	37.17	+7.91 [+7.69, +8.12]
mafand	35.77	40.44	+4.67 [+4.39, +4.92]	40.78	+5.01 [+4.76, +5.23]
all 93 cells	26.59	29.97	+3.38	34.55	+7.96

Table 52: E-03: finished and loop rates (the reward's loop flag; clean rows; language average).

system	FLORES+ finished	FLORES+ loop	NTREX finished	NTREX loop	MAFAND finished	MAFAND loop	FLORES+ length ratio
SFT	0.829	0.209	0.890	0.122	0.963	0.043	3.03
control C1 (6,000)	0.854	0.178	0.937	0.072	0.990	0.014	2.68
stage A	0.983	0.068	0.996	0.016	0.999	0.003	1.33
B1 best	0.993	0.054	0.999	0.009	1.000	0.001	1.17
mixed best	0.995	0.048	0.999	0.006	1.000	0.001	1.14

Table 53: Finished rate / loop rate (reward's flag) / hypothesis-to-reference length, clean rows, language macro.

system	FLORES+ finished / loop / hyp-to-ref length	NTREX	MAFAND
SFT	0.829 / 0.209 / 3.03	0.890 / 0.122 / 2.35	0.963 / 0.043 / 1.48
RL stage A	0.983 / 0.068 / 1.33	0.996 / 0.016 / 1.07	0.999 / 0.003 / 1.00
RL B1 best	0.993 / 0.054 / 1.17	0.999 / 0.009 / 1.02	1.000 / 0.001 / 0.99
RL mixed best	0.995 / 0.048 / 1.14	0.999 / 0.006 / 1.02	1.000 / 0.001 / 0.98

Table 54: RL minus SFT over the 93 cells under four ways of removing the finishing effect (clean rows; 95% CI).

row set	chrF++ RL - SFT [95% CI]	AfriCOMET RL - SFT [95% CI] (76 cells)
all clean rows	+7.96 [+7.82, +8.12]	+0.0631 [+0.0620, +0.0643]
SFT output cut at its first loop (RL as is)	+4.06 [+3.97, +4.15]	not scorable
rows where the SFT finished and did not loop	+2.41 [+2.33, +2.49]	+0.0366 [+0.0357, +0.0374]
rows where both finished and neither loops	+2.61 [+2.54, +2.70]	+0.0370 [+0.0362, +0.0379]
rows where the SFT failed (21% of FLORES+ rows)	+24.57 [+24.26, +24.90] (SFT 5.95, RL 30.53)	+0.3342 [+0.3295, +0.3391]

Table 55: The pre-registered decision rule applied to mixed best against the 6,000-step control.

registered condition	value	met
lower 95% bound of d (chrF++, 93 cells) $> +1.0$	$+4.58$ $[+4.45, +4.70]$	yes
lower bound of the AfriCOMET difference > 0	$+0.0183$ $[+0.0172, +0.0192]$	yes
$d > 0$ on the 19 TranslatePsy languages	-0.03 $[-0.13, +0.09]$	**no**
$d > 0$ on the other languages	$+9.29$ $[+9.09, +9.49]$	yes

G Science and long-document results

Per-language science results

Table 65: AfriScience-MT test, sentence level (970 sentences per language), plain chrF per language for every stage; macro over the six languages. Columns: SFT = SFT baseline; RL 1st / 2nd stage = first and second RL stage (best checkpoint); RL model = final general RL model; sci. SFT = science fine-tuning ($\alpha = 0.5$); sci. RL = science RL (best, final); E-14 = the same science RL started from the RL model without the science fine-tuning step.

ISO	Language	SFT	RL 1st stage	RL 2nd stage	RL model	sci. SFT	sci. RL best	sci. RL final	E-14
amh	Amharic	20.30	28.75	28.85	29.10	29.90	30.19	30.69	29.48
hau	Hausa	47.33	54.44	55.35	55.18	56.10	56.07	56.74	55.40
lug	Luganda	23.31	36.92	37.22	38.15	39.27	39.75	39.87	38.77
nso	Sepedi	41.27	51.98	52.15	52.12	54.72	54.66	54.69	52.00
yor	Yoruba	25.36	37.55	37.72	37.66	38.11	38.32	38.62	37.89
zul	Zulu	58.42	60.72	60.53	60.20	61.71	61.78	61.97	60.42
Macro		36.00	45.06	45.30	45.40	46.63	46.79	47.10	45.66

Table 66: AfriScience-MT test, chrF++ per language for every stage (same systems and order as Table 65).

ISO	Language	SFT	RL 1st stage	RL 2nd stage	RL model	sci. SFT	sci. RL best	sci. RL final	E-14
amh	Amharic	18.52	26.42	26.59	26.77	27.60	27.86	28.37	27.16
hau	Hausa	45.65	52.36	53.19	53.01	54.11	54.05	54.71	53.24
lug	Luganda	20.51	33.01	33.27	34.14	35.29	35.77	35.88	34.76
nso	Sepedi	39.36	49.80	49.93	49.88	52.65	52.57	52.55	49.69
yor	Yoruba	24.19	36.53	36.69	36.56	37.07	37.25	37.51	36.78
zul	Zulu	52.75	54.94	54.70	54.29	56.10	56.09	56.29	54.51
Macro		33.50	42.18	42.40	42.44	43.80	43.93	44.22	42.69

Table 67: AfriScience-MT test, held-out AfriCOMET-STL per language for every stage (same systems and order as Table 65).

ISO	Language	SFT	RL 1st stage	RL 2nd stage	RL model	sci. SFT	sci. RL best	sci. RL final	E-14
amh	Amharic	0.5186	0.6002	0.5951	0.6002	0.6061	0.6083	0.6154	0.6049
hau	Hausa	0.6473	0.6825	0.6839	0.6802	0.6927	0.6913	0.6941	0.6828
lug	Luganda	0.5161	0.6212	0.6188	0.6244	0.6312	0.6341	0.6378	0.6276
nso	Sepedi	0.6296	0.7098	0.7113	0.7129	0.7156	0.7168	0.7205	0.7133
yor	Yoruba	0.6199	0.7564	0.7583	0.7613	0.7594	0.7615	0.7647	0.7622
zul	Zulu	0.7068	0.7232	0.7205	0.7178	0.7227	0.7237	0.7268	0.7191
Macro		0.6064	0.6822	0.6813	0.6828	0.6879	0.6893	0.6932	0.6850

Table 68: AfriScience-MT test, SSA-COMET-QE per language for every stage. **Reward family: not independent evidence for the RL stages** (same systems and order as Table 65).

ISO	Language	SFT	RL 1st stage	RL 2nd stage	RL model	sci. SFT	sci. RL best	sci. RL final	E-14
amh	Amharic	0.5354	0.5815	0.5781	0.5806	0.5830	0.5852	0.5888	0.5847
hau	Hausa	0.5664	0.5827	0.5819	0.5808	0.5845	0.5838	0.5841	0.5821
lug	Luganda	0.5276	0.5972	0.5972	0.6016	0.5999	0.6044	0.6059	0.6056
nso	Sepedi	0.6056	0.6533	0.6540	0.6561	0.6518	0.6532	0.6562	0.6569
yor	Yoruba	0.5728	0.6434	0.6450	0.6489	0.6408	0.6428	0.6477	0.6519
zul	Zulu	0.6037	0.6127	0.6115	0.6100	0.6107	0.6119	0.6123	0.6105
Macro		0.5686	0.6118	0.6113	0.6130	0.6118	0.6135	0.6158	0.6153

Table 69: E-14 (science RL without the science fine-tuning) minus other models on AfriScience-MT test, per language and macro, with 95% bootstrap intervals over sentences. Negative = E-14 lower. SSA-COMET-QE is of the reward family and not independent evidence.

E-14 minus	language	chrF	AfriCOMET	SSA-COMET-QE (reward family)
science RL final	amh	−1.21 [−1.61, −0.80]	−0.0105 [−0.0163, −0.0054]	−0.0041 [−0.0079, −0.0008]
	hau	−1.34 [−1.84, −0.89]	−0.0114 [−0.0158, −0.0072]	−0.0020 [−0.0046, +0.0006]
	lug	−1.11 [−1.64, −0.53]	−0.0102 [−0.0157, −0.0049]	−0.0003 [−0.0047, +0.0037]
	nso	−2.70 [−3.01, −2.34]	−0.0072 [−0.0107, −0.0038]	+0.0007 [−0.0017, +0.0031]
	yor	−0.72 [−1.07, −0.27]	−0.0024 [−0.0056, +0.0006]	+0.0042 [+0.0019, +0.0066]
	zul	−1.55 [−1.94, −1.15]	−0.0078 [−0.0112, −0.0043]	−0.0018 [−0.0039, +0.0003]
	macro	−1.44 [−1.61, −1.25]	−0.0082 [−0.0101, −0.0065]	−0.0006 [−0.0018, +0.0007]
science fine-tuning ($\alpha = 0.5$)	amh	−0.42 [−0.96, +0.13]	−0.0012 [−0.0078, +0.0050]	+0.0017 [−0.0026, +0.0058]
	hau	−0.70 [−1.42, +0.02]	−0.0099 [−0.0147, −0.0046]	−0.0024 [−0.0052, +0.0006]

E-14 minus	language	chrF	AfriCOMET	SSA-COMET-QE (reward family)
	lug	−0.51 [−1.06, +0.02]	−0.0037 [−0.0093, +0.0018]	+0.0057 [+0.0014, +0.0099]
	nso	−2.73 [−3.13, −2.25]	−0.0023 [−0.0062, +0.0020]	+0.0051 [+0.0023, +0.0080]
	yor	−0.21 [−0.65, +0.31]	+0.0029 [−0.0009, +0.0066]	+0.0112 [+0.0084, +0.0139]
	zul	−1.29 [−1.73, −0.82]	−0.0036 [−0.0075, +0.0002]	−0.0002 [−0.0026, +0.0024]
	macro	−0.97 [−1.20, −0.75]	−0.0030 [−0.0050, −0.0009]	+0.0035 [+0.0022, +0.0047]
RL model (its own start)	amh	+0.38 [−0.05, +0.83]	+0.0047 [−0.0007, +0.0096]	+0.0041 [+0.0010, +0.0071]
	hau	+0.22 [−0.12, +0.58]	+0.0026 [−0.0011, +0.0061]	+0.0013 [−0.0009, +0.0035]
	lug	+0.62 [+0.24, +1.07]	+0.0032 [−0.0012, +0.0077]	+0.0039 [+0.0001, +0.0073]
	nso	−0.13 [−0.34, +0.10]	+0.0004 [−0.0020, +0.0028]	+0.0007 [−0.0011, +0.0024]
	yor	+0.23 [+0.02, +0.47]	+0.0009 [−0.0013, +0.0032]	+0.0030 [+0.0013, +0.0049]
	zul	+0.22 [−0.13, +0.61]	+0.0013 [−0.0019, +0.0045]	+0.0005 [−0.0014, +0.0026]
	macro	+0.26 [+0.12, +0.39]	+0.0022 [+0.0007, +0.0036]	+0.0023 [+0.0013, +0.0033]
second RL stage (final)	amh	+0.49 [+0.13, +0.86]	−0.0013 [−0.0067, +0.0036]	−0.0025 [−0.0060, +0.0006]
	hau	−0.04 [−0.35, +0.27]	−0.0033 [−0.0070, +0.0006]	−0.0022 [−0.0046, +0.0003]
	lug	+0.55 [+0.17, +1.00]	+0.0031 [−0.0018, +0.0077]	+0.0035 [−0.0005, +0.0072]
	nso	−0.03 [−0.32, +0.30]	+0.0018 [−0.0013, +0.0054]	+0.0024 [+0.0000, +0.0048]
	yor	+0.04 [−0.30, +0.44]	+0.0012 [−0.0021, +0.0045]	+0.0054 [+0.0030, +0.0079]
	zul	+0.23 [−0.13, +0.59]	+0.0011 [−0.0025, +0.0047]	+0.0005 [−0.0016, +0.0026]
	macro	+0.21 [+0.07, +0.36]	+0.0004 [−0.0012, +0.0020]	+0.0012 [+0.0000, +0.0023]

Table 70: Document level per language (168 test documents, 28 per language; greedy, 4,096 new tokens in whole mode; chunked = sentence by sentence with best-of-4 repair): chrF, finished rate, loop rate and tail coverage in whole-document mode, and chrF, finished rate and tail coverage in chunked mode, for every document-level model. One run each; no intervals.

model	language	whole: chrF	finished	loop	tail coverage	chunked: chrF	finished	tail coverage
RL model (no science step)	amh	9.2	0.46	0.89	0.11	38.0	1.00	0.40
	hau	27.4	0.79	0.71	0.35	65.7	1.00	0.72
	lug	20.1	0.82	0.96	0.23	54.0	1.00	0.57
	nso	27.8	0.82	0.89	0.34	64.8	1.00	0.72
	yor	17.5	0.57	0.96	0.22	49.8	1.00	0.57
	zul	35.0	0.79	0.75	0.37	71.4	1.00	0.70
science fine-tuning ($\alpha = 0.5$)	amh	10.9	0.21	0.89	0.13	39.3	1.00	0.42
	hau	32.6	0.75	0.68	0.41	67.4	1.00	0.74
	lug	23.6	0.43	1.00	0.26	55.6	1.00	0.59
	nso	29.4	0.68	0.96	0.37	67.3	1.00	0.75
	yor	18.5	0.39	0.96	0.22	50.6	1.00	0.58
	zul	34.6	0.61	0.64	0.40	72.7	1.00	0.72
earlier 270M RL-final	amh	22.7	0.93	0.39	0.27	34.9	1.00	0.37
	hau	49.9	0.79	0.25	0.61	69.1	1.00	0.76
	lug	33.1	0.43	0.79	0.39	60.6	1.00	0.64
	nso	35.8	0.75	0.86	0.45	67.8	1.00	0.75
	yor	24.3	0.64	0.86	0.30	49.5	1.00	0.57
	zul	43.3	0.64	0.54	0.47	73.7	1.00	0.73
science RL (best)	amh	9.8	0.25	0.89	0.12	39.1	1.00	0.42
	hau	38.3	0.86	0.64	0.46	67.1	1.00	0.74
	lug	22.2	0.61	0.96	0.26	55.4	1.00	0.59
	nso	25.5	0.75	0.93	0.32	67.2	1.00	0.75
	yor	22.6	0.54	0.96	0.28	50.6	1.00	0.58
	zul	41.9	0.75	0.68	0.44	72.7	1.00	0.72
science RL (final)	amh	13.3	0.43	0.82	0.16	39.3	1.00	0.42
	hau	32.5	0.75	0.68	0.40	67.3	1.00	0.74
	lug	25.3	0.64	0.96	0.29	55.5	1.00	0.59
	nso	29.2	0.82	0.93	0.36	67.0	1.00	0.75
	yor	21.6	0.68	0.96	0.26	50.6	1.00	0.58
	zul	42.9	0.86	0.71	0.45	72.7	1.00	0.71
long-document RL (best)	amh	15.1	0.50	0.86	0.17	39.4	1.00	0.42
	hau	42.3	0.86	0.50	0.51	67.6	1.00	0.74
	lug	22.9	0.79	0.89	0.28	55.7	1.00	0.59
	nso	34.0	0.93	0.75	0.41	67.2	1.00	0.75
	yor	23.5	0.96	0.89	0.29	50.6	1.00	0.58
	zul	47.4	0.89	0.50	0.49	72.7	1.00	0.72
long-document RL (final)	amh	14.9	0.68	0.68	0.17	39.4	1.00	0.42
	hau	40.5	0.93	0.54	0.47	67.5	1.00	0.74
	lug	24.4	0.82	0.82	0.28	55.7	1.00	0.59
	nso	33.3	0.93	0.79	0.41	67.2	1.00	0.75
	yor	20.9	0.86	0.89	0.27	50.6	1.00	0.58
	zul	44.2	0.89	0.61	0.45	72.7	1.00	0.72

Table 57: Weight interpolation sweep of the science fine-tuning step, AfriScience-MT test (macro over six languages), with NTREX and MAFAND-MT chrF++ (last two columns). SSA-QE = SSA-COMET-QE (headline); AfriC. = AfriCOMET-STL.

α	SSA-QE	chrF	chrF++	BLEU	spBLEU	AfriC.	NTREX	MAFAND
0	0.6130	45.40	42.44	17.72	21.89	0.6828	37.16	40.78
0.3	0.6128	46.28	43.40	18.73	23.25	0.6865	37.13	41.06
0.5 (chosen)	0.6118	46.63	43.80	19.27	23.87	0.6879	36.90	41.05
0.7	0.6092	46.56	43.81	18.98	23.83	0.6859	36.43	40.83
1.0	0.5986	44.92	42.30	16.78	20.87	0.6708	34.71	39.58

Table 58: Science RL without the science SFT (E-14): AfriScience-MT test, sentence level, macro over 6 languages.

model	chrF	chrF++	AfriCOMET-STL (held out)	SSA-COMET-QE (reward family)
mixed best (= alpha 0; E-14's start)	45.40	42.44	0.6828	0.6130
B1 final (stage-B1 RL)	45.45	42.50	0.6846	0.6141
E-14 final (science RL, no science SFT)	**45.66**	42.69	**0.6850**	0.6153
science SFT alpha 0.5	46.63	43.80	0.6879	0.6118
science RL final (from the science SFT)	47.10	44.22	0.6932	0.6158

Table 59: E-14 final minus X, 95% bootstrap (independent weights per language).

E-14 final minus	chrF	AfriCOMET-STL (held out)	SSA-COMET-QE (reward family)
science RL final (from the science SFT)	**-1.44 [−1.61, −1.25]**	**-0.0082 [−0.0101, −0.0065]**	−0.0006 [−0.0018, +0.0007]
science SFT alpha 0.5	−0.97 [−1.20, −0.75]	−0.0030 [−0.0050, −0.0009]	+0.0035 [+0.0022, +0.0047]
mixed best (its own start)	+0.26 [+0.12, +0.39]	+0.0022 [+0.0007, +0.0036]	+0.0023 [+0.0013, +0.0033]
B1 final	+0.21 [+0.07, +0.36]	+0.0004 [−0.0012, +0.0020]	+0.0012 [+0.0000, +0.0023]

Table 60: Chunked document decoding (sentence by sentence with best-of-4 repair), 168 test documents.

model (chunked)	SSA-COMET-QE, 4-sentence windows	chrF	AfriCOMET window	BLEU
385M start (alpha 0)	0.574	57.3	0.687	20.0
385M science SFT (alpha 0.5)	0.574	58.8	0.692	22.0
old 270M RL-final (earlier 270M work)	0.593	59.3	0.703	22.5

Table 61: Document level, 168 test documents (28 papers × 6 languages), macro over the six languages, whole-document and chunked modes (long-document RL evaluation).

model	mode	**SSA-COMET-QE window**	chrF	AfriCOMET window	SSA-MTL(ref) window	SSA-QE concat (first 512 tokens)	AfriCOMET concat	SSA-MTL(ref) concat	chrF++	BLEU	spBLEU	finished	loop	length ratio	tail coverage	concat COMET x100 - chrF
385M start (alpha 0, no science SFT)	whole	-	22.8	-	-	0.378	0.470	0.214	20.0	4.0	5.4	0.71	0.86	1.03	0.27	15.0
385M start (alpha 0, no science SFT)	chunked	**0.574**	57.3	0.687	0.448	0.436	0.661	0.219	53.0	20.0	25.8	1.00	0.14	0.99	0.61	−13.7
385M science SFT (alpha 0.5)	whole	-	24.9	-	-	0.381	0.495	0.208	22.1	5.2	6.5	0.51	0.86	1.30	0.30	13.2
385M science SFT (alpha 0.5)	chunked	**0.574**	58.8	0.692	0.448	0.434	0.661	0.213	54.7	22.0	28.5	1.00	0.23	1.01	0.63	−15.4
old 270M RL-final (rl2c, earlier 270M work)	whole	-	34.9	-	-	0.420	0.599	0.268	30.9	6.9	10.2	0.70	0.61	1.10	0.41	7.2
old 270M RL-final (rl2c, earlier 270M work)	chunked	**0.593**	59.3	0.703	0.468	0.457	0.682	0.244	55.0	22.5	29.3	1.00	0.11	1.00	0.64	−13.6
385M science RL (best)	whole	-	26.7	-	-	0.388	0.519	0.216	23.8	5.7	7.9	0.62	0.85	1.19	0.31	12.1
385M science RL (best)	chunked	**0.574**	58.7	0.692	0.450	0.437	0.661	0.216	54.5	21.9	28.3	1.00	0.18	1.00	0.63	−14.9
385M science RL (final)	whole	-	27.4	-	-	0.398	0.531	0.228	24.4	5.8	8.1	0.70	0.85	1.12	0.32	12.4
385M science RL (final)	chunked	**0.577**	58.7	0.696	0.455	0.440	0.669	0.217	54.6	21.9	28.3	1.00	0.20	1.00	0.63	−14.8
385M long-document RL (best, step 50)	whole	-	30.9	-	-	0.403	0.564	0.231	27.8	7.3	9.9	0.82	0.73	0.96	0.36	9.5
385M long-document RL (best, step 50)	chunked	**0.579**	58.9	0.698	0.458	0.442	0.675	0.220	54.7	22.0	28.3	1.00	0.17	0.99	0.63	−14.7
385M long-document RL (final, step 80)	whole	-	29.7	-	-	0.403	0.552	0.225	26.7	6.9	9.2	0.85	0.72	0.81	0.34	10.6
385M long-document RL (final, step 80)	chunked	**0.579**	58.8	0.698	0.458	0.441	0.676	0.220	54.7	22.0	28.3	1.00	0.13	0.99	0.63	−14.7

Table 62: Chunked mode: sentence coverage and repairs.

model	sentences	unfinished before repair (repaired)	unfinished after	sentence coverage
385M start (alpha 0, no science SFT)	5,820	7 repaired	0	1.000
385M science SFT (alpha 0.5)	5,820	29 repaired	0	1.000
old 270M RL-final (rl2c, earlier 270M work)	5,820	0 repaired	0	1.000
385M science RL (best)	5,820	12 repaired	0	1.000
385M science RL (final)	5,820	6 repaired	0	1.000
385M long-document RL (best, step 50)	5,820	1 repaired	0	1.000
385M long-document RL (final, step 80)	5,820	0 repaired	0	1.000

Table 63: Per language, chunked mode: chrF / SSA-COMET-QE window (SSA-COMET-QE is of the reward family) for every document-level model.

model	amh	hau	lug	nso	yor	zul
385M start (alpha 0, no science SFT)	38.0 / 0.531	65.7 / 0.536	54.0 / 0.563	64.8 / 0.621	49.8 / 0.619	71.4 / 0.575
385M science SFT (alpha 0.5)	39.3 / 0.538	67.4 / 0.538	55.6 / 0.565	67.3 / 0.618	50.6 / 0.607	72.7 / 0.577
old 270M RL-final (rl2c, earlier 270M work)	34.9 / 0.497	69.1 / 0.571	60.6 / 0.625	67.8 / 0.633	49.5 / 0.638	73.7 / 0.592
385M science RL (best)	39.1 / 0.536	67.1 / 0.537	55.4 / 0.564	67.2 / 0.620	50.6 / 0.612	72.7 / 0.577
385M science RL (final)	39.3 / 0.539	67.3 / 0.539	55.5 / 0.571	67.0 / 0.621	50.6 / 0.615	72.7 / 0.578
385M long-document RL (best, step 50)	39.4 / 0.541	67.6 / 0.541	55.7 / 0.572	67.2 / 0.626	50.6 / 0.617	72.7 / 0.577
385M long-document RL (final, step 80)	39.4 / 0.542	67.5 / 0.539	55.7 / 0.575	67.2 / 0.624	50.6 / 0.616	72.7 / 0.578

Table 64: Per language, AfriScience-MT test: science RL final minus science fine-tuning ($\alpha = 0.5$), chrF and AfriCOMET with 95% bootstrap intervals.

iso	chrF alpha 0.5	chrF RL final	diff [CI]	AfriCOMET alpha 0.5	AfriCOMET RL final	diff [CI]
amh	29.90	30.69	+0.79 [+0.34, +1.22]	0.6061	0.6154	+0.0093 [+0.0038, +0.0147]
hau	56.10	56.74	+0.64 [+0.01, +1.28]	0.6927	0.6941	+0.0015 [−0.0026, +0.0055]
lug	39.27	39.87	+0.60 [+0.01, +1.20]	0.6312	0.6378	+0.0066 [+0.0015, +0.0120]
nso	54.72	54.69	−0.03 [−0.34, +0.34]	0.7156	0.7205	+0.0049 [+0.0016, +0.0085]
yor	38.11	38.62	+0.51 [+0.21, +0.83]	0.7594	0.7647	+0.0053 [+0.0025, +0.0082]
zul	61.71	61.97	+0.26 [−0.09, +0.60]	0.7227	0.7268	+0.0042 [+0.0010, +0.0073]

Table 71: AfriScience-MT test, sentence level (six languages, one run per stage). SSA-QE = SSA-COMET-QE, of the reward family for every RL row.

model	chrF	AfriCOMET	SSA-QE
SFT baseline	36.00	0.6064	0.5686
first RL stage	45.06	0.6822	0.6118
RL model (mixed best)	45.40	0.6828	0.6130
science RL without science fine-tuning (E-14)	45.66	0.6850	0.6153
science fine-tuning ($\alpha = 0.5$)	46.63	0.6879	0.6118
science RL final	47.10	0.6932	0.6158
long-document RL final	47.16	0.6936	0.6165

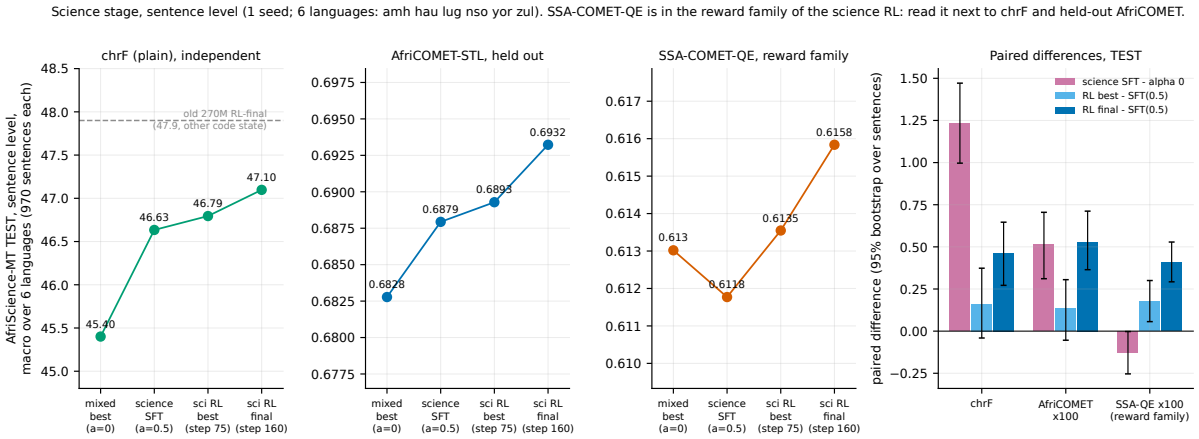


Figure 12: Science stage, AfriScience-MT TEST, sentence level (970 sentences per language, macro over amh hau lug nso yor zul; greedy; one seed): plain chrF (independent of the reward; dashed: the earlier 270M work's 270M RL-final, other code state), held-out AfriCOMET-STL (independent), and SSA-COMET-QE (reward family of the science RL) for the mixed best (alpha 0), the science SFT (alpha 0.5), and the science RL best (step 75) and final (step 160); right: paired differences with 95% bootstrap over sentences. Sources: `reports/data/eval_science_rl_2026-11-04/sci_pairs.json`, `reports/data/eval_science_sft_2026-11-04/tables.md`.

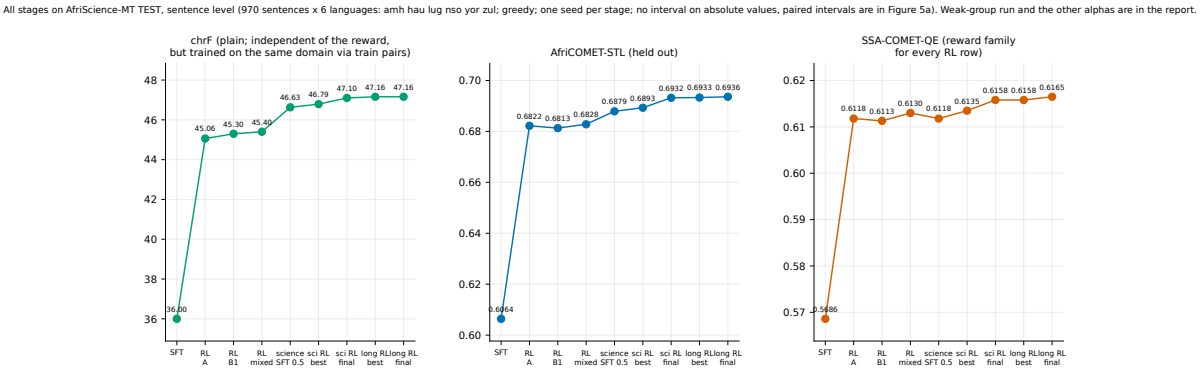


Figure 13: All stages on AfriScience-MT TEST, sentence level (970 sentences per language, macro over amh hau lug nso yor zul; greedy; one seed per stage): plain chrF, held-out AfriCOMET-STL and SSA-COMET-QE (reward family for every RL row) for the SFT, RL stages A, B1 and mixed, the science SFT (alpha 0.5), the science RL and the long-document RL. The weak-group run and the other interpolation weights are in the source table. Source: `reports/data/e05_ssa_all_stages_2026-11-04/tables.md`.

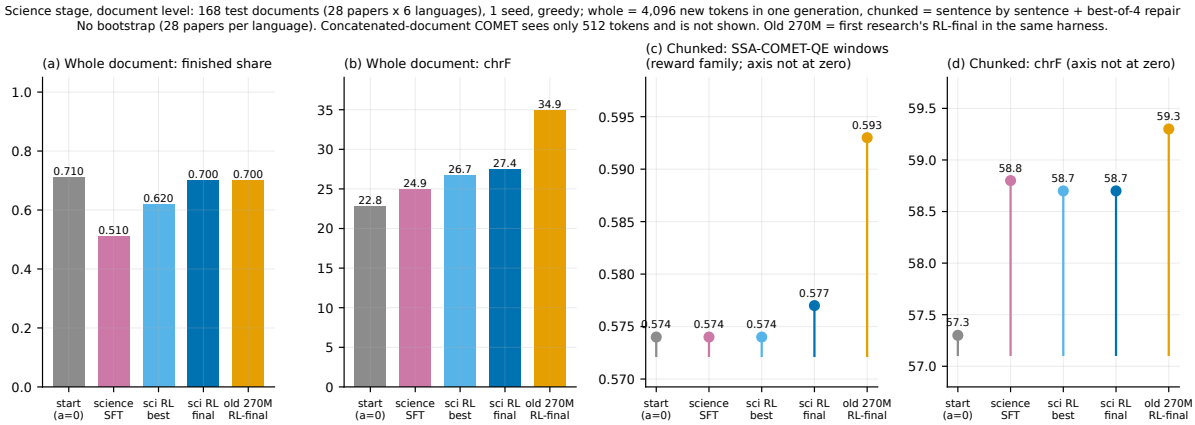


Figure 14: Science stage, document level: 168 test documents (28 papers × 6 languages), greedy, one seed, no bootstrap. Whole-document generation (4,096 new tokens): finished share and chrF. Chunked decoding (sentence by sentence plus best-of-4 repair): SSA-COMET-QE on 4-sentence windows (reward family) and chrF; axes of (c) and (d) do not start at zero. Source: `reports/data/eval_science_rl_2026-11-04/doc_tables.md`.

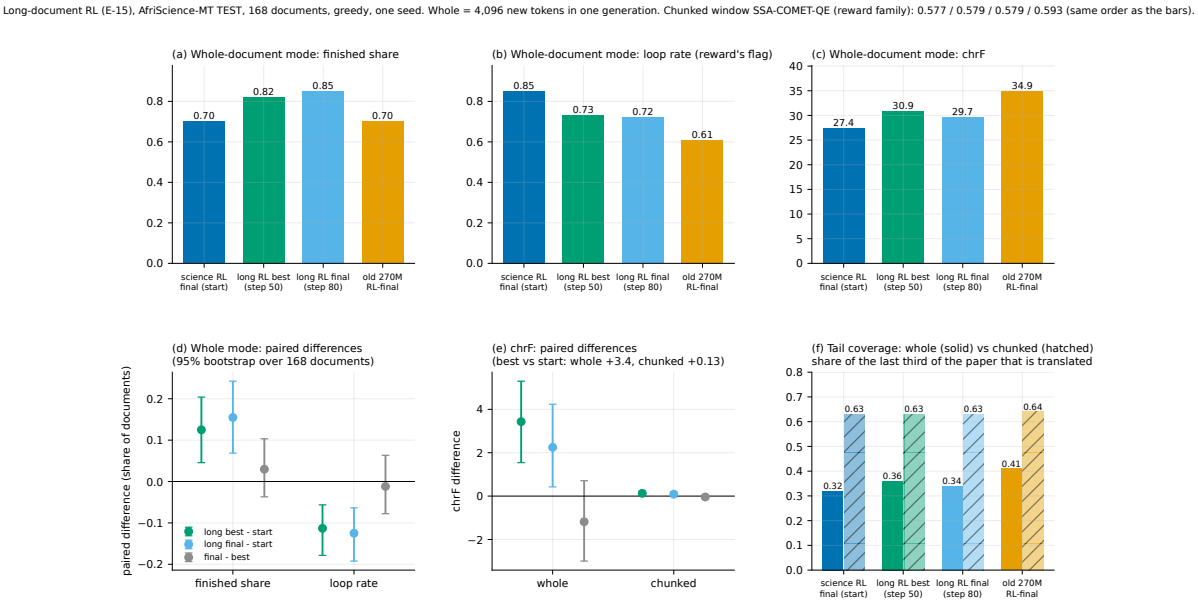


Figure 15: Long-document RL (E-15), AfriScience-MT TEST, 168 documents (28 papers × 6 languages), greedy, one seed. (a-c) Whole-document mode (4,096 new tokens in one generation): finished share, loop rate (the reward's flag) and chrF for the science RL final (the start), the long-document RL best (step 50) and final (step 80), and the earlier 270M work's 270M RL-final. (d, e) Paired differences with 95% bootstrap over documents (Poisson weights): finished share, loop rate and chrF in whole mode, chrF in chunked mode. (f) Tail coverage, the share of the last third of the paper that is translated, whole against chunked mode. Chunked window SSA-COMET-QE (reward family) is in the figure title. Sources: `reports/data/eval_long_rl_2026-11-04/{doc_tables.md,doc_pairs.json}`.

H Audit of the repetition check

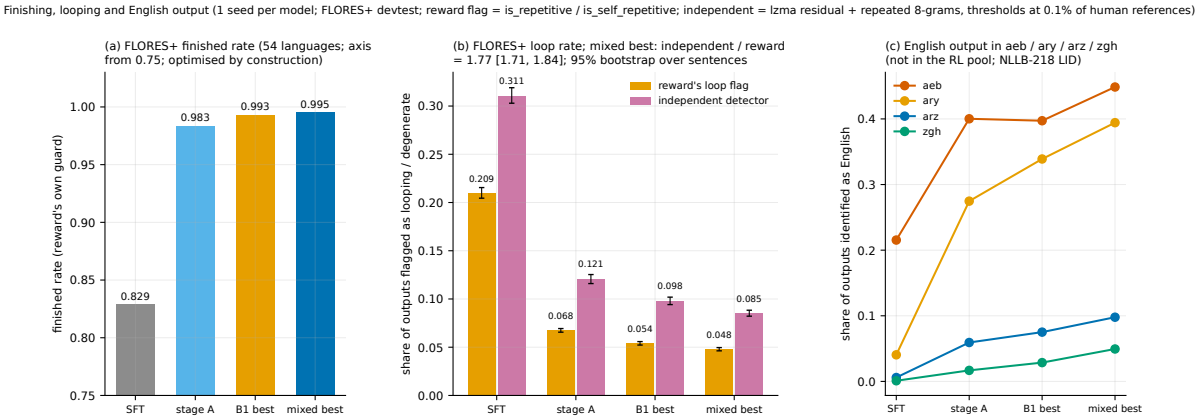


Figure 16: Finishing, repetition and English output on FLORES+ (language averages, one run per model). (a) Share of outputs that finish (axis from 0.75). (b) The reward's repetition check against the independent detector; error bars are 95% intervals. (c) Share of English outputs in four languages that were not among the RL prompts.

Table 72: Loop flag of the reward against the independent detector, language macro (95% bootstrap in brackets).

system	FLORES+ reward flag	FLORES+ independent flag	NTREX reward / independent	MAFAND reward / independent
SFT	0.2095	0.3107 [0.3029, 0.3190]	0.1218 / 0.1662	0.0431 / 0.0754
RL stage A	0.0675	0.1207 [0.1160, 0.1253]	0.0165 / 0.0309	0.0026 / 0.0092
RL B1 best	0.0541	0.0980 [0.0942, 0.1019]	0.0092 / 0.0190	0.0013 / 0.0056
RL mixed best	0.0480 [0.0462, 0.0496]	0.0851 [0.0822, 0.0883]	0.0064 / 0.0143	0.0009 / 0.0039
human references	0.0001	0.0018	0.0000 / 0.0018	0.0000 / 0.0013

Table 73: Share of FLORES+ outputs identified as English (NLLB-218 top-1, $p \geq 0.5$), by system.

language	SFT	stage A	B1 best	mixed best	weak final
aeb	0.215	0.400	0.397	0.449	0.498
ary	0.041	0.275	0.339	0.394	0.488
arz	0.006	0.059	0.075	0.098	0.151
zgh	0.001	0.017	0.029	0.049	0.074
macro, 54 languages	0.0055	0.0146	0.0164	0.0191	0.0235

I Hyper-parameters, recipe and reward details

This appendix collects the SFT and RL settings that Sections 3 and 3 summarise.

I.1 SFT recipe and learning-rate schedule

Recipe. Samples are packed into 4,096-token packs (one language per pack, block-diagonal attention) at 32 packs = 131,072 tokens per step; each pack is used once. Optimiser: Muon (momentum 0.95, 5 Newton-Schulz steps) on the 2-D decoder matrices and AdamW (betas 0.9/0.95) on embeddings and norms, weight decay 0.01, gradient clipping 1.0, 3% linear warm-up then cosine decay, seed 1234. The learning-rate schedule, read from the executed per-shard configs, is: SFT-1 (`g1a`, 1,500 steps) peak 5e-5, cosine to 50% of the peak (2.5e-5); shard 1 (`g1b_s1`) peak 5e-5, cosine to 80% (4e-5); shards 2-5 (`g1b_s2`..`g1b_s5`) peak 3e-5 (lowered from 5e-5 on 2026-11-01 because of forgetting in tiny languages), cosine to 80% (2.4e-5); shard 6 (`g1b_s6`) peak 3e-5, cosine to 10%, so the last step runs at 3.0e-6. Every shard starts a fresh Muon state with a 3% linear warm-up. SFT-2 runs in six shards of about 4,800 to 5,300 steps, with tail-language replay from shard 2. Executed stage B: 31,353 steps, 4.109 B tokens, 5.14 s per step on an A100 (44.8 h); 32,853 steps in all with SFT-1. Final validation loss: macro 1.3412, micro 1.2846 over 78 languages (end of SFT-1: 1.7847). The macro rises from 1.78 to 1.98 at the end of shard 1 because new languages enter the validation set at stage 2.

I.2 RL: optimiser, stages, prompts, reward and features

The science SFT before the science RL used AdamW at 3e-6 (Section 5.6). Dynamic-sampling, std-floor and reference-domain settings differ by stage as well (Appendix K).

Stages. Stage A (400 steps, short English windows, bin b64), B1 (135 steps run, b256; best checkpoint step 50 by dev chrF++), mixed (60 steps, b64 and b256), then a continued-RL run on eight weak languages (85 steps), a science SFT with weight interpolation, a science RL and a long-document RL (Appendix K). B1 started from stage A's final model (step 400), mixed from B1's best checkpoint (step 50), the weak-group run from the mixed best, the science SFT from the mixed best. The dev set has **720 sentences = 18 FLORES+ dev languages** $\times$ **40 sentences** (afr ewe fon hau ibo kin nya orm sna som sot swh tsn twi wol xho yor zul). B1's step-135 weights (Hub branch `stage-b1-final`) were evaluated after the fact (E-20, Section 5.5) and equal B1 best within noise.

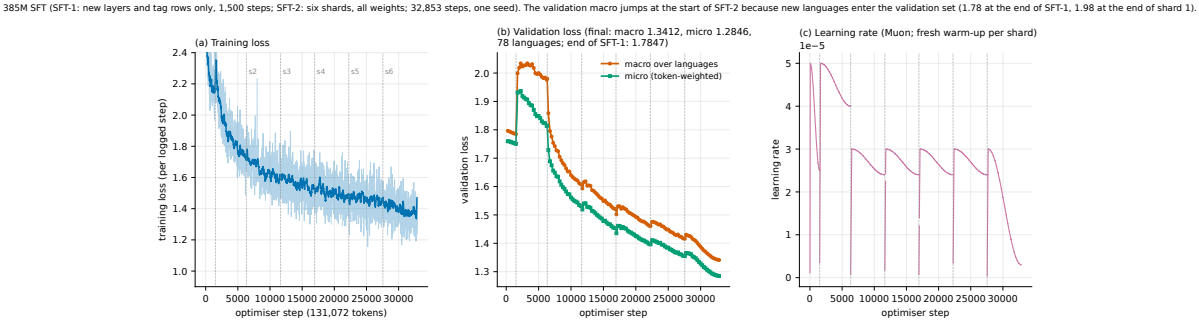


Figure 17: Training of the 385M SFT (one seed): (a) training loss per logged step (thin) and 20-point mean (thick), (b) validation loss (macro over languages and token-weighted micro) and (c) learning rate over the 32,853 optimiser steps of SFT-1 (1,500 steps, new layers and tag rows only) and the six SFT-2 shards (dotted lines); the validation macro jumps at the start of SFT-2 because new languages enter the validation set, and the learning-rate panel shows the per-shard schedule of Section 3. Final validation loss: macro 1.3412, micro 1.2846 over 78 languages. Source: `reports/data/e23_hygiene_2026-11-04/sft_385m_curve.csv`.

Table 74: RL hyper-parameters by stage.

stage	learning rate	KL coefficient (k3)	group size	guide member	steps (cap) × prompts
A, B1, mixed	2e-6 (10 warm-up steps)	0.05	16 (15 + greedy)	beam/MBR, fraction 0.25 (from step 41 in A)	400 × 64; 150 (135 run) × 48; 60 × 42
science RL	1.0e-6	0.1	16	beam/MBR, fraction 0.25	150 in the yaml, cap extended to 190 by the extension rule at step 120; executed 160 (stopped at step 150 by dev plateau, then 10 wind-down steps; `summary.json`, Appendix K) × 42
long-document RL	1.0e-6	0.1	8 (7 + greedy)	off	80 × 32 (51% long prompts)

J SFT against A0 and the earlier 270M work (details)

J.1 Weak and added languages

Table 77: Weak-group RL (continued RL on eight weak West-African languages, 85 steps) against its start, the RL model, per language and benchmark (all rows): chrF++ before and after, and the change in chrF++ and spBLEU with 95% paired-bootstrap intervals. One run, no matched control; the targets are marked.

benchmark	ISO	Language	weak-group target	RL model	after weak-group RL	change in chrF++	change in spBLEU
FLORES+	aeb	aeb		4.1	3.2	−0.94 [−1.30, −0.58]	−0.06 [−0.20, +0.08]
FLORES+	afr	Afrikaans		64.1	64.0	−0.16 [−0.36, +0.05]	−0.28 [−0.54, +0.00]
FLORES+	amh	Amharic		26.6	26.5	−0.11 [−0.43, +0.24]	−0.01 [−0.40, +0.33]
FLORES+	ary	ary		6.9	5.6	−1.24 [−1.63, −0.84]	−0.16 [−0.31, −0.01]
FLORES+	arz	arz		9.2	9.2	−0.02 [−0.61, +0.55]	+0.08 [−0.03, +0.21]
FLORES+	bam	Bambara	target	27.2	27.3	+0.10 [−0.10, +0.30]	−0.08 [−0.26, +0.11]
FLORES+	bem	Bemba		33.9	34.1	+0.22 [−0.26, +0.65]	+0.22 [+0.00, +0.44]
FLORES+	cjk	cjk		23.0	23.0	+0.00 [−0.16, +0.17]	+0.04 [−0.03, +0.13]
FLORES+	dik	dik		14.1	13.6	−0.46 [−0.79, −0.14]	−0.38 [−0.52, −0.24]
FLORES+	dyu	Dyula	target	19.0	19.4	+0.45 [+0.28, +0.64]	+0.08 [−0.01, +0.18]
FLORES+	ewe	Ewe	target	31.2	31.1	−0.11 [−0.35, +0.12]	−0.11 [−0.36, +0.14]
FLORES+	fon	Fon	target	22.2	22.3	+0.09 [−0.05, +0.22]	−0.10 [−0.22, +0.01]
FLORES+	hau	Hausa		46.5	46.4	−0.05 [−0.37, +0.26]	+0.11 [−0.20, +0.43]
FLORES+	ibo	Igbo		40.1	40.0	−0.12 [−0.35, +0.10]	−0.12 [−0.39, +0.16]
FLORES+	kab	Kabyle		26.7	26.4	−0.27 [−0.45, −0.08]	−0.14 [−0.34, +0.04]
FLORES+	kam	Kamba		25.9	26.2	+0.25 [+0.04, +0.47]	+0.11 [−0.07, +0.28]
FLORES+	kbp	Kabiye	target	24.9	24.9	+0.04 [−0.28, +0.37]	+0.04 [−0.19, +0.27]
FLORES+	kea	Kabuverdianu		39.2	39.5	+0.34 [+0.11, +0.57]	+0.12 [−0.13, +0.37]
FLORES+	kik	Kikuyu		33.3	33.3	−0.01 [−0.26, +0.25]	−0.06 [−0.33, +0.22]
FLORES+	kin	Kinyarwanda		38.7	38.8	+0.12 [−0.17, +0.44]	−0.07 [−0.39, +0.24]
FLORES+	kmb	Kimbundu		26.6	26.7	+0.08 [−0.08, +0.25]	+0.09 [−0.03, +0.20]
FLORES+	knc	Kanuri		14.2	15.0	+0.74 [+0.34, +1.12]	+0.18 [+0.07, +0.29]
FLORES+	lin	Lingala		44.3	44.2	−0.16 [−0.43, +0.10]	−0.23 [−0.53, +0.08]
FLORES+	lua	Luba-Kasai		33.4	33.4	+0.04 [−0.19, +0.30]	+0.05 [−0.17, +0.28]
FLORES+	lug	Luganda		31.7	31.9	+0.13 [−0.27, +0.53]	+0.19 [−0.19, +0.56]
FLORES+	luo	Luo		34.5	34.8	+0.27 [+0.02, +0.52]	+0.16 [−0.07, +0.40]
FLORES+	mfe	mfe		37.9	39.0	+1.08 [+0.56, +1.73]	+1.35 [+0.66, +2.16]
FLORES+	mlg	Malagasy		44.7	44.7	−0.00 [−0.30, +0.29]	−0.08 [−0.37, +0.19]
FLORES+	mos	Mossi	target	22.8	22.9	+0.15 [−0.09, +0.39]	+0.09 [−0.11, +0.27]
FLORES+	nso	Sepedi		49.0	49.0	−0.03 [−0.28, +0.22]	−0.09 [−0.39, +0.25]
FLORES+	nus	Nuer		24.8	24.7	−0.16 [−0.37, +0.05]	−0.34 [−0.60, −0.06]
FLORES+	nya	Chichewa		42.2	41.9	−0.27 [−0.47, −0.06]	−0.25 [−0.48, −0.02]
FLORES+	orm	Oromo		30.2	30.1	−0.10 [−0.36, +0.18]	−0.20 [−0.45, +0.05]
FLORES+	run	Kirundi		33.9	33.9	−0.01 [−0.25, +0.23]	−0.08 [−0.35, +0.20]
FLORES+	sag	Sango		34.3	34.2	−0.07 [−0.36, +0.21]	+0.05 [−0.21, +0.33]

benchmark	ISO	Language	weak-group target	RL model	after weak-group RL	change in chrF++	change in spBLEU
FLORES+	seh	seh		29.6	29.7	+0.05 [−0.15, +0.29]	+0.08 [−0.10, +0.28]
FLORES+	sna	Shona		37.6	37.2	−0.40 [−0.60, −0.18]	−0.38 [−0.64, −0.14]
FLORES+	som	Somali		39.7	39.6	−0.14 [−0.33, +0.05]	−0.18 [−0.41, +0.06]
FLORES+	sot	Sesotho		42.7	42.6	−0.16 [−0.36, +0.07]	−0.24 [−0.46, +0.01]
FLORES+	ssw	Swati		37.9	37.8	−0.03 [−0.29, +0.21]	−0.05 [−0.34, +0.22]
FLORES+	swh	Swahili		58.4	58.2	−0.21 [−0.47, +0.03]	−0.15 [−0.49, +0.18]
FLORES+	taq	taq		11.1	10.9	−0.19 [−0.55, +0.17]	−0.11 [−0.26, +0.03]
FLORES+	tir	Tigrinya		18.7	18.7	−0.09 [−0.31, +0.14]	−0.08 [−0.32, +0.16]
FLORES+	tsn	Setswana		45.1	45.0	−0.04 [−0.26, +0.19]	−0.01 [−0.30, +0.27]
FLORES+	tso	Tsonga		46.9	46.6	−0.23 [−0.46, −0.01]	−0.26 [−0.54, +0.06]
FLORES+	tum	Tumbuka		34.3	34.1	−0.15 [−0.36, +0.09]	−0.08 [−0.31, +0.14]
FLORES+	twi	Twi	target	35.1	34.7	−0.32 [−0.76, +0.02]	+0.11 [−0.50, +0.35]
FLORES+	umb	Umbundu		25.5	25.6	+0.10 [−0.11, +0.35]	+0.18 [+0.05, +0.32]
FLORES+	vmw	vmw		27.1	27.1	−0.05 [−0.22, +0.12]	+0.11 [−0.03, +0.27]
FLORES+	wol	Wolof	target	30.6	30.4	−0.14 [−0.37, +0.06]	−0.10 [−0.32, +0.12]
FLORES+	xho	Xhosa		45.4	44.9	−0.50 [−0.75, −0.26]	−0.85 [−1.17, −0.52]
FLORES+	yor	Yoruba		30.3	30.4	+0.08 [−0.10, +0.28]	+0.16 [−0.07, +0.40]
FLORES+	zgh	zgh		4.1	3.1	−0.97 [−1.36, −0.66]	−0.26 [−0.42, −0.13]
FLORES+	zul	Zulu		48.3	48.0	−0.30 [−0.56, −0.05]	−0.54 [−0.87, −0.20]
NTREX	afr	Afrikaans		63.9	63.9	+0.00 [−0.16, +0.15]	−0.09 [−0.31, +0.11]
NTREX	amh	Amharic		18.0	18.1	+0.05 [−0.15, +0.27]	−0.14 [−0.32, +0.03]
NTREX	bem	Bemba		34.8	34.9	+0.13 [−0.13, +0.39]	+0.20 [+0.05, +0.35]
NTREX	ewe	Ewe	target	31.6	31.7	+0.10 [−0.07, +0.27]	+0.10 [−0.07, +0.25]
NTREX	hau	Hausa		48.6	48.4	−0.17 [−0.37, +0.03]	−0.23 [−0.43, −0.04]
NTREX	ibo	Igbo		44.9	44.8	−0.03 [−0.20, +0.16]	+0.17 [−0.15, +0.51]
NTREX	kin	Kinyarwanda		37.1	37.0	−0.09 [−0.28, +0.11]	−0.14 [−0.31, +0.04]
NTREX	mlg	Malagasy		40.8	40.7	−0.13 [−0.29, +0.04]	−0.22 [−0.40, −0.03]
NTREX	nde	nde		33.0	33.1	+0.14 [−0.01, +0.30]	−0.03 [−0.18, +0.12]
NTREX	nso	Sepedi		32.7	32.8	+0.07 [−0.04, +0.17]	+0.09 [−0.03, +0.19]
NTREX	nya	Chichewa		44.2	43.9	−0.32 [−0.48, −0.14]	−0.30 [−0.49, −0.09]
NTREX	orm	Oromo		27.3	27.7	+0.38 [+0.03, +0.83]	+0.09 [−0.10, +0.29]
NTREX	shi	shi		1.2	1.2	−0.01 [−0.04, +0.03]	+0.00 [−0.02, +0.02]
NTREX	sna	Shona		39.2	38.9	−0.30 [−0.46, −0.13]	−0.32 [−0.48, −0.13]
NTREX	som	Somali		44.2	44.1	−0.13 [−0.35, +0.08]	−0.01 [−0.21, +0.18]
NTREX	ssw	Swati		38.8	38.7	−0.06 [−0.28, +0.14]	−0.22 [−0.42, −0.02]
NTREX	swh	Swahili		58.7	58.6	−0.08 [−0.33, +0.22]	−0.16 [−0.38, +0.14]
NTREX	tir	Tigrinya		17.0	16.9	−0.06 [−0.22, +0.12]	−0.08 [−0.26, +0.11]
NTREX	tsn	Setswana		48.7	48.5	−0.22 [−0.43, −0.03]	−0.21 [−0.43, +0.02]
NTREX	ven	ven		38.8	38.8	−0.02 [−0.20, +0.16]	+0.22 [+0.01, +0.43]
NTREX	wol	Wolof	target	30.4	30.5	+0.15 [−0.06, +0.43]	−0.10 [−0.23, +0.13]
NTREX	xho	Xhosa		42.4	42.1	−0.27 [−0.42, −0.11]	−0.39 [−0.59, −0.19]
NTREX	yor	Yoruba		32.1	32.0	−0.09 [−0.24, +0.07]	−0.12 [−0.29, +0.05]
NTREX	zul	Zulu		43.5	43.2	−0.27 [−0.42, −0.11]	−0.51 [−0.73, −0.29]
MAFAND-MT	amh	Amharic		18.3	18.3	−0.08 [−0.31, +0.15]	−0.19 [−0.46, +0.04]
MAFAND-MT	hau	Hausa		32.8	32.7	−0.15 [−0.43, +0.04]	−0.09 [−0.23, +0.04]
MAFAND-MT	ibo	Igbo		44.9	44.3	−0.60 [−0.80, −0.39]	−0.70 [−0.99, −0.41]
MAFAND-MT	kin	Kinyarwanda		46.4	46.2	−0.24 [−0.53, +0.05]	−0.34 [−0.69, +0.01]
MAFAND-MT	lug	Luganda		39.6	39.6	+0.02 [−0.31, +0.32]	−0.06 [−0.52, +0.22]
MAFAND-MT	luo	Luo		32.6	32.7	+0.06 [−0.19, +0.29]	+0.17 [+0.01, +0.33]
MAFAND-MT	nya	Chichewa		41.0	40.7	−0.22 [−0.45, +0.01]	−0.03 [−0.31, +0.17]
MAFAND-MT	pcm	pcm		53.3	53.5	+0.21 [+0.02, +0.40]	+0.13 [−0.07, +0.34]
MAFAND-MT	sna	Shona		32.9	32.7	−0.24 [−0.43, −0.03]	−0.06 [−0.26, +0.12]
MAFAND-MT	swh	Swahili		51.8	51.5	−0.26 [−0.47, −0.07]	−0.18 [−0.40, +0.04]
MAFAND-MT	tsn	Setswana		53.1	53.0	−0.07 [−0.30, +0.15]	+0.07 [−0.25, +0.39]
MAFAND-MT	twi	Twi	target	34.1	34.1	−0.05 [−0.23, +0.11]	−0.01 [−0.16, +0.17]
MAFAND-MT	xho	Xhosa		51.8	51.3	−0.53 [−0.85, −0.22]	−0.67 [−1.07, −0.24]
MAFAND-MT	yor	Yoruba		33.4	33.3	−0.14 [−0.32, +0.05]	−0.30 [−0.49, −0.12]
MAFAND-MT	zul	Zulu		45.6	45.3	−0.29 [−0.50, −0.08]	−0.53 [−0.79, −0.27]

Table 75: chrF++ of the 385M SFT against A0 (language macro, all rows; 95% paired bootstrap). Language counts differ between benchmarks.

benchmark (languages)	chrF++ 385M SFT	chrF++ A0	delta [95% CI]
FLORES+ devtest (46)	25.52	20.20	+5.31 [+5.13, +5.50]
NTREX-128 (24)	29.25	23.01	+6.24 [+6.02, +6.45]
MAFAND-MT test (15)	35.77	25.56	+10.21 [+9.90, +10.54]
SmolSent (45)	22.53	17.18	+5.36 [+5.14, +5.60]
SALT (5)	35.49	25.13	+10.36 [+9.29, +11.44]
GlotStoryBook (8)	20.22	14.62	+5.59 [+4.92, +6.21]
IbomMT (2)	12.04	9.87	+2.16 [+1.58, +2.74]
FLORES-MOZ (1)	21.54	21.08	+0.46 [−0.72, +1.73]

Table 76: chrF++ of the earlier 270M work's 270M SFT and RL-final (rl2c) re-run in the 385M harness (E-13), against the 385M SFT and RL (mixed best), on rows clean for both generations; language macro.

benchmark (languages)	old 270M SFT	385M SFT	385M SFT - old SFT	old 270M RL (rl2c)	385M RL	385M RL - old RL
FLORES+ devtest (45)	24.69	22.34	−2.35 [−3.25, −1.38]	31.68	32.03	+0.35 [+0.08, +0.62]
NTREX (24)	31.46	27.33	−4.13 [−4.40, −3.85]	35.52	35.55	+0.02 [−0.07, +0.11]
MAFAND (15)	38.26	34.71	−3.54 [−4.22, −2.75]	41.80	40.25	−1.55 [−1.72, −1.39]

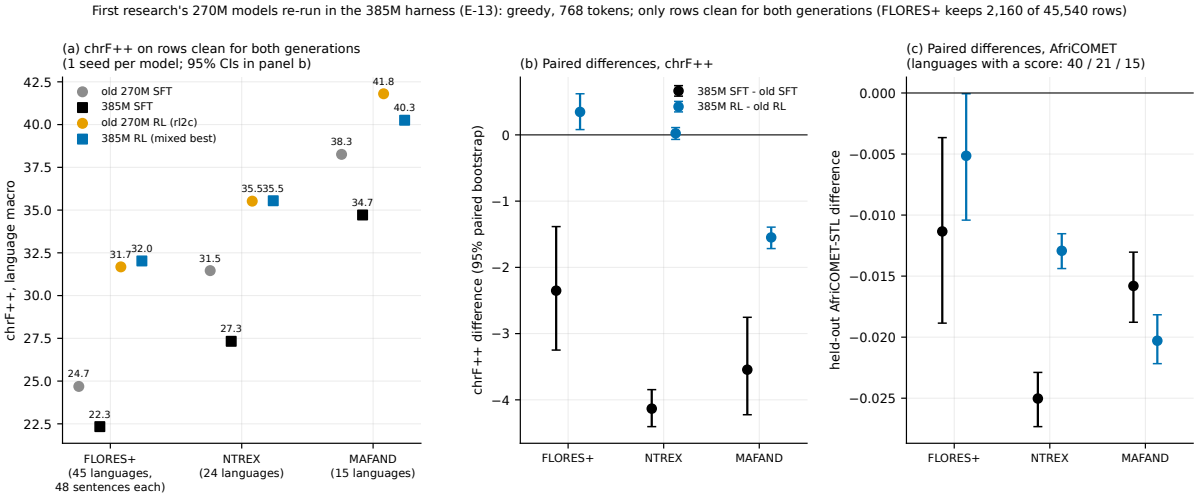


Figure 18: The earlier 270M work's 270M models re-run in the 385M harness (E-13): (a) language-macro chrF++ of the old 270M SFT and RL-final (rl2c) and the 385M SFT and RL (mixed best) on FLORES+ devtest (45 languages, 48 sentences each), NTREX (24) and MAFAND (15), rows clean for both generations only; (b) paired differences with 95% bootstrap over sentences (Poisson weights; 1,000 resamples); (c) the same for held-out AfriCOMET-STL (languages with a score: 40 / 21 / 15). One seed per model. Source: `reports/data/e13_old270m_same_harness_2026-11-04/evidence_e13.json`.

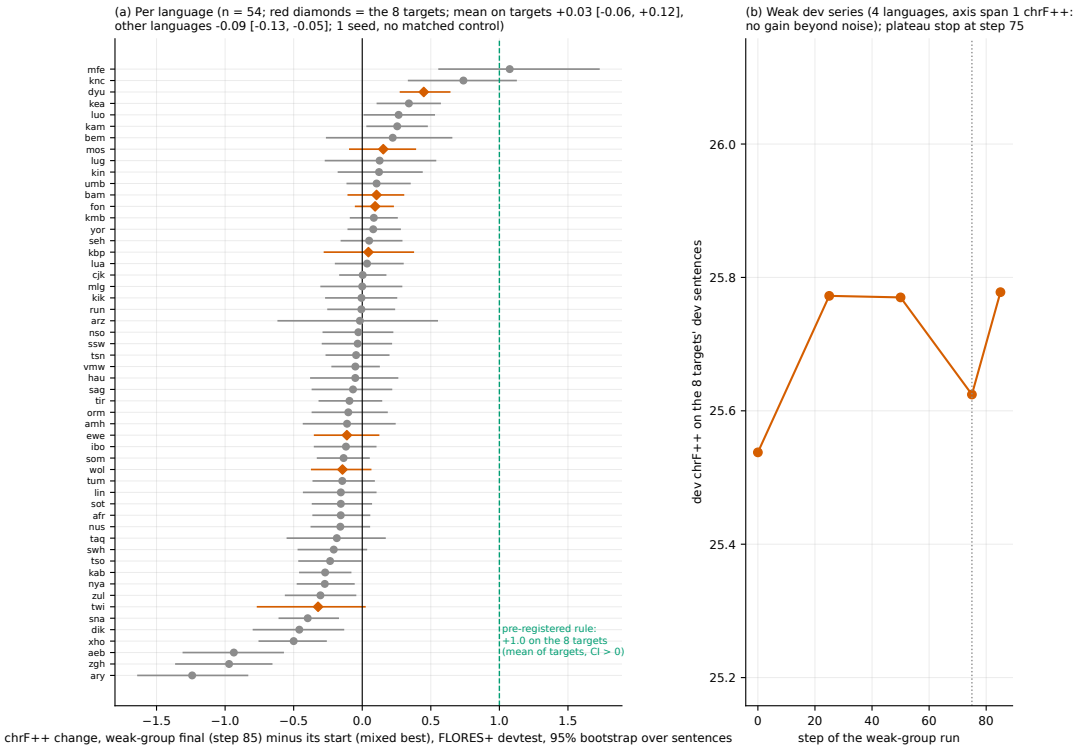


Figure 19: (a) Continued RL on eight weak West-African languages (mos fon kbp bam wol ewe twi dyu; 85 steps, 2.1 GPU-hours): chrF++ change of the final (step 85) against its start, the mixed best, per FLORES+ language (n = 54; 1,012 sentences each; 95% bootstrap over sentences; one seed, no matched control). Red diamonds are the targets: mean +0.03 [−0.06, +0.12] against the pre-registered +1.0; the 46 other languages move by −0.09 [−0.13, −0.05]. (b) The run's own dev series on the targets' dev sentences (axis span 1 chrF++; dev evaluation noise is sd 0.09 to 0.10). Sources: `reports/data/eval_weak_group_2026-11-03/scores/bootstrap_by_language.csv`, `research_archive_2026-11-03/clean_rerun_misc/runs_385m_weak/mixed/metrics.jsonl`.

K RL chain: stage summary, B1 final and training records

Table 78: Stage chain summary (not all rows are FLORES+ stages).

stage	steps run	cost	result on FLORES+ chrF++ (vs previous stage)
SFT-1 / SFT-2 (A0 → 385M)	1,500 + 31,353	44.8 h for stage B (A100)	22.85 (SFT, 54 languages)
continued-SFT control (E-03; not an RL stage)	6,000	8.4 h (5.05 s/step), RL chain 8.66 h	FLORES+ 25.38 (+2.53 over the SFT [+2.35, +2.71]); 93 cells 29.97 against 34.55 for the mixed best
RL stage A	400	15,920 s (4.42 h)	30.81 (+7.96)
RL B1	135 (best step 50)	10,479 s	31.35 (+0.54)
RL mixed	60	4,786 s	31.66 (+0.31)
weak-group RL (8 languages)	85 (plateau stop at 75)	2.1 GPU-h	targets +0.03 [−0.06, +0.12]; other languages −0.09 [−0.13, −0.05]
science SFT	283 (65,536 tokens)	17 min	AfriScience chrF +1.2; SSA-COMET-QE flat
science RL	160 steps executed (cap 150 in the yaml, extended to 190 at step 120; dev plateau at step 150 plus 10 wind-down steps; `summary.json` stop reason `dev_plateau`; best at step 75)	not stated in the evaluation report	AfriScience test (6 languages), final vs start: chrF +0.46 [+0.27, +0.65], held-out AfriCOMET +0.0053 [+0.0036, +0.0071], SSA-COMET-QE +0.0041 (reward family); NTREX +0.42, MAFAND +0.10 chrF++; no document-level gain (not a FLORES+ stage)
science RL without the science SFT (E-14)	160 (150 + 10 wind-down; one OOM restart from step 20; final only)	about 5 h wall-clock (15:18 to 20:17 UTC; 14,395 s in the last process)	AfriScience test chrF 45.66 (start 45.40; science RL from the science SFT 47.10); 93-cell chrF++ +0.13 over the mixed best (not a FLORES+ stage)
long-document RL	80 (best at step 50; no extension)	7.7 h wall-clock, of which about 4.5 h a crash loop (Appendix P)	whole-document finished +0.125, loop −0.113, chrF +3.4; chunked and sentence level unchanged (not a FLORES+ stage)

Table 79: B1 final (step 135) against B1 best (step 50) and neighbouring stages; chrF++, language macro, 95% paired bootstrap.

comparison	FLORES+ chrF++	NTREX chrF++	MAFAND chrF++	all 93 cells chrF++
B1 final minus B1 best	+0.04 [+0.00, +0.08]	−0.00 [−0.05, +0.05]	−0.09 [−0.15, −0.01]	+0.01 [−0.02, +0.04]
B1 final minus stage A	+0.58 [+0.53, +0.64]	+0.30 [+0.24, +0.36]	−0.09 [−0.17, +0.00]	
mixed best minus B1 final	+0.26 [+0.22, +0.30]	+0.19 [+0.14, +0.23]	+0.05 [−0.01, +0.12]	

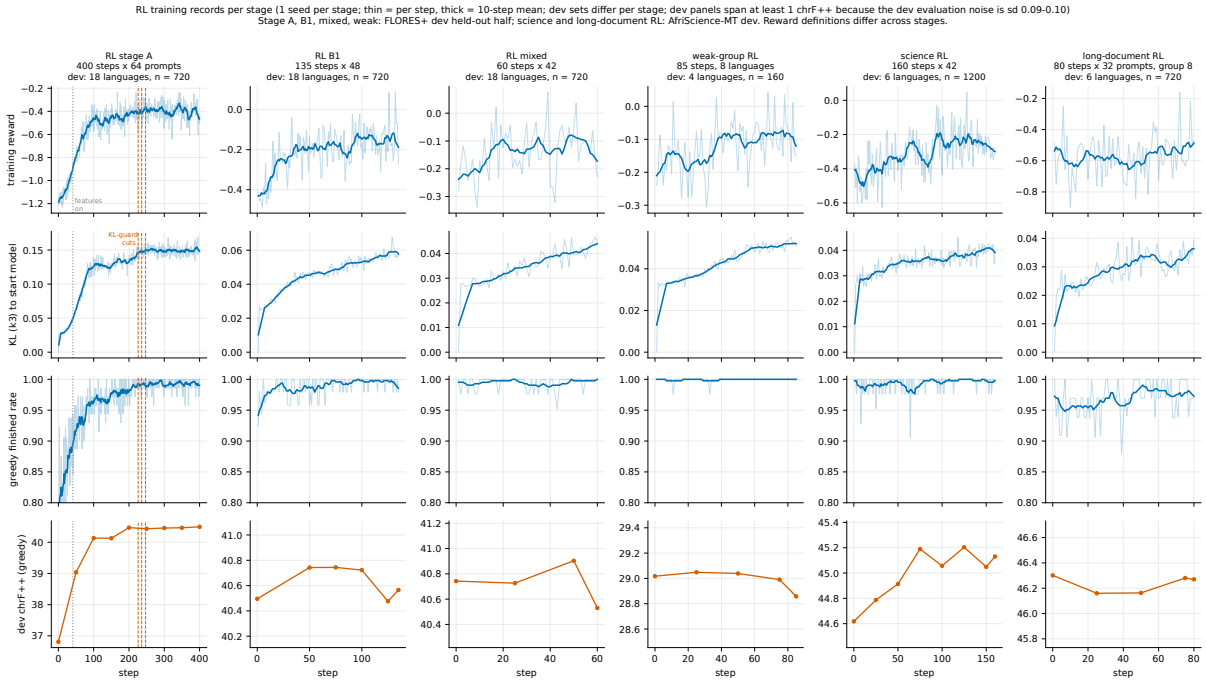


Figure 20: Training records of every RL stage from the run logs (one seed per stage): training reward, KL (k3) to the stage's start model, finished rate of the greedy group member, and dev chrF++ against the step (thin lines per step, thick lines 10-step means). Stage A: dotted line at step 41 (four features on), dashed lines at the three KL-guard cuts (226, 236, 247). The dev sets differ by stage (FLORES+ dev held-out half for stage A, B1, mixed and weak; AfriScience-MT dev for science and long-document RL; the number of languages and sentences is in each title) and the reward definitions differ, so columns are not comparable; dev panels span at least 1 chrF++ because the dev evaluation noise is sd 0.09 to 0.10. The long-document RL log shows the 80 planned steps; step 25 was evaluated repeatedly after the crash restarts of Appendix P and the last evaluation is shown (steps 21 to 25 were repeated by the restarts). Sources: `research_archive_2026-11-03/runs_385m/{A,B1,mixed}/metrics.jsonl`, `clean_rerun_misc/runs_385m_{weak/mixed,sci/final,long/final}/metrics.jsonl`.

L Family, script and resource level (exploratory, fragile)

Table 81: RL gain by Glottolog family, regression family and script (FLORES+ devtest, 54 languages, chrF++): mean SFT and RL scores, mean gain with a bootstrap interval over languages (none for one-language groups), and member languages. Exploratory and observational; the main result changes with the languages included.

grouping	group	n	SFT	RL	RL gain [bootstrap CI over languages]	languages
glottolog_top_family	Nilotic	3	12.9	24.5	+11.58 [+10.30, +13.13]	dik luo nus
glottolog_top_family	Saharan	1	3.8	14.2	+10.41	knc
glottolog_top_family	Indo-European	3	36.9	47.1	+10.23 [+1.01, +20.27]	afr kea mfe
glottolog_top_family	Atlantic-Congo	33	26.2	35.3	+9.14 [+7.58, +10.79]	bem cjk ewe fon ibo kam kbp kik kin kmb lin lua lug mos nso nya run sag seh sna sot ssw swh tsn tso tum twi umb vmw wol xho yor zul
glottolog_top_family	Mande	2	14.6	23.1	+8.42 [+7.78, +9.06]	bam dyu
glottolog_top_family	Afro-Asiatic	11	13.4	20.3	+6.89 [+4.29, +9.89]	aeb amh ary arz hau kab orm som taq tir zgh
glottolog_top_family	Austronesian	1	39.0	44.7	+5.68	mlg
family_reg	Nilotic	3	12.9	24.5	+11.58 [+10.30, +13.13]	dik luo nus
family_reg	Indo-European (creoles, Afrikaans)	3	36.9	47.1	+10.23 [+1.01, +20.27]	afr kea mfe
family_reg	Atlantic-Congo: Narrow Bantu	24	27.4	37.3	+9.91 [+7.96, +11.71]	bem cjk kam kik kin kmb lin lua lug nso nya run seh sna sot ssw swh tsn tso tum umb vmw xho zul
family_reg	Mande	2	14.6	23.1	+8.42 [+7.78, +9.06]	bam dyu
family_reg	Other (Austronesian, Saharan, Songhay, Ta-Ne-Omotic)	2	21.4	29.5	+8.05 [+5.68, +10.41]	knc mlg
family_reg	Atlantic-Congo: non-Bantu	9	23.1	30.2	+7.06 [+4.76, +9.61]	ewe fon ibo kbp mos sag twi wol yor
family_reg	Afro-Asiatic	11	13.4	20.3	+6.89 [+4.14, +9.63]	aeb amh ary arz hau kab orm som taq tir zgh
family_branch	Indo-European: Romance (Portuguese/French-lexifier creoles)	2	23.7	38.6	+14.84 [+9.42, +20.27]	kea mfe
family_branch	Atlantic-Congo: Cameroun-Ubangian	1	19.7	34.3	+14.62	sag
family_branch	Nilotic	3	12.9	24.5	+11.58 [+10.30, +13.13]	dik luo nus
family_branch	Afro-Asiatic: Cushitic	2	23.8	34.9	+11.18 [+4.24, +18.12]	orm som
family_branch	Saharan	1	3.8	14.2	+10.41	knc
family_branch	Atlantic-Congo: Narrow Bantu	24	27.4	37.3	+9.91 [+7.90, +11.73]	bem cjk kam kik kin kmb lin lua lug nso nya run seh sna sot ssw swh tsn tso tum umb vmw xho zul
family_branch	Atlantic-Congo: Gur	2	14.3	23.8	+9.54 [+8.70, +10.37]	kbp mos
family_branch	Mande	2	14.6	23.1	+8.42 [+7.78, +9.06]	bam dyu
family_branch	Afro-Asiatic: Berber	3	7.1	13.9	+6.79 [+1.78, +10.34]	kab taq zgh
family_branch	Afro-Asiatic: Chadic	1	39.9	46.5	+6.58	hau
family_branch	Austronesian	1	39.0	44.7	+5.68	mlg
family_branch	Atlantic-Congo: Benue-Congo	2	29.6	35.2	+5.67 [+2.33, +9.01]	ibo yor
family_branch	Afro-Asiatic: Semitic	5	7.8	13.1	+5.29 [+2.36, +8.22]	aeb amh ary arz tir
family_branch	Atlantic-Congo: Kwa Volta-Congo	3	24.6	29.5	+4.94 [+2.07, +7.53]	ewe fon twi
family_branch	Atlantic-Congo: North-Central Atlantic	1	26.9	30.6	+3.72	wol
family_branch	Indo-European: Germanic (Afrikaans, English-lexifier creoles)	1	63.1	64.1	+1.01	afr
script	LATIN	48	24.8	34.2	+9.32 [+8.03, +10.68]	afr bam bem cjk dik dyu ewe fon hau ibo kab kam kbp kea kik kin kmb knc lin lua lug luo mfe mlg mos nso nus nya orm run sag seh sna som sot ssw swh taq tsn tso tum twi umb vmw wol xho yor zul
script	ETHIOPIC	2	13.6	22.7	+9.13 [+8.26, +10.00]	amh tir
script	ARABIC	3	4.0	6.7	+2.73 [+1.36, +4.59]	aeb ary arz
script	TIFINAGH	1	2.3	4.1	+1.78	zgh
our_group	Creole / Pidgin	1	18.9	39.2	+20.27	kea
our_group	Nilo-Saharan	3	12.9	24.5	+11.62 [+10.41, +13.13]	knc luo nus
our_group	Bantu	23	28.1	37.9	+9.81 [+7.80, +11.88]	bem kam kik kin kmb lin lua lug nso nya run seh sna sot ssw swh tsn tso tum umb vmw xho zul
our_group	Afroasiatic	6	21.8	31.4	+9.59 [+6.54, +13.48]	amh hau kab orm som tir
our_group	Mande	2	14.6	23.1	+8.42 [+7.78, +9.06]	bam dyu
our_group	Atlantic-Congo (non-Bantu)	9	23.1	30.2	+7.06 [+4.51, +9.69]	ewe fon ibo kbp mos sag twi wol yor
our_group	family unlabelled	8	7.5	13.8	+6.28 [+3.61, +9.22]	aeb ary arz cjk dik mfe taq zgh
our_group	Other (Indo-European, Austronesian)	2	51.1	54.4	+3.34 [+1.01, +5.68]	afr mlg
africomet	1	40	27.1	35.9	+8.83 [+7.39, +10.28]	afr amh bam bem ewe fon hau ibo kab kam kbp kik kin kmb lin lua lug luo mlg mos nso nus nya orm run sna som sot ssw swh tir tsn tso tum twi umb wol xho yor zul
africomet	0	14	10.8	19.6	+8.73 [+6.18, +11.38]	aeb ary arz cjk dik dyu kea knc mfe sag seh taq vmw zgh

Table 80: RL gain (mixed best minus SFT, FLORES+ devtest chrF++, 54 languages) by Glottolog family, script and regression family; bootstrap over languages.

grouping	group	n	mean SFT chrF++	mean gain [bootstrap CI over languages]
Glottolog top family	Atlantic-Congo	33	26.20	+9.14 [+7.58, +10.79]
	Afro-Asiatic	11	13.45	+6.89 [+4.29, +9.89]
	Nilotic	3	12.87	+11.58 [+10.30, +13.13]
	Indo-European (creoles, Afrikaans)	3	36.85	+10.23 [+1.01, +20.27]
	Mande	2	14.65	+8.42 [+7.78, +9.06]
regression family	Narrow Bantu	24	27.36	+9.91 [+7.96, +11.71]
	other Atlantic-Congo	9	23.11	+7.06 [+4.76, +9.61]
script	Latin	48	24.84	+9.32 [+8.03, +10.68]
	Ethiopic (amh tir)	2	13.55	+9.13 [+8.26, +10.00]
	Arabic (aeb ary arz)	3	4.00	+2.73 [+1.36, +4.59]
	Tifinagh (zgh)	1	2.29	+1.78

M Contamination audit and follow-ups

6.4 Contamination. The audit scanned both sides of 25,911,095 training samples (A0 packs and the six stage-B shards) against all benchmark rows: **no benchmark sentence of 3 or more words is in the training text**; FLORES+ devtest 0 of 54,648 rows, NTREX 120 of 47,928 (0.25%), MAFAND 94 of 20,009 (0.47%), SmolSent 45 of 38,833, GlotStoryBook 15 of 3,426, all one- or two-word strings; the FLORES+ dev train half is found in 23,706 of 23,760 rows (positive control); `all`, `clean`, `no_pair` and `clean_strict` macros agree within 0.01 chrF++.

Follow-ups (CPU, 2026-11-04).

- **Phrase level.** A lexical fuzzy scan of 56,975,939 pair rows and 4,277,612 instruction rows (reachable pair and instruction files, before gating and packing) finds no benchmark string of 8 or more tokens as a contiguous run. Strings of 4 to 7 tokens do occur inside longer training text (English side: MAFAND 134 of 17,428 strings, GlotStoryBook 30 of 898, NTREX 18 of 1,973, SALT 12 of 498; FLORES+ devtest, SmolSent, IbomMT and FLORES-MOZ 0); the audit's run rule and the intake filter test such short strings only for whole-sentence equality, so "no benchmark sentence of 3 or more words" holds for whole sentences, not for phrases. 263 (string, benchmark) cases of 8+ token English strings have a character-5-gram Jaccard ≥ 0.5 with some training text (MAFAND 184, SALT 27, NTREX 19, GlotStoryBook 15, SmolSent 6, FLORES+ devtest 4 of 1,012). These are candidates, not proof of leakage; their effect on scores was not measured.
- **Prompts of the RL.** Stage A drew 6 flagged prompts of 28,120 (0.021%); 6 of 1,345 prompts drawn for the seven languages whose prompt files had not been filtered (0.45%), all distinct, in steps 72-364, source side only (this replaces the earlier estimate of about 0.15%). The pool of the long-document RL (241,679 rows) has 0 rows flagged by the protected index.
- **AfriScience-MT train versus the SFT text.** 305 of 34,007 train rows (0.9%; hau 64, yor 116, zul 125) have their African-language sentence verbatim in the A0 pair data, all from `HuggingFaceFW/finetranslations`; 3 rows (yor 2, zul 1) as a pair. The packed shards were not scanned. The test papers are untouched (0 of 5,820 rows).
- **A0 text.** The files support that A0's pack set is the first 138,873 packs of the audited g0 (identical per-language pack counts for 84 languages); pack content was not hash-compared.
- **Not covered:** embedding or MT-round-trip paraphrase contamination, Gemma-3 pretraining, packed-shard coverage of the AfriScience train overlap and of the phrase scan (not run: these scans need the data box; the Limitations section item 15).

Benchmarks were evaluated many times; checkpoint choice used only the dev set (the Limitations section).

N Metric independence: full table

Table 82: Metric independence (full). IND = not in any reward; RWD = in the reward; LOOP = optimised by construction.

metric	in the RL reward?	use in this paper
`masakhane/africomet-qe-stl-1.1` (reference-free)	yes	not used as evidence
`McGill-NLP/ssa-comet-mtl` (reference-free; reference-based in the science RL)	yes	not used as evidence of RL gains
`McGill-NLP/ssa-comet-qe` (AfriScience-MT headline)	no, but the same family as the reward member	headline for SFT-only comparisons; for RL models shown only next to plain chrF and held-out AfriCOMET (all stages in Section 5.6)
TranslatePsy quality filters (AfriCOMET, SSA-COMET, MetricX)	not a reward; they filtered the training data of the E-03 control	neural metrics favour the control; chrF and chrF++ are the evidence (Section 5.4)
consensus utility (sample agreement in chrF++)	yes	chrF++ gains are partly in the optimised family
NLLB-3.3B pseudo-reference chrF++	yes, 5 languages (kea kbp nus bam kab)	chrF++ on those 5 languages is not independent; relevant to the NLLB comparison (Section 5.5)
chrF++ against a human reference, weight 0.35	yes, 7 languages (reference rows only; AfriScience TRAIN pairs)	chrF++ on those 7 languages can be partly trained; the test sets differ
language-ID, copy, repetition, length, end-of-turn guards	yes	finished and loop rates are optimised by construction
independent degeneration detector (lzma residual, repeated character 8-grams, repeated word 3-grams; thresholds from the 0.1% tail of human references)	no	measures the same idea as the loop guard with different code and constants (Section 5.3)
NLLB-218 language identification (`facebook/fasttext-language-identification`)	no	independent check of the language guard; evaluation use only (cc-by-nc-4.0)
`masakhane/africomet-stl-1.1` (reference-based)	no (held out; monitor and alarm)	the independent neural check; same family as the QE member
chrF / chrF++ / spBLEU against human references of the benchmarks	no	the main lexical evidence

O Disclosures: full statement

Rule: data and signals that were actually used are stated, including those that are non-commercial or NLLB-derived. `DISCLOSURES.md` is the source of this section and holds the full tables.

O.1 Training data and licences

- **NLLB-derived share of the SFT.** NLLB-mined bitext (OPUS-NLLB, 43 languages, 13.46 M rows, plus a Fulah NLLB-mined set of 0.45 M rows) is **14.4% of the 3.990 B stage-B pack tokens** (13.3% of the 4.109 B executed; 10.84% of the 5.043 B tokens drawn in the whole SFT); adding NLLB-Seed (tzm) and the aligned part of the Aya collection (NLLB-3.3B outputs; 169 M tokens in 12 languages) gives **18.7%** (746 M tokens; executed 17.2%; 14.36% of the whole SFT); adding the machine-translated Tumbuka instruction sets (system not stated, possibly NLLB) gives **22.8%** (909 M tokens; 17.86% of the whole SFT). By rows, 13.91 M of 42.31 M stage-B translation pairs (32.9%) are NLLB-mined. A0 contains none of the three. The authors allowed NLLB-mined rows with a per-language cap of 150k, ODC-BY with special attribution. NLLB models and outputs carry CC-BY-NC terms and the authors cannot use NLLB for production translation; a licence statement must accompany any release of the 385M weights. Not counted and needing a legal decision: the `*_mt560` OPUS sets (69% of the main pair output of the clean rebuild, 16.2 M of 23.4 M pairs, mostly Jehovah's Witnesses and Bible text, licence unknown, labelled NLLB-derived by an old converter docstring); `African-Languages-Lab/multi-open` (3,748,964 pairs, "license: other"); AfroLingu-MT (gated, authors' exception); other machine-translated instruction sets with unstated systems (`african_ultrachat`, `kiya_sft`, `alpaca_*`, `amharic_*`); all four families are in the instruction pack stream (`packed_instr`: every source of the instruction intake except `muri_it`, 4,190,976 samples, 1.232 B tokens; `reports/intake2_instr_packs_2026-11-01.md`); `african_ultrachat` is UltraChat machine-translated into 11 languages, `alpaca_*` (twi, swh, zul, wol, som) are translated Alpaca sets, `amharic_*` and

kiya_sft (om, yo, sw, am rows) are machine-translated sets, and the translation system is not stated in the dataset cards or in the intake records; the packs carry no source field, so the tokens of each source drawn in stage B are not recorded.

- **Unspecified-licence sources** in the main output: KwekuAinoo/eng_twi 551,358; sentence-transformers/parallel-sentences-opus-100 210,479; UBC-NLP/AfroLingu-MT 207,796; cibfaye/wolof-english-bible 11,415; webfaq_v2; bambara_lm_qa (381k, 83% of Bambara instruction rows). Religious text dominates many low-resource languages (28 of the 81 RL languages are religious-only in the post-editor selection); the Efik NT sheet (7,936 verse pairs) has unconfirmed provenance and licence.
- **IbomMT** is CC-BY-NC-4.0: a disclosed non-commercial exception in the SFT (once) and a human-reference source of the RL reference domain (efi, ibb).
- **Quarantined and not used:** yigagilbert/synthetic-parallel-external, fiifinketia/pristine-twi-english, muri-it, the Aya parallel pairs.
- **Gemma base model:** google/gemma-3-270m-it; the Gemma terms of use apply to the derived weights; pretraining text is unknown, so exposure of benchmarks through pretraining cannot be excluded.
- **FLORES+ dev train half** (440 ids; 23,706 of 23,760 rows found in the audit) is in the SFT on purpose; the held-out half (557 ids) is protected.

O.2 RL prompts and reward-side data

- General prompts: FineWeb-Edu (ODC-By) and Wikipedia (CC BY-SA). News prompts (30% of the draw): BBC News 2017-2025 (RealTimeData/bbc_news_alltime, **no licence declared, BBC copyright**), permutans/c4-bbc-news (ODC-By label inherited from C4, BBC copyright applies), Wikinews (CC BY 2.5), Global Voices (CC BY 3.0 per the sister card), Wikivoyage (CC BY-SA 4.0; 188 prompts drawn in stage A) and SetFit/bbc-news (37 pool rows, none drawn, licence not recorded). The prompts are RL **input** text only; the source is never a training target.
- Seven new-language prompt files of stage A (tll mfe guw lub crs zne lue) had not been filtered; stage A drew 6 flagged prompts of 28,120 (0.021%), 6 of 1,345 for those languages, source side only. B1 was restarted from its step-20 checkpoint with the filtered pool.
- Reference domain (stage A from step 41, B1, mixed, science RL): human pairs from AfriScience-MT **train** (Apache-2.0; amh hau lug nso yor zul) and IbomMT train (CC-BY-NC-4.0; efi, ibb), anchors for 7 languages only. Test papers are in no pool. 0.9% of the AfriScience train rows have their African-language sentence in the A0 pair data (Appendix M).
- **NLLB-3.3B pseudo-references (CC-BY-NC) are a reward signal** for kea kbp nus bam kab (weight 0.25, max 0.35), reward only; no NLLB text is a training target in RL.
- Reward models: masakhane/africomet-qe-stl-1.1, McGill-NLP/ssa-comet-mtl; held-out monitor masakhane/africomet-stl-1.1 (apache-2.0 verified); GlotLID language guard. Licences as declared on the Hub (read 2026-11-05): McGill-NLP/ssa-comet-mtl and ssa-comet-qe apache-2.0; masakhane/africomet-qe-stl-1.1 apache-2.0; cis-lmu/glotlid license: other with license_name: apache-2.0-plus-notices (card: Apache License 2.0 plus notices).
- Science stages: science RL prompts clean_sci_v2 (PMC, arXiv CC BY/CC0, AfriScience-MT train English); **LibreTexts was excluded** from the science RL (licence conflict CC BY against CC BY-NC-SA, 10,257 prompts). **LibreTexts is not absent from the long-document RL pool**: 3,428 common-pile/libretexts rows (17.8% of science_b2048). The rows were removed before the stage started; the long-document RL did not use them. The E-14 run (science RL started from the mixed best) used the same pool and reward as the executed science RL. Science SFT: AfriScience-MT train (Apache-2.0; 28,750 windows) plus replay cut from the general SFT shards (so the NLLB, Aya and Tumbuka shares apply to the replay); 54 further catalogue candidates were scanned and none used.

O.3 TranslatePsy: used only in a research control

qvac/TranslatePsy-AfriSLM-Synthetic-Mix (Gritta et al.; CC-BY-NC-4.0; targets are facebook/nllb-200-3.3B translations of MADLAD-400 sentences, filtered by AfriCOMET, SSA-COMET and MetricX) was authorised by the authors on 2026-11-04 **for the matched-compute**

continued-SFT control (E-03) only. Its English-to-X rows (19 languages: afr amh hau ibo kin lin lug mlg nya orm sna som sot swh tsn wol xho yor zul) enter that control run, scanned against the protected index (any hit on either side removed; 0.67% of the 950,000-row sample); **no TranslatePsy row is in the SFT, any RL stage, the science or long-document stages, or any released weight.** The control checkpoints are research artefacts (CC-BY-NC-4.0 and NLLB terms), are not pushed to any public repository and are not used in production. The comparison RL against control must state two things: the TranslatePsy quality filters are the reward family of the RL, so neural metrics favour the control while chrF and chrF++ do not; and TranslatePsy targets are NLLB-3.3B-like, so a good control result partly means "distilling NLLB". Status: run (preparation `reports/sft_control_prep_2026-11-04.md`; execution record and result `reports/e03_sft_control_2026-11-05.md`): 277,900 pairs per language for the 19 languages (5.28 M pairs, 96,138 packs, each seen once) after removing every row with a hit against the protected index; the control checkpoint is a research artefact under CC-BY-NC-4.0 and the NLLB terms, kept on the authors' storage, never released and never used in a released model. The comparison states that the TranslatePsy filters are the reward family of the RL, so AfriCOMET and SSA-COMET-QE favour the control, and that its targets are NLLB-3.3B outputs (Section 5.4).

O.4 Analysis decisions and data-audit facts

- The 385M SFT text contains no benchmark sentence of 3 or more words; one- and two-word strings, numerals and names occur (NTREX 0.25%, MAFAND 0.47%, SmolSent 0.12%, GlotStoryBook 0.44% of rows); phrase-level exposure is in Appendix M.
- The RL dev set is a FLORES+ dev subset (720 sentences, 18 languages). The weak-language choice used devtest-derived per-language tables (selection on a test set).
- Benchmarks were evaluated many times; checkpoint choice used only the 720-sentence dev set. The science alpha rule was written before the headline metric was fixed; the headline then changed from SSA-COMET-MTL (reference-based) to SSA-COMET-QE after the calibration. The post-hoc RL-versus-SFT decomposition ran after the results were known.
- The earlier 270M pipeline (not the clean 385M rerun) trained on text that held the English side of 95% of FLORES+ devtest; the old 270M models; their numbers appear only on their clean rows (48 sentences per language on FLORES+) and are descriptive; the earlier 270M work's FLORES+ results are not reused.
- FLORES+ and FLORES-200 references differ in places; the NLLB comparison uses published FLORES-200 numbers.

O.5 Process disclosures

- **Gates overridden by the authors.** The RL preflight gate failed (gradient norm 3.40 > 3.0, greedy loop 0.109 > 0.02, greedy finished 0.906 < 0.97) and was overridden; the stage-A gate failed (max KL 0.169 > 0.15, three KL-guard cuts, greedy loop 0.016 > 0.01) and B1 was started on the authors' instruction.
- **Mid-run changes.** Stage A was stopped at step 40 and resumed with the four features; B1 was restarted at its step-20 checkpoint (filtered prompt pool, reference domain, oversample 1.0, KL guard 0.25) and again at step 40 (MBR guide, dev early stop); the mixed stage was restarted at its step-20 checkpoint with the patch-3 monitor, and the authors loosened the hacking alarm (chrF++ 0.3 $\rightarrow$ 0.75; AfriCOMET 0.005 $\rightarrow$ 0.03). Stage A's KL guard cut the learning rate three times and it never recovered.
- **Start checkpoints.** Mixed started from B1 best (step 50), not B1 final; B1 final exists on the Hub and was evaluated after the fact (E-20): it equals B1 best within noise (+0.01 chrF++ [-0.02, +0.04] over 93 cells). The weak-group run never produced a `best/` checkpoint; its final (step 85) was evaluated. The `masakhane/africomet-stl-1.1` monitor influenced early stopping in the mixed stage and later (never a reward).
- **Incident.** The mixed stage was auto-started once by a process outside the launch agent, stopped within a minute without a step, and relaunched from B1 best. A first science-SFT pipeline pass wrote an empty alpha and launched the science RL with an invalid model path, which stopped itself at start. The GPU idled for about 2 h 10 min between stage A and B1. The E-14 run (science RL from the mixed best) had one CUDA out-of-memory in the COMET reward worker at about step 38; the launcher lowered the

micro-batch from 12,288 to 6,144 tokens and resumed from the step-20 checkpoint, so steps 21 to 38 were replayed (same prompt schedule, vLLM sampling not bit-reproducible); `metrics.jsonl` holds two records for those steps and the later one is used.

- **Science RL length.** The science-RL yaml names 150 steps a hard cap, but the trainer's extension rule raised the cap to 190 steps at step 120 while dev metrics were still improving; the run then stopped on a dev plateau at step 150 and, after a 10-step wind-down, executed 160 steps (`summary.json`). The long-document RL has the extension rule off.
- **Checkpoint storage.** Stage checkpoints were written to RAM-backed `/dev/shm`; weights are in anonymised repositories, optimiser states are not.
- **Evaluation-only tool under a non-commercial licence:** the NLLB-218 language-ID model (cc-by-nc-4.0) was used for measurement only.
- **Compute.** A100 80 GB (vast.ai) for SFT stage B (44.8 h), RL (8.66 GPU-h for A + B1 + mixed), evaluation and the science stage; smaller boxes for data preparation and the post-editor. GPU-hours by component, estimated from logged durations, are in Appendix (compute table): the sized components sum to about 94 A100-hours; A0, SFT-1 and several evaluation passes are not sized; no cost figure is reported.

O.6 Intended use and risks

Outputs are machine translations of unverified quality for languages without any human evaluation, with documented failure modes: Arabic dialects, Tamazight, Dinka, Tamasheq and Kanuri are not usable (chrF++ 2 to 14 after RL), RL increases English output there, and the model should be used sentence by sentence, not on whole documents. Religious text dominates the data of many low-resource languages, which may shift register and vocabulary toward it. The model must not be presented as a replacement for NLLB-class systems: it is 3.26 chrF++ below NLLB-3.3B on 43 shared languages. The authors' use case is production translation; the licences of NLLB and TranslatePsy prevent using those systems there, which is why we state exactly where NLLB-derived text and signals entered this model.

P Reproducibility statement

- **Code and configs.** `gemma_mt/` (trainer `train_sft.py`, RL `gemma_mt/rl/`, evaluation scripts); SFT configs `reports/data/e23_hygiene_2026-11-04/sft_configs/`; RL configs `.../rl_configs/` (note: the archived stage-A `config.json` is the resumed configuration); reports and per-language tables under `reports/` and `reports/data/` as keyed in `CLAIMS_LEDGER.md`; analysis scripts for every new section in `reports/data/{nllb_comparison_rl_models,family_script_resource_analysis,degeneration_audit_independent,stage_A_feature_split,contamination_followups_385m,e23_hygiene}_2026-11-04/`.
- **Seeds and hyper-parameters.** Seed 1234 throughout; SFT: Muon, peak 5e-5 (SFT-1, shard 1) and 3e-5 (shards 2-6) with cosine floors 50% (SFT-1), 80% (shards 1-5) and 10% (shard 6), 131,072 tokens per step (Section 3); RL stages A to mixed: Muon, 2e-6, group 16, KL 0.05 (reference = each stage's start model), guard 0.15/0.25, temperature 1.0; **science RL: Muon, 1.0e-6, KL 0.1, group 16, guide 0.25, 160 steps executed (yaml cap 150, extended to 190, stopped by dev plateau); long-document RL: Muon, 1.0e-6, KL 0.1, group 8, no guide, 80 steps $\times$ 32 prompts** (Table 74); science SFT AdamW 3e-6, alpha 0.5. The science-RL yaml is in `reports/data/rl_configs_as_executed_2026-11-04/sci_rl_F_final.yaml`.
- **Hardware and software.** One A100 80 GB; vLLM 0.32.0 bf16 greedy, 768 new tokens, for every headline number. Outputs are not bit-identical across engines (9.6% identical between vLLM bf16 and HF bf16 on 240 sentences); a reader reproducing with another engine should expect small, unsystematic differences (240 sentences, HF bf16 minus vLLM bf16: +0.54 [−0.67, +1.79] chrF++ for the SFT and −0.23 [−0.73, +0.23] for the RL model; the SFT in fp32 against bf16 on the full set: −0.04 [−0.13, +0.07] over 93 cells, E-09b, which supersedes the earlier +0.24 [−0.07, +0.55] on 4,000 sentences and +1.81 on 240 sentences). sacreBLEU 2.6.0 and the chrF++/spBLEU settings are given in Section 4; the full signature strings are not recorded; exact commands per table are in the archived scripts named in

the ledger.

- **Weights and branches.** RL repository https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl: branch main (mixed best, the final general RL model), latest, stage-a-b64 (stage A step 200), stage-a-final (step 400), stage-b1-b256 (B1 best, step 50), stage-b1-final (step 135), stage-g-weak-latest, science-sft, science-sft-alpha1, science-rl, science-rl-latest, science-rl-final, long-rl, long-rl-latest (long-document RL checkpoints) and tag rl-step-50. SFT final: https://huggingface.co/AletheiaAING/gemma-0.4B-translate-sft (branch main, step 32,853). Experiment weights: https://huggingface.co/AletheiaAING/gemma-0.4B-translate-rl-experiments, branches e04-seed2-step100, e04-seed3-step100, e04-a_base-step100, e04-c_noref-step100, e04-d_noguide-step100, e14-science-rl-only-final. The control checkpoint of E-03 is not published (CC-BY-NC data). No licence field or model card is declared on these repositories **[OWNER: confirm release licence]**. Run configs, logs, per-sentence outputs and scripts of E-02, E-04, E-11 and E-14 are in the research archive and the reports directory named in the ledger.
- **Data.** The decontaminated SFT and RL prompt data and the protected index (protected-2026-10-28.3) are in anonymised storage named in the reports and released on acceptance; the NC and unspecified-licence parts cannot be redistributed as they are (Appendix O); TranslatePsy rows are not released (Appendix O). The state, prompts, eval and reports repositories were private at the last check (Hub API 2026-11-04). [OWNER: which parts of the data are released.]
- **Long-document RL incident.** The stage took 7.7 h of wall-clock time (10:03 to 17:44 UTC), of which about 4.5 h were a crash loop: the trainer stopped (ENOSPC, root disk full) at the first step-25 evaluation and Hub-staging save and was restarted 18 times by the supervisor, each time resuming from last/ at step 20 and repeating steps 21 to 25; the staging directories were real directories on the root disk instead of symlinks to /dev/shm. Fixed at 15:35 UTC; the model path of the 80 steps is otherwise the planned one (one seed). About 3.2 h were productive. The repeated step-20-to-25 segments are visible as repeated step-25 evaluations in the metrics log (Figure of the training records).
- **Not reproducible from what is stored:** stage-A steps 1-40 hyper-parameters except through the stage report; B1 final's evaluation; per-sentence NLLB-3.3B outputs on NTREX and MAFAND (only corpus-level aggregates survive); the training seed spread; the SFT with another seed (stage B alone is 44.8 A100-hours).

Q Prompt format and data statement

Prompt format. Every translation request is the fixed prompt Translate the text to {language}: {english} (two spaces after the colon) inside the Gemma 3 chat template; the model answers <tag> {translation}<end_of_turn> with the single-token tag of the target language; the loss covers the answer only (Section 3). **RL prompts** are English windows of consecutive sentences from FineWeb-Edu and Wikipedia (70%) and from news (30%), filtered against the protected index; 81 languages with higher draw weights for the 29 weakest (Section 3).

Data statement. Training data are English-to-African-language pairs and instruction data in 125 languages from the sources listed in Appendix O and Appendix O; many low-resource languages are dominated by religious text; stage B includes NLLB-mined, Aya-aligned and machine-translated instruction data; evaluation data are human references of FLORES+ devtest, NTREX-128, MAFAND-MT, SmolSent, SALT, GlotStoryBook, IbomMT, FLORES-MOZ and AfriScience-MT. No personal data was collected by us; no human subjects were involved; no human evaluation was run.

R Responsible NLP checklist: items filled from the paper

- **Scientific artefacts used and licences.** Base model: Gemma-3-270m-it (Gemma terms of use apply to the derived weights). Reward and monitor models, as declared on the Hub (read 2026-11-05): McGill-NLP/ssa-comet-mtl and ssa-comet-qe, masakhane/africomet-qe-stl-1.1 and africomet-stl-1.1: apache-2.0; GlotLID (cis-lmu/glotlid): license: other,

`license_name: apache-2.0-plus-notices`. Training and reward-side data and their licences are listed in Ethical considerations and Appendix O: NLLB-mined and NLLB-derived data (CC-BY-NC terms), IbomMT (CC-BY-NC-4.0), AfriScience-MT train (Apache-2.0), FineWeb-Edu (ODC-By), Wikipedia (CC BY-SA), several sources with unspecified licences; TranslatePsy (CC-BY-NC-4.0, NLLB-3.3B targets) only in the research control. Evaluation uses the public benchmarks named in Section 4; the NLLB-218 language-ID model (CC-BY-NC) was used for measurement only. Release licence of the weights: the Hub repositories declare none **[OWNER: confirm release licence]**.

- **Parameters and compute.** 385,144,448 parameters; one A100 80 GB at a time. Table 83 gives A100-hours per component, *estimated from logged durations* (training seconds from the run summaries, or the logged start and end times of the job); no cost figure is reported. The sized components add up to about 94 A100-hours, a lower bound. **Not sized** (no duration in any report): the A0 stage (5,625 steps) and SFT-1 (1,500 steps); E-01 (guard grid and guarded full evaluations); E-09 and E-09b (the fp32 generation ran 17:46 to 18:38 UTC beside another job at a 0.15 GPU share); E-13 and E-20; the SSA-COMET scoring of E-05; the full evaluation passes of every stage (122,585 rows each) and the AfriScience and document evaluations; data preparation and scoring boxes; aborted runs and the compute of the earlier 270M models.
- **Number of runs and seeds.** One training run per model and stage (seed 1234); two extra seeds (20261105, 20261106) of the stage-A recipe at 100 steps; three ablation arms, one run each; the control and the E-14 run are one run each. No multiple-run statistics beyond that pair; test-sentence bootstrap intervals elsewhere.
- **Hyper-parameters and selection.** Appendix I (all SFT and RL settings, per-shard schedules, reward and features). Checkpoints were selected on a 720-sentence FLORES+ dev subset; the decode-time guard setting by a rule fixed in advance on the SFT dev grid.
- **Evaluation.** Metrics defined in Section 4; sacreBLEU 2.6.0 (from the install log of the evaluation machine; the full signature strings were not recorded), chrF++ = `CHRF(word_order=2)` with the defaults, spBLEU = BLEU with `tokenize=flores200`, NFC-normalised hypotheses and references; vLLM 0.32.0, bf16, greedy, 768 new tokens; intervals are 95% paired bootstrap, 1,000 resamples (language bootstrap, 10,000 resamples, for NLLB). Exact commands per table are in the archived scripts named in the ledger.
- **Human subjects and annotators.** None were involved; no human judgement of system outputs was collected; no personal data was collected by us.
- **Code, data and checkpoints.** See the links in Ethical considerations and the branch list in the reproducibility statement, subject to the licence limits above. NLLB-derived and non-commercial parts cannot be redistributed as they are, and TranslatePsy rows are not released **[OWNER: which parts of the data are released]**.

Table 83: A100-hours by component, estimated from logged durations (lower bound; unsized components are listed above).

component	A100-h	basis
SFT stage B (31,353 steps)	44.8	5.14 s per step
RL stage A, B1, mixed (400 + 135 + 60 steps)	8.66	15,920 + 10,479 + 4,786 s
weak-group RL (85 steps)	2.1	7,642 s
science SFT (283 steps)	0.3	17 min
science RL (160 steps)	4.8	17,153 s
long-document RL (80 steps)	7.7	wall-clock 10:03–17:44, of which about 4.5 h crash loop
E-03 control (6,000 steps and two probes)	10.5	GPU session 9.75 h, evaluation 0.55 h, SSA-COMET pass 0.2 h
E-02 beam-4 and MBR-16	2.6	logged GPU time
E-04 two seed runs and their evaluations	3.2	wall-clock 08:51–12:05
E-11 three ablation arms and evaluations	2.9	36 + 54 + 42 min of training, 3 × about 15 min of evaluation
E-14 science RL without science SFT	6.0	training 15:18–20:17, evaluation about 1 h
sized components, total	about 94	lower bound